

DNA/polysome phase separation and cell width confinement couple nucleoid segregation to cell growth in *Escherichia coli*

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eLife Assessment

This **important** study presents **compelling** observational data supporting a role for transcription and polysome accumulation in the separation of newly replicated bacterial chromosomes. The study is generally thorough and rigorous in nature, although there are several instances where revisions would help clarify for the reader that the evidence is primarily circumstantial in nature and that a direct causal relationship between polysome accumulation has yet to be tested. With regard to the latter, the model's predictions could possibly be tested by examining the impact of translation inhibitors on nucleoid organisation. The authors could also compare the radial dimensions of the nucleoid with cell width to confirm that the nucleoid is radially confined across all conditions, a critical assumption of the model.

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Abstract

Chromosome segregation is essential for cellular proliferation. Unlike eukaryotes, bacteria lack cytoskeleton-based machinery to segregate their chromosomal DNA (nucleoid). The bacterial ParABS system segregates the duplicated chromosomal regions near the origin of replication. However, this function does not explain how bacterial cells partition the rest (bulk) of the chromosomal material. Furthermore, some bacteria, including *Escherichia coli*, lack a ParABS system. Yet, *E. coli* faithfully segregates nucleoids across various growth rates. Here, we provide theoretical and experimental evidence that polysome production during chromosomal gene expression helps compact, split, segregate, and position nucleoids in *E. coli* through phase separation, inherently coupling these processes to biomass growth across nutritional conditions. Halting polysome formation immediately stops sister nucleoid migration while ensuing polysome depletion gradually reverses nucleoid segregation.

Redirecting gene expression away from the chromosome and toward plasmids arrests nucleoid segregation and causes ectopic polysome accumulations that drive aberrant nucleoid dynamics. Cell width perturbations show that radial confinement of polysomes and nucleoids spatially controls their phase separation to ensure that nucleoids split along the cell width and segregate along the cell length. Our findings suggest a built-in mechanism for coupling chromosome segregation to cell growth and highlight the importance of cell width regulation in nucleoid segregation.

Introduction

All cells must segregate their replicated chromosomes to propagate genetic information to daughter cells at division. Remarkably, this essential process is far less understood in bacteria than eukaryotes. Eukaryotic cells use the mitotic spindle, a sophisticated cytoskeleton-based machine. No equivalent structure has been identified in bacteria, which fold their chromosomal material into a membrane-less organelle called the nucleoid. Many bacterial species use a ParABS system to segregate a specific DNA region proximal to the chromosomal origin of replication (*oriC*) (Figge et al., 2003; Ireton et al., 1994; Jalal and Le, 2020; Lim et al., 2014; Livny et al., 2007; Mohl and Gober, 1997). However, this initial step does not explain how the rest of the replicated megabase-long chromosomes partition. Furthermore, molecular dynamics simulations suggest that the segregation of the duplicated *oriC* regions is dispensable for the segregation of the entire chromosome (Harju et al., 2024). Consistent with this idea, the chromosomally encoded *para* and/or *parB* gene are often not essential for cell viability (Bartosik et al., 2009; Charaka and Misra, 2012; Donczew et al., 2016; Donovan et al., 2010; Du et al., 2016; Ireton et al., 1994; Jakimowicz et al., 2007a, 2007b; Jecz et al., 2015; Kadoya et al., 2011; Kawalek et al., 2020; Kim et al., 2000; Lagage et al., 2016; Lee and Grossman, 2006; Lewis et al., 2002; Li, 2019; Li et al., 2015; Minnen et al., 2011; Santi and McKinney, 2015; Takacs et al., 2022; Yamaichi et al., 2007; Yu et al., 2010). In fact, some bacteria do not even encode a ParABS system for *oriC* partitioning (Livny et al., 2007), with *Escherichia coli* being a prime example. Yet, *E. coli* segregates its duplicated nucleoids faithfully. *E. coli* mutants in which the *mukB* gene is deleted exhibit moderate nucleoid segregation defects linked to the organization and condensation of the nucleoid (Danilova et al., 2007; Mäkelä et al., 2021; Sawitzke and Austin, 2000). However, even in these mutants, most cells successfully segregate and partition their nucleoids between the daughter cells. Furthermore, topoisomerase I mutations or *seqA* deletion suppress the chromosome segregation defects associated with the *mukB* deletion (Danilova et al., 2007; Sawitzke and Austin, 2000; Weitao et al., 1999), suggesting that at least one other mechanism is at play.

Nucleoid segregation must somehow be coordinated with cell growth to accommodate the various growth rates that *E. coli* can display depending on the environmental conditions. In the presence of a poor-quality carbon source, *E. coli* initiates nucleoid segregation late in the cell division cycle. As the quality of the carbon source improves and the growth rate increases, nucleoid segregation occurs at an increasingly earlier stage of the cell division cycle until it initiates in the preceding cycle to accommodate faster division times (Bates and Kleckner, 2005; Govers et al., 2024; Tiruvadi-Krishnan et al., 2022). Over the years, multiple mechanisms have been proposed to explain nucleoid segregation in *E. coli*. In 1963, Jacob et al. hypothesized that sister chromosomes migrate from mid-cell to quarter-cell positions through membrane attachment of the sister chromosomes and zonal growth of the cell envelope between the attachment points (Jacob et al., 1963). This early model was largely abandoned following the discovery that *E. coli* cells elongate via a dispersed mode of cell wall growth along the cell length up to the final stage of the division cycle (Cooper and Hsieh, 1988; Navarro et al., 2022; Wientjes and Nanninga, 1989; Woldringh et al., 1987). Moreover, the segregation of chromosomal loci is faster than the rate of cell elongation (Kuwada et al., 2013). DNA replication has also been proposed to facilitate chromosome partitioning through a DNA polymerase-dependent extrusion mechanism (Lemon and

Grossman, 2000 [\[1\]](#)). Simulations of coarse-grained polymer models have also suggested that the conformational entropy of confined circular nucleoid chains could provide an unmixing force that separates two mixed DNA polymers (Jun and Mulder, 2006 [\[2\]](#); Jun and Wright, 2010 [\[3\]](#); Pande et al., 2023 [\[4\]](#)). While these proposed mechanisms likely contribute to the nucleoid splitting into two objects, they cannot explain how separated sister nucleoids move away from each other. Additionally, the DNA-demixing models consider the DNA polymers in an empty cylinder, ignoring any potential effects from crowders. Furthermore, recent modeling work has argued that entropic forces generated from excluded volume interactions between partially replicated chromosomes inhibit (rather than promote) their segregation (Harju et al., 2023 [\[5\]](#)). This argues for the existence of one or more additional segregation mechanisms, which could involve mechanical stress between overlapping sister nucleoids (Fisher et al., 2013 [\[6\]](#)), DNA loop extrusion (Harju et al., 2023 [\[5\]](#)), or excluded volume effects between the DNA meshwork and cytoplasmic crowders (Miangolarra et al., 2021 [\[7\]](#); Wu et al., 2019b [\[8\]](#)). While it is likely that multiple factors contribute to nucleoid segregation and positioning inside the *E. coli* cell, the dominant mechanism underlying the diffusional bias must be tightly coupled to cellular metabolism to explain how the timing of nucleoid segregation changes with growth rate. How this is achieved remains unknown.

In contrast to eukaryotes, bacteria do not insulate their DNA in membrane-bound compartments. As a result, the chromosomal meshwork can freely interact with the macromolecules present in the cytoplasm, which exhibits a high level of polydispersity and crowding (McGuffee and Elcock, 2010 [\[9\]](#); Zimmerman and Trach, 1991 [\[10\]](#)). Experiments and modeling have suggested that macromolecular crowders exert compaction forces on the compressible nucleoids through steric (i.e., excluded-volume) interactions (Pelletier et al., 2012 [\[11\]](#); Wu et al., 2019b [\[8\]](#); Yang et al., 2020 [\[12\]](#); Zhang et al., 2009 [\[13\]](#)). In this context, polysomes, which consist of mRNAs loaded with translating ribosomes, constitute a sizeable and abundant cytoplasmic crowder (McGuffee and Elcock, 2010 [\[9\]](#)). The proteome fraction allocated to ribosomes increases with growth rate across nutrient conditions (Chure and Cremer, 2023 [\[14\]](#); Dai et al., 2017 [\[15\]](#); Dourado and Lercher, 2020 [\[16\]](#); Hu et al., 2020 [\[17\]](#); Molenaar et al., 2009 [\[18\]](#); Scott et al., 2014 [\[19\]](#); Si et al., 2017 [\[20\]](#)), reaching up to 20% of the cell mass during fast growth (Chure and Cremer, 2023 [\[14\]](#)). The proportion of ribosomes engaged in translation also increases with improving nutrient quality, attaining 70-80% under fast growth (Dai et al., 2017 [\[15\]](#); Forchhammer and Lindahl, 1971 [\[21\]](#); Mohapatra and Weisshaar, 2018 [\[22\]](#); Sanamrad et al., 2014 [\[23\]](#); Varricchio and Monier, 1971 [\[24\]](#)). These polysomes form structures (Brandt et al., 2009 [\[25\]](#)) of comparable or larger size than the average ~50 nm mesh size of the nucleoid (Xiang et al., 2021 [\[26\]](#)). Consistent with a steric clash between polysomes and the nucleoid meshwork, fluorescently-labeled ribosomes accumulate in DNA-free regions such as the cell poles and between segregated nucleoids (Azam et al., 2000 [\[27\]](#); Bakshi et al., 2015 [\[28\]](#), 2012 [\[29\]](#); Chai et al., 2014 [\[30\]](#); Gray et al., 2019 [\[31\]](#); Lewis et al., 2000 [\[32\]](#); Mohapatra and Weisshaar, 2018 [\[22\]](#); Robinow and Kellenberger, 1994 [\[33\]](#); Sanamrad et al., 2014 [\[23\]](#); Xiang et al., 2021 [\[26\]](#)).

Interestingly, theoretical work suggests that excluded-volume effects alone may be sufficient for the DNA to spontaneously phase separate from polysomes and compact into its observed nucleoid form (Mondal et al., 2011 [\[34\]](#)). Furthermore, a recent nonequilibrium statistical physics model proposes that nucleoid positioning and segregation could potentially be explained by considering the steric interaction (i.e., repulsion) between polysomes and DNA and the nonequilibrium effects associated with mRNA synthesis and degradation (Miangolarra et al., 2021 [\[7\]](#)). In this theoretical model, polysome accumulation in the middle of the nucleoid emerges due to polysomes born in the middle of the nucleoid taking longer to diffuse out of the nucleoid than those born closer to the nucleoid edges. Polysome accumulation beyond a certain concentration threshold drives phase separation between the DNA and polysomes at the mid-nucleoid location, resulting in nucleoid splitting. However, we lack experimental validation of this model. Furthermore, the coupling between nucleoid segregation and growth rate has not been addressed either experimentally or theoretically.

Here, we quantitatively characterize the temporal and spatial dynamics of nucleoids and polysomes in *E. coli* under different conditions and perturbations. In this work, we refer to nucleoid segregation as a series of events observable by microscopy (**Figure 1A**): (i) the initiation of nucleoid splitting, marked by the depletion of a DNA marker near the mid position of the nucleoid, (ii) the end of nucleoid splitting, which results in the generation of two separable nucleoid objects, and (iii) the migration of the sister nucleoids away from each other, marked by the increasing distance between the centroid of the nucleoids. This sequence of events can initiate in the preceding cell division cycle under nutrient-rich (fast-growth) conditions. Our experimental findings, combined with modeling, provide evidence that out-of-equilibrium and asymmetric polysome rearrangements in the cell result in both DNA compaction and the choreography of the nucleoid segregation cycle during growth across cells and nutrient conditions. The dual involvement of polysomes in protein synthesis and nucleoid dynamics ensures self-regulating coordination between cell growth and chromosome segregation across a wide range of growth rates, even under conditions where *E. coli* divides faster than it can replicate its chromosome. We also show that cell width plays a critical role in the directional patterning of nucleoid segregation by spatially regulating the phase separation between polysomes and chromosomes.

Results

The rate of polysome accumulation correlates with the rate of nucleoid segregation across cells

While the negative correlation between ribosomes and DNA has been well established from snapshot images (Azam et al., 2000; Bakshi et al., 2015, 2012; Chai et al., 2014; Gray et al., 2019; Lewis et al., 2000; Mohapatra and Weisshaar, 2018; Robinow and Kellenberger, 1994; Sanamrad et al., 2014; Xiang et al., 2021), the relative dynamics of these two macromolecules have not been quantified. To examine these dynamics with high temporal resolution (every 1 or 3 min, **Figure 1 – figure supplement 1A-E**), we ran time-lapse microfluidic experiments using an *E. coli* strain carrying the 50S ribosomal protein RplA fused to GFP and the nucleoid-associated protein HupA tagged with mCherry (Xiang et al., 2021). Cells were grown at 30°C in M9 buffer supplemented with glucose, casamino acids and thiamine (M9gluCAAT). While the ribosome signal displayed the expected polar and inter-nucleoid enrichments, the accrual of ribosome signal in the middle of the cells was particularly strong compared to the polar regions, as shown by representative individual cells (**Figure 1B** and Movie S1) and average kymographs (**Figure 1C**, $n = 4122$ cell division cycles). Cell constriction contributed to the apparent depletion of ribosomal signal from the mid-cell region at the end of the cell division cycle (**Figure 1B-C** and Movie S1). In predivisional cells, the ribosomal signal accumulated in the middle of the segregated nucleoids near the $\frac{1}{4}$ and $\frac{3}{4}$ cell positions, as shown in average one-dimensional (1D) and two-dimensional (2D) fluorescence profiles ($n = 1907$ predivisional cells, **Figure 1D**). As expected (Bakshi et al., 2012; Sanamrad et al., 2014), the RplA-GFP signal became homogeneous within cells following their treatment with rifampicin (**Figure 1 – figure supplement 1F**), a transcription inhibitor that causes polysome depletion over time (Blundell and Wild, 1971; Campbell et al., 2001; Hartmann et al., 1967). This indicates that all ribosomal enrichments consist of polysomes and therefore will be referred to as polysome accumulations hereafter.

Under our relatively nutrient-rich growth conditions (M9gluCAAT), nucleoid splitting occurred early in the division cycle and even sometimes initiated in the segregated nucleoids from the preceding division cycle (Movie S1 and **Figure 1B**). Given the variability in nucleoid segregation timing across cells (**Figure 1 – figure supplement 2A**), we developed a computational method to track nucleoid dynamics independently of the cell division cycle (see Methods and **Figure 1 – figure supplement 2B-C**). Specifically, we focused on the nucleoid cycle—defined as the period between the ends of two nucleoid splitting events (**Figure 1 – figure supplement 2D**)—instead

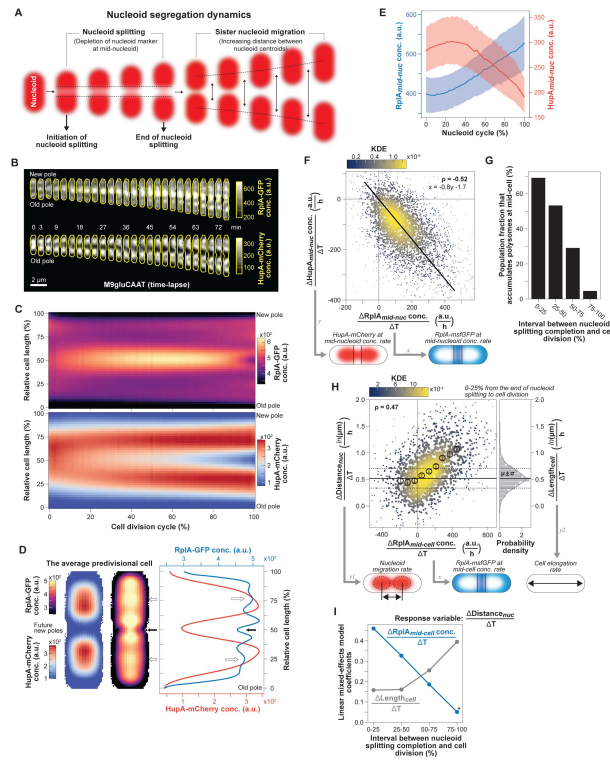


Figure 1

Correlations between polysome and nucleoid dynamics at the single-cell level.

CJW7323 cells were grown in in M9gluCAAT in a microfluidic device. **A.** Schematic illustrating observable nucleoid segregation events. **B.** Fluorescence images of RplA-GFP and HupA-mCherry for a representative cell (CJW7323) from birth to division. **C.** Ensemble kymographs of the average RplA-GFP and HupA-mCherry fluorescence during the cell division cycle (>300,000 segmented cell instances from 4122 complete cell division cycles). **D.** Two-dimensional projections of the average RplA-GFP and HupA-mCherry fluorescence signals in predivisional cells (4564 cells with two nucleoid objects, from 1907 cell division cycles, 95-100% into the cell division cycle) and their intensity profiles. White arrows indicate RplA-GFP enrichments at the quarter cell positions, while the black arrow indicates the site of cell constriction. **E.** Plot showing the dynamics of RplA-GFP accumulation and HupA-mCherry depletion at mid-nucleoid (median $\pm$ IQR) during the nucleoid cycle (see **Figure 1 – Figure supplement 2D**). Data from 3240 nucleoid segregation cycles are shown (40 nucleoid cycle bins, 2512-4823 segmented nucleoids per bin). **F.** Correlation (Spearman $\rho = -0.52$, p-value $< 10^{-10}$) between the rate of RplA-GFP accumulation in the middle of the nucleoid ($\frac{\Delta RplA_{mid-nuc} conc.}{\Delta T}$) and the rate of HupA-mCherry depletion in the same region ($\frac{\Delta HupA_{mid-nuc} conc.}{\Delta T}$) between the initiation of nucleoid splitting and just before the end of nucleoid splitting (3214 complete nucleoid cycles). The colormap indicates the Gaussian kernel density estimation (KDE). Solid black line indicates the linear-regression fit to the data. **G.** Percentage of cells that continue to accumulate polysomes in the middle of the cell ($\frac{\Delta RplA_{mid-cell} conc.}{\Delta T}$) during four relative time bins (1335 to 1957 cell division cycles per bin) covering the period from the end of nucleoid splitting until cell division. **H.** Correlation (Spearman $\rho = 0.47$, p-value $< 10^{-10}$) between the rate of RplA-GFP accumulation at mid-cell ($\frac{\Delta RplA_{mid-cell} conc.}{\Delta T}$) and the rate of distance increase between the sister nucleoids ($\frac{\Delta Distance_{nuc}}{\Delta T}$) during the first quartile (0-25%) of the period between the end of nucleoid splitting and cell division (1376 cell division cycles). The black markers correspond to nine bins (mean $\pm$ SEM, 75 to 177 cell division cycles per bin) within the 5th-95th percentiles of x-axis range. Also shown is the distribution of the cell elongation rates ($\frac{\Delta Length_{cell}}{\Delta T}$) during the same time interval, with the mean and SD shown by the solid and dashed lines, respectively. **I.** Plot showing the coefficients of a linear mixed-effects model (see **eq. 3** in Methods and **Figure 1 – figure supplement 3B**) for four interval bins between the completion of nucleoid splitting and cell division. The coefficients quantify the relative contribution of polysome accumulation at mid-cell ($\frac{\Delta RplA_{mid-cell} conc.}{\Delta T}$) and cell elongation ($\frac{\Delta Length_{cell}}{\Delta T}$) to the rate of sister nucleoid migration ($\frac{\Delta Distance_{nuc}}{\Delta T}$). All coefficients are significant (Prob($<|Z|$) $< 10^{-9}$), except for the one marked with an asterisk that is marginally significant (Prob($<|Z|$) = 0.02).

of the division cycle. By tracking the accumulation of RplA-GFP and the depletion of HupA-mCherry in the middle of nucleoids, we found that polysome accumulation and nucleoid splitting correlated in time (**Figure 1E**). Furthermore, the rate of polysome accumulation at mid-nucleoid ($\Delta RplA_{mid-nuc}/\Delta T$) correlated with the rate of DNA depletion in the same region ($\Delta HupA_{mid-cell}/\Delta T$) at the single-cell level (**Figure 1F**). This indicates that cells that accumulated polysomes faster also split their nucleoids faster. Importantly, the fitted linear regression had an intercept close to zero for both axes (**Figure 1F**), indicating that when the rate of polysome accumulation approached zero, so did the rate of nucleoid splitting.

To examine what happens when sister nucleoids move away from each other, we divided the time between the end of nucleoid splitting and cell division into four bins. We found that right after nucleoid splitting (0-25% bin), most (~70%) cells continued to accumulate polysomes at mid-cell, i.e., between the sister nucleoids (**Figure 1G**). For these cells, the rate of polysome accumulation ($\Delta RplA_{mid-cell}/\Delta T$) positively correlated with the rate of nucleoid migration ($\Delta Distance_{nuc}/\Delta T$) (**Figure 1H**). Thus, the faster that cells accumulated polysomes at mid-cell, the faster the sister nucleoids migrated apart (and vice versa).

Cell elongation may also contribute to sister nucleoid migration near the end of the division cycle

We noted that there was a substantial percentage (~30%) of cells with decreasing RplA-GFP signal at mid-cell right after nucleoid splitting ($\Delta RplA_{mid-cell}/\Delta T \leq 0$, **Figure 1H**). Interestingly, in these cells, nucleoid migration did not stop; instead, its average rate was similar to the average rate of cell elongation ($\Delta Length_{cell}/\Delta T$) (**Figure 1H**). In fact, when the cell elongation rate was subtracted from the nucleoid migration rate for each single cell (**Figure 1-figure supplement 3A**), the positive correlation between polysome accumulation and nucleoid migration rates remained, but now the average rate of nucleoid migration was near zero in cells with no polysome accumulation ($\Delta RplA_{mid-cell}/\Delta T \leq 0$). This finding may suggest a mixed contribution between polysome accumulation and cell elongation to nucleoid migration. This was interesting considering that polysomes became less enriched at mid-cell but more enriched in the middle of sister nucleoids in predivisional cells (**Figure 1B-D**), with the average ribosome concentration remaining constant during the cell division cycle (**Figure 1 – figure supplement 1E**). The spatial change in polysome enrichments was accompanied with a steady decline in the percentage of cells that continued to accumulate polysomes at mid-cell between the end of nucleoid splitting and cell division (**Figure 1G**). Concurrent with this decline, the migration rate of sister nucleoids became less correlated with the rate of polysome accumulation and more correlated with the rate of cell elongation (**Figure 1 – figure supplement 3B**). To examine the relative correlation of polysome accumulation and cell elongation with nucleoid migration over time, we used a linear mixed-effects model to analyze each relative nucleoid migration interval (see Methods). The coefficients of the fitted mixed linear regressions suggest the following hypothesis: Early during nucleoid migration (0-25% between the completion of nucleoid splitting and cell division), polysome accumulation contributes most to the measured variance in the displacement of the sister nucleoids (**Figure 1I**). This contribution progressively decreases over time while that of cell elongation increases (**Figure 1I**). Such a switch in relative contribution would be consistent with the spatiotemporal dynamics of polysome accumulation and cell wall synthesis, as described in the Discussion.

Polysome accumulation at mid-cell correlates with the relative timing of nucleoid segregation across nutrient conditions and growth rates

If polysome production plays a role in nucleoid segregation, it predicts that the timing and amount of polysome accumulation at mid-cell will correlate with the timing of nucleoid segregation across nutrient conditions. To test this prediction, we analyzed images of fluorescently labeled ribosomes

and nucleoids in cells grown under 30 different carbon source conditions (Table S1) that vary the doubling times (~ 40 min to ~ 4 h) and average cell areas (~ 1.9 to $\sim 3 \mu\text{m}^2$) of *E. coli*. This dataset included both previously published (Gray et al., 2019) and new microscopy snapshots from our laboratory. Demographs generated from these images revealed that the polysome accumulation at mid-cell was reproducible across all conditions and strains, irrespective of the ribosomal subunit protein (RplA or RpsB) or the fluorescent tag (msfGFP, mEos2 or GFP) used to mark ribosomes (Figure 2A and Figure 2 – figure supplement 1). In these experiments, nucleoids were typically imaged using DAPI (rather than a fluorescent fusion to HupA), indicating that polysome accumulation at mid-nucleoid was independent of the DNA labeling method. Importantly, and consistent with our prediction, the richer the growth conditions (i.e., the larger the average cell area), the earlier the polysome accumulation and the nucleoid splitting occurred in the division cycle based on relative cell lengths (Figure 2A and Figure 2 – figure supplement 1). In addition, the polysome accumulation at mid-cell was more pronounced in nutrient-rich media (e.g., M9malaCAAT) compared to nutrient-poor ones (e.g., M9mann) where the nucleoid segregated later in the division cycle (Figure 2A and Figure 2 – figure supplement 1).

To quantify these phenotypes across strains and nutrient conditions, we extracted and correlated population-level polysome and nucleoid statistics. We found a strong correlation (Spearman correlation $\rho = -0.85$) between the amplitude of the average ribosomal signal accumulation and the average nucleoid signal depletion at mid-cell (Figure 2B). Since the average cell area (colormap in Figure 2B) correlated with the growth rate of the population (Schaechter et al., 1958), this plot also confirmed that faster growing populations displayed stronger polysome accumulation and greater DNA depletion at mid-cell on average (Figure 2B).

We wondered whether the correlation between polysome and nucleoid dynamics could also be seen across cells growing in the exact same nutritional condition but at different growth rates due to cell-to-cell variability. To examine this possibility, we divided our timelapse microfluidics data into growth rate bins (Figure 2C) and compared their temporal polysome and nucleoid patterns. We found that, compared to the slowest growing subpopulation (bin 1 with a mean growth rate = 0.28 h^{-1}), the fastest growing subpopulation (bin 13 with = 0.74 h^{-1}) displayed a more pronounced and earlier accumulation of polysomes that was associated with earlier nucleoid segregation (Figure 2D). The subpopulation with an intermediate growth rate (bin 7 with = 0.51 h^{-1}) also displayed an intermediate pattern of polysome accumulation and nucleoid segregation. In fact, when we calculated the average polysome accumulation and nucleoid depletion at mid-cell for each growth rate bin, we found that the two statistics correlated (Figure 2E). Faster growing cells accumulated more polysomes at mid-cell and split their nucleoids earlier during the cell division cycle compared to slower growing cells.

Both the average proteome fraction dedicated to ribosomes and the average fraction of ribosomes engaged in translation are known to correlate with growth rate across nutrient conditions (Chure and Cremer, 2023; Dai et al., 2017; Dourado and Lercher, 2020; Hu et al., 2020; Molenaar et al., 2009; Scott et al., 2014; Si et al., 2017). Thus, a role for polysome production in nucleoid segregation may provide a mechanistic link between growth rate and the relative timing of nucleoid splitting both across cells in a given nutrient condition as well as across populations in different nutrient conditions.

Spatial polysome asymmetry correlates with nucleoid positioning

A surprising result was the apparent higher polysome accumulation at mid-cell relative to the cell pole regions (Figures 1B-C and 2A), which was not artificially created by the smaller cytoplasmic volumes at the cell poles or constriction sites due to membrane curvature (Figure 2 – figure supplement 2). This uneven distribution of polysomes suggested limited diffusion-driven equilibration of polysome concentration between the DNA-free regions. At division, such a disequilibrium could lead to a higher concentration of polysomes at the new cell pole relative to the old pole in daughter cells through inheritance. To examine this possibility, we turned back to

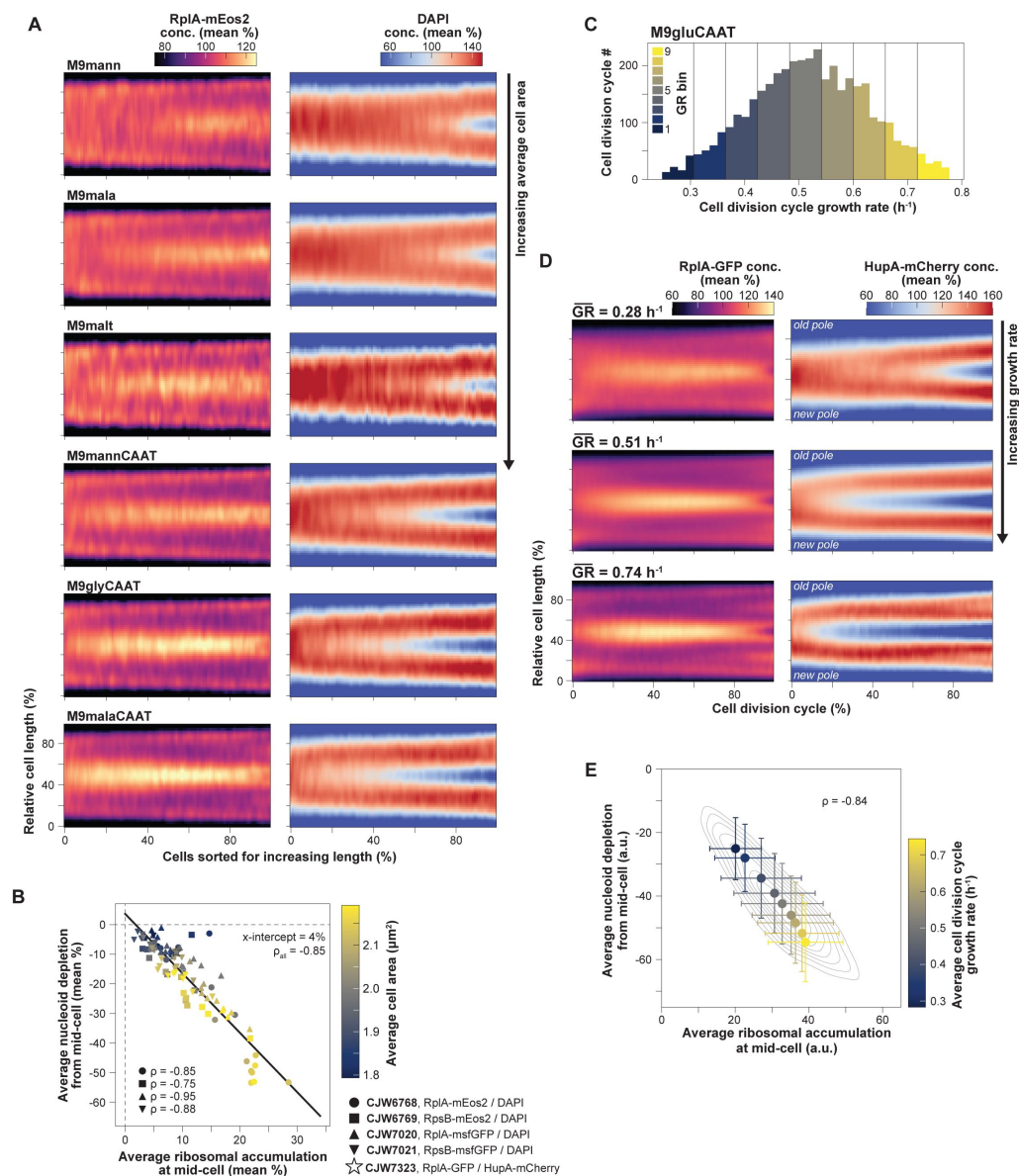


Figure 2

Correlation of the extent and relative timing of polysome accumulation with nucleoid segregation at the population and single-cell levels.

A. RplA-mEos2 and DAPI (scaled by the whole cell average) demographs constructed from snapshots of DAPI-stained CJW6768 cells (815 to 2771 cells per demograph) expressing RplA-mEos2 and growing in different nutrient conditions (see Table S1 for abbreviations). The demographs were arranged from smallest (M9mann, top) to biggest average cell area (M9malaCAAT, bottom). Additional demographs for different ribosomal reporters and nutrient conditions are shown in **Figure 2 – figure supplement 1**. **B.** Correlation between average polysome accumulation and average nucleoid depletion at mid-cell for all tested strains (Spearman $\rho_{\text{all}} = -0.85$, p-value $< 10^{-10}$) with different ribosomal and nucleoid reporters, and within each strain ($-0.75 < \text{Spearman } \rho_{\text{strain}} < -0.95$, p-values $< 10^{-3}$) across nutrient conditions. A linear regression was fitted to all the data. **D.** Plot showing the variability in growth rate (GR) across cell division cycles for cells growing in microfluidics in M9gluCAAT (4114 cell division cycles, 9 growth rate bins with 93 to 860 cell division cycles each). **E.** Ensemble kymographs of the RplA-GFP and HupA-mCherry concentration normalized by the average fluorescence for the slowest, intermediate and fastest growing population bins shown in panel D. **F.** Plot showing the correlation between the average RplA-GFP accumulation and HupA-mCherry depletion at mid-cell across the growth rate bins (mean $\pm$ SD, 93 to 860 cell division cycles per bin) shown in panel D.

the microfluidic experiments in which we traced cell lineages from mother to daughter cells (**Figure 1 – figure supplement 1A** [↗](#)), determined the pole identity (new vs. old) of each tracked cell (**Figure 1 – figure supplement 2B** [↗](#)), and compared the polysome accumulations between the new and old poles (see Methods and **Figure 3 – figure supplement 1A-D** [↗](#)). Old mother cells (located at the end of the microfluidic channels) and their daughters were excluded from our analysis to avoid cell aging effects (Chao et al., 2023 [↗](#); Coquel et al., 2013 [↗](#); Koleva and Hellweger, 2015 [↗](#); Łapińska et al., 2019 [↗](#); Lindner et al., 2008 [↗](#); Proenca et al., 2019 [↗](#)). Consistent with the absence of rapid equilibration of polysome concentration between DNA-free regions through diffusion, we found that newborn cells had more polysomes at the new pole compared to the old one on average (**Figure 3A** [↗](#)).

Across cells, the polysome distribution asymmetry between cell poles negatively correlated with the position of the nucleoid in newborn daughter cells (**Figure 3B** [↗](#), cells always oriented with the new pole to the right). The x and y intercepts of a fitted linear regression to the data were not zero, indicating that a nucleoid positioned precisely at the cell center did not equate with an even distribution of polysomes between poles. Instead, newborn cells with centrally located nucleoids tended to have more polysomes at the new pole compared to the old pole (example #1 in **Figure 3B-C** [↗](#)) whereas cells born with symmetric polysome enrichments between poles tended to display an off-center nucleoid closer to the new pole (example #2 in **Figure 3B-C** [↗](#)).

Spatial polysome asymmetry correlates with asymmetric nucleoid compaction in newborn cells

Construction of average 2D cell projections of the RplA-GFP signal in cells sorted based on their relative timing to cell division confirmed the polysome asymmetry between poles in newborn cells (**Figure 3D** [↗](#) and Movie S2). Strikingly, the corresponding 2D projections of the HupA-mCherry signal revealed another spatial asymmetry, this time, in the average DNA mass distribution along the nucleoid length (axial asymmetry). The average HupA-mCherry signal concentration was higher toward the new pole right after birth or, correspondingly, toward the middle of the cell (future new pole) in the period prior to division (**Figure 3D** [↗](#) and Movie S2). Quantification of this axial asymmetry in newborn cells revealed that the HupA-mCherry concentration is ~20% higher toward the new pole on average (**Figure 3E** [↗](#)), suggesting that the DNA density is uneven along the nucleoid length. This nucleoid mass asymmetry emerged late in the nucleoid migration cycle, typically before cell division such that it was inherited by newborn cells (**Figure 3D** [↗](#)).

To examine if features of polysome accumulations (e.g., position, amplitude, or fraction of cell length covered) correlate with the asymmetric nucleoid density and its variability among cells, we used a linear mixed-effects model (see Methods). We found that in newborn cells, the positions of polysome enrichments in the old-pole and mid-cell regions significantly correlated with the HupA-mCherry density asymmetry (**Figure 3 – figure supplement 1E-G** [↗](#)). Guided by this finding, we hypothesized that the positions of the accumulating polysomes along the cell length determine the space available for the chromosome to occupy, and thereby dictate the DNA density distribution along the nucleoid. To examine this hypothesis, we combined the correlated polysome accumulation characteristics into a compound statistic that describes the relative polysome-free space between two cell halves (see Methods). Compared to other polysome statistics, the relative availability of cell space depleted of polysomes (RplA-GFP signal) correlated most strongly with the asymmetric distribution of DNA (HupA-mCherry signal) in newborn cells (**Figure 3F** [↗](#)). In other words, the DNA concentration was higher in cell regions with more space available between polysome enrichments. In most (~60%) cells, the distance between polysome enrichments, and thus the DNA concentration along the nucleoid, was greater between mid-cell and the new pole (i.e., x and y values >1, example #1 in **Figure 3F-G** [↗](#)). The opposite pattern was true for a small fraction (~14%) of cells, where the larger distance between polysome enrichments was located between mid-cell and the old pole (x and y values < 1, example #2 in **Figure 3F-G** [↗](#)). The correlated variability in polysome and nucleoid patterning across cells suggests that the size of the

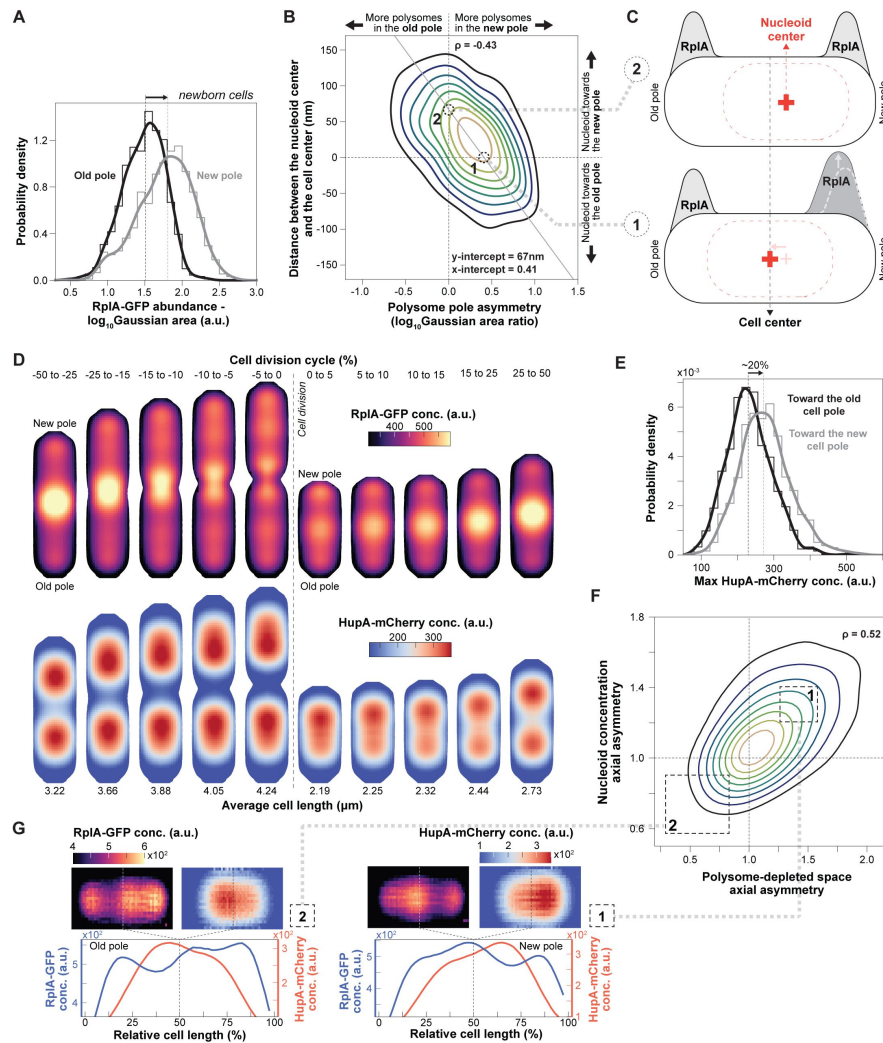


Figure 3

Correlations between polysome and nucleoid asymmetries.

A. Distributions of RplA-GFP concentration in the new (grey) and the old (black) pole regions of newborn cells (0-2.5% of the cell division cycle, $n = 912$ cell division cycles). The histograms were smoothed using Gaussian kernel density estimations. **B.** Correlation (Spearman $\rho = -0.41$, p -value $< 10^{-10}$) between the polar polysome asymmetry and the position of the nucleoid centroid around the cell center of cells at the beginning of the division cycle (0-10%, $n = 1179$ cell division cycles). The contour plot consists of 9 levels with a lower data density threshold of 25%. The polar polysome profile for cells with correlations indicated by the numbers 1 and 2 are schematically illustrated in the next panel. A linear regression (solid grey line) was fitted to the data. **C.** Schematic illustrating the effects of the relative polysome abundance between the poles on the position of the nucleoid. **D.** Average 2D projections of the RplA-GFP and HupA-mCherry concentration (conc.) at different cell division cycle intervals (~9440 to 47240 cell images per cell division cycle interval from 4122 cell division cycles). The dotted line indicates the boundary between two cell division cycles. **E.** Density plot comparing the distribution of the HupA-mCherry maximum concentration toward the new pole (gray) to that toward the old pole (black) in newborn cells (0-2.5% of the division cycle, $n = 912$ cell division cycles). The histograms were smoothed using Gaussian kernel density estimations. **F.** Correlation (Spearman $\rho = 0.52$, p -value $< 10^{-10}$) between the nucleoid density asymmetry and the relative availability of polysome-free space between the two cell halves early in the cell division cycle (0-10% of the division cycle, $n = 2150$ cell division cycles). The contour plot consists of 9 levels with a lower data density threshold of 25%. Values above 1 on the x-axis indicate more polysome-free space toward the new pole, and values below 1 correspond to cells with more polysome-free space toward the old pole. On the y-axis, values above 1 indicate higher DNA density toward the new pole and values below 1 indicate higher DNA density toward the old pole. **G.** Average 2D projections of newborn cells (0-10% into the cell division cycle) from the lower-left quartile in panel C (region 2, $n = 223$ cell division cycles) and the upper right quartile in panel C (region 1, $n = 557$ cell division cycles) and their 1D intensity profiles.

polysome-depleted spaces helps determine where the chromosomal DNA is most concentrated along the cell length. This patterning is likely reinforced through the displacement of the polysomes away from the DNA dense region.

A minimal reaction-diffusion model generates experimentally observed cellular asymmetries and growth rate-dependent nucleoid segregation

Our single-cell studies provided experimental support that phase separation between polysomes and DNA contributes to nucleoid segregation. However, it remained unclear whether the same mechanism could also explain the growth rate-dependent trends and cellular asymmetries that we observed (**Figures 2** and **3**). Therefore, we built a minimal reaction-diffusion model (see Methods) to describe the dynamics of polysomes and DNA during the cell cycle based on previous findings (Miangolarra et al., 2021). The model takes into account two ingredients important for nucleoid segregation: steric effects (repulsion) between DNA and polysomes (described by Cahn-Hilliard theory) and the nonequilibrium processes of polysome synthesis and degradation (described by linear reaction kinetics). The model is based on realistic parameters of polysome diffusion, production, and degradation (see Methods). We assume polysome production (i.e., transcription) to be uniform within the nucleoid and polysome degradation (i.e., mRNA decay) to be uniform across the entire cell (from pole to pole). These assumptions consider the most trivial cases (see Discussion for other cases). We also assume that the cell grows exponentially and that the nucleoid expands in size proportionally to the cell during growth, which has been experimentally verified (Campos et al., 2014; Govers et al., 2024). Since *E. coli* grows along its long axis and polysomes do not readily go around the nucleoid to equilibrate (**Figure 3A-D** and **Figure 2 – figure supplement 2**), we reduced the problem to one dimension, the cell length. Importantly, the system is driven out of equilibrium by the continuous production and degradation of polysomes. It operates at a nonequilibrium steady state even at fixed cell length.

Initialized from the steady state of a single compacted nucleoid in the middle of the cell with symmetric polysome distributions at the poles, our model shows that polysomes accumulate in the middle of the nucleoid during growth (**Figure 4A** and Movie S3, left panel). This is followed by division of the nucleoid into two entities, which then move apart from each other as polysomes accumulated between them. The model provides a minimal mechanism for nucleoid segregation: At any given point, polysomes that form in the middle of the nucleoid have a lower probability of escaping the nucleoid through diffusion compared to polysomes born at the edge of the nucleoid (**Figure 4B**). Thus, the polysome concentration rises monotonically towards the center of the nucleoid, its peak increasing with nucleoid length (quadratically in the quasi-steady-state limit). Nucleoid elongation during cell growth promotes polysome accumulation in the middle of the nucleoid. Phase separation driven by the steric repulsion between DNA and polysomes results in the nucleoid splits in two (**Figure 4A** and Movie S3, left panel).

We wondered if this simple reaction-diffusion mechanism could also explain the correlation between growth rate and the relative timing of nucleoid segregation. Therefore, we performed simulations for different growth rates, matching the cell and nucleoid length at birth with population-level measurements (**Figure 4 – figure supplement 1A**). To initialize each simulation in a realistic fashion, we used the last time point (i.e., half of the predivisional cell) of the previous simulation as initial conditions, capturing the new/old pole identity as well as any cellular asymmetries inherited between generations. The relative timing of nucleoid splitting was measured as nucleoid depletion at mid-cell beyond a threshold of 0.1 arbitrary units (**Figure 4A** and **Figure 4 – figure supplement 1B**). We found that our model successfully captures the negative trend between the growth rate and the relative timing of nucleoid splitting. Simulated cells that grew faster also split their nucleoids earlier from birth to division (**Figure 4C** and Movie S3), in agreement with population-level data (**Figure 2 – figure supplement 1**).

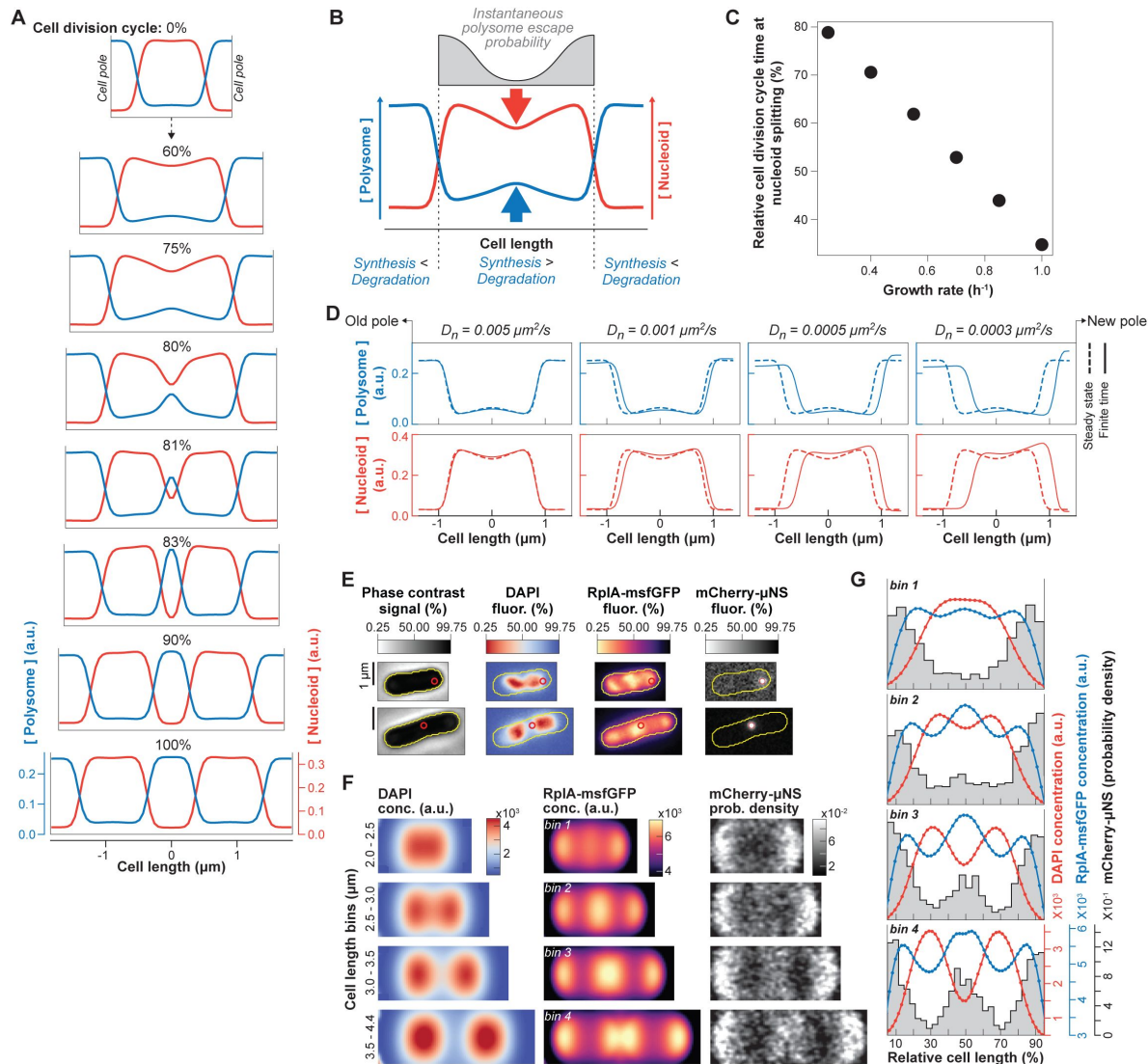


Figure 4

Simulation results of the reaction-diffusion model for different growth rates or nucleoid diffusion rates, and comparison of the nonequilibrium polysome accumulation with freely diffusing particles.

A. 1D simulation of polysome (blue) and nucleoid (red) dynamics during slow growth (growth rate = 0.25 h^{-1} , $D_n = 0.001 \mu\text{m}^2/\text{s}$, cell length at birth = $1.84 \mu\text{m}$) at different relative cell division cycle timepoints. The simulation was initialized from the equilibrium polysome and nucleoid distribution (at 0%). **B.** Schematic summarizing how polysomes accumulate in the middle of the elongating nucleoid, causing nucleoid splitting. **C.** Correlation between the relative timing of nucleoid splitting and the growth rate as captured by our reaction-diffusion model ($D_n = 0.001 \mu\text{m}^2/\text{sec}$) across six growth rate bins. The cell and nucleoid lengths for each growth rate bin matched previously published data shown in **Figure 4 – figure supplement 1A**. **D.** Deviation between the steady state after infinite relaxation time (dashed curves) and the polysome or nucleoid profiles in newborn cells after one simulation round (solid curves) for increasing nucleoid diffusion constants during fast growth (1 h^{-1}). **E.** Phase contrast and fluorescence (fluor.) images of two representative single cells (CJW7651). The red circles indicate the position of the mCherry- μNS particle in each cell. **F.** Two-dimensional average cell projections of the DAPI concentration (conc.) and the RplA-msfGFP concentration, and 2D histogram of the mCherry- μNS particle density for four cell length bins of CJW7651 cells (~ 2580 cells per bin) grown in M9gluCAAT and spotted on an agarose pad containing the same medium. Since the cell pole identity cannot be inferred from snapshot images, pole assignment was random. **G.** Average 1D profiles of the scaled DAPI and RplA-msfGFP concentrations and the mCherry- μNS probability density.

To examine a potential origin for the asymmetries in polysome distribution and nucleoid compaction that we observed (**Figure 3**), we examined the effect of the nucleoid diffusion coefficient D_n , which is a model parameter describing how fast the nucleoid relaxes towards its equilibrium configuration. Large D_n represents the quasi-steady-state limit where the nucleoid relaxation time scale is much shorter than the cell doubling time, and the nucleoid always assumes its (symmetric) equilibrium state. Conversely, small D_n (i.e., slower relaxation time) can lead to asymmetric concentration profiles that will be inherited by the daughter cells. Consistent with this expectation, we found that during relatively fast growth (~40 min doubling time), a lower nucleoid diffusion coefficient results in a larger deviation from the equilibrium concentration profiles for the nucleoid and polysomes (dotted curves vs. solid curves, **Figure 4D**). In fact, at diffusion coefficients below $0.001 \mu\text{m}^2/\text{s}$, the model (**Figure 4D**) reproduced the experimentally observed asymmetries, including the nucleoid position offset towards the new pole at birth, the higher polysome concentration at the new pole compared to the old one, and the asymmetric nucleoid compaction (**Figure 3**). Our minimal model thus suggests that the material properties of the nucleoid (e.g., stiffness) may contribute to the observed nucleoid and polysome asymmetries in *E. coli* (see Discussion).

Polysomes accumulate at mid-nucleoid in DNA regions inaccessible to freely diffusing particles of similar sizes

In the model, the early polysome accumulation in the middle of nucleoid is caused by the nonequilibrium processes of polysomes being born within the nucleoid while being degraded uniformly across the cell (due to mRNA turnover). It predicts that the early mid-nucleoid enrichment of polysome signal observed in our experiments is the product of such nonequilibrium processes rather than of polysomes simply diffusing into undetected DNA-free space. If this is correct, freely diffusing objects of similar sizes to polysomes should accumulate at mid-cell after polysomes accumulate there, i.e., after free DNA-space has been generated through nucleoid splitting. To test this expectation, we compared the average distribution of RplA-msfGFP with that of freely diffusing mCherry-labeled μNS particles from snapshot images of DAPI-stained cells (**Figure 4E**). These μNS particles consist of a fragment of a mammalian reovirus protein (Broering et al., 2005, 2002) that self-assembles into a particle, typically one per cell, when produced orthogonally in *E. coli* (Parry et al., 2014). They have sizes between 50 and 200 nm (Parry et al., 2014; Xiang et al., 2021) similar to polysomes (Brandt et al., 2009; Slayter et al., 1968) and are therefore largely excluded by the nucleoid mesh (Xiang et al., 2021). After sorting cells by length into four bins, the positions of mCherry-labeled μNS particles from approximately 2580 cells per bin were superimposed using the relative cellular coordinates to construct particle density maps. Cells were randomly oriented in this analysis (meaning that asymmetries between poles cannot be observed), as the pole identity cannot be assigned from snapshot images.

We found that in the shortest (i.e., newborn) cells, RplA-msfGFP-labeled polysomes had already accumulated within the nucleoid (**Figure 4F**, bin 1). In contrast, mCherry- μNS particles were restricted to the cell poles and were not able to access the mid-cell region until after nucleoid splitting was clearly visible (**Figure 4F**, bins 2-4). This was also shown in the corresponding 1D average concentration and particle density profiles (**Figure 4G**). These observations support the notion that the early mid-nucleoid accumulation of RplA-msfGFP is caused by nonequilibrium effects associated with polysome synthesis and degradation rather than polysome diffusion into DNA-free space.

Arrest of polysome production immediately stops nucleoid segregation while polysome depletion gradually reverses it

Our correlative analyses and model (**Figures 1–4**) support the hypothesis that the interactions and ensuing phase separation between polysomes and nucleoid promote nucleoid segregation and macromolecular asymmetries along the cell length. To probe causality, we first aimed to disrupt the proposed mechanism using rifampicin. Rifampicin treatment is known to homogenize ribosome distribution and expand the nucleoid over time through polysome depletion (Bakshi et al., 2014; Dworsky and Schaechter, 1973; Koch and Gross, 1979; Pettijohn and Hecht, 1974; Xiang et al., 2021). However, our hypothesis predicts a faster effect on nucleoid segregation. Blocking transcription should instantly reduce the rate of polysome production to zero, causing an immediate arrest of nucleoid segregation. Then, gradual depletion of the existing polysomes due to mRNA decay should cause, on a slower time scale, a dissipation of the phase separation.

To test these predictions, we subjected cells growing in M9gluCAAT in microfluidic channels to two rounds of rifampicin treatment (**Figure 5A** and Movie S4). Rifampicin resulted in growth rate inhibition (**Figure 5A**) and changes in the nucleoid area and nucleoid-to-cell area (NC) ratio (**Figure 5 – figure supplement 1**), as previously described (Bakshi et al., 2014; Dworsky and Schaechter, 1973; Koch and Gross, 1979; Pettijohn and Hecht, 1974; Xiang et al., 2021). In cells that completed their division cycle before antibiotic addition (squares and fitted blue curve in **Figure 5B**), the distance between the intensity peak of each sister nucleoid increased monotonically between birth and division, displaying the dynamics of normal, unperturbed nucleoid segregation. Cells born 22 min to 12 min before the treatment (circles in **Figure 5B**) experienced the same nucleoid segregation dynamics up to the time of rifampicin addition. Exposure to rifampicin lead to the near-immediate arrest of nucleoid segregation (**Figure 5B**), consistent with our prediction.

Interestingly, the distance between the sister nucleoids remained the same for close to 30 min into rifampicin treatment, after which it started to decrease (**Figure 5B**). Average 1D and 2D cell projections of the scaled (divided by the whole cell average) RplA-GFP and HupA-mCherry concentration suggest that the time delay between the arrest in nucleoid segregation and its reversal is likely due to the compressible nature of the nucleoid, which has been demonstrated in vitro (Pelletier et al., 2012). As the polysomes in the middle of the cells started to visibly deplete (> 6 min after rifampicin addition), the DNA signal expanded to fill the emerging available space without affecting the distance between the peaks of the sister nucleoids (white crosses, **Figure 5C**). This is consistent with the removal of a compaction force exerted by the accumulating polysomes on the soft nucleoid. About 30 min after rifampicin addition and further polysome depletion, the peak signals of the sister nucleoids (white crosses, **Figure 5C**) started migrating closer to each other. Eventually, after 1 h of rifampicin treatment, when the RplA-GFP fluorescence was homogeneous and there were no polysome accumulations detected, the two nucleoid objects fused into one (**Figure 5C**), consistent with the dissipation of phase separation.

These results support the notion that in untreated cells, polysome accumulation effectively exerts a force on the compressible nucleoid, which translates into its observed compaction and translocation (hence, segregation). Gradual polysome depletion through rifampicin treatment progressively decreased this effect, reversing the process. This reversal became obvious when we plotted the correlation between the relative polysome accumulation and nucleoid depletion at mid-cell for two cell lineages that experienced rifampicin at different times after birth (**Figure 5D**). Irrespective of their birth time (12 or 3 min before the addition of rifampicin), the negative correlation between the two variables was reversed ~9 min after the addition of the antibiotic, following the same path as for the untreated cells but in the opposite direction (**Figure 5D**). Polysome depletion during rifampicin treatment also resulted in correlated loss of asymmetric

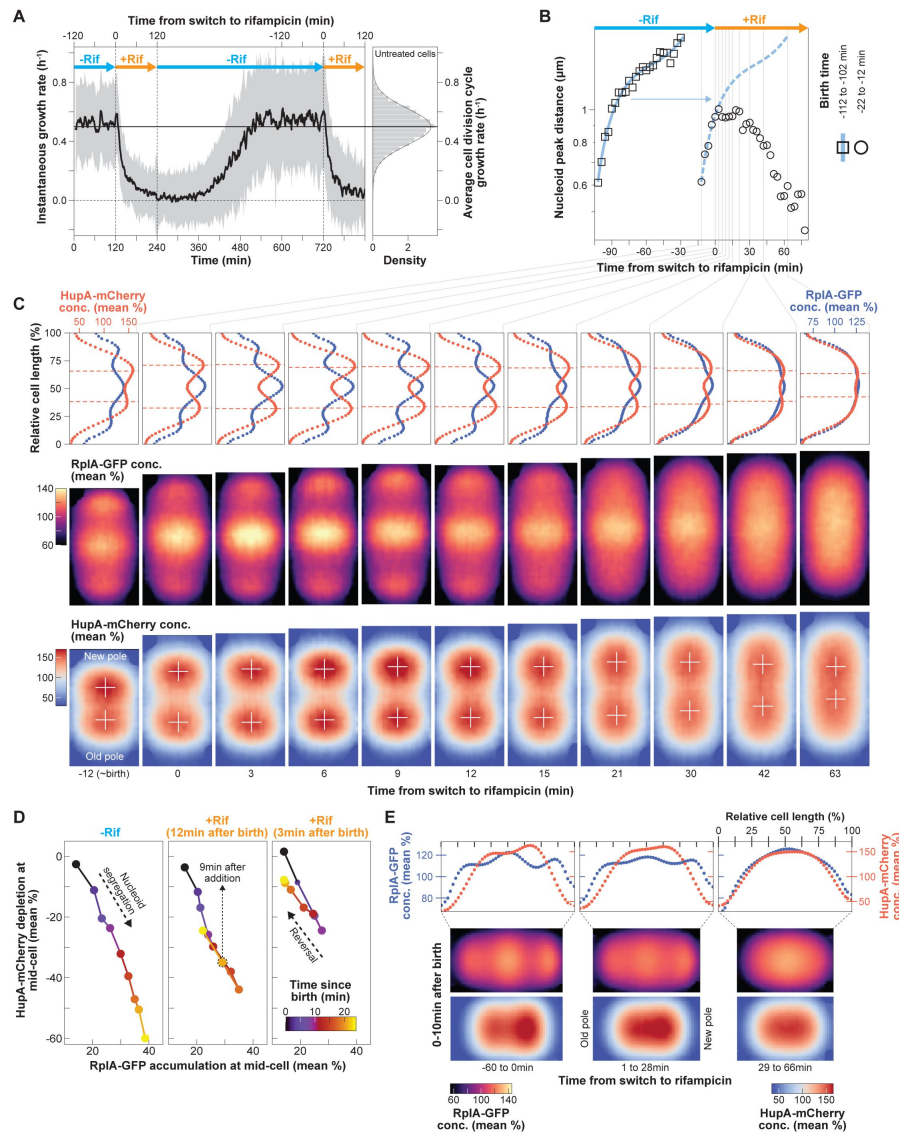


Figure 5

Effects of rifampicin treatment and polysome depletion on nucleoid segregation and compaction.

A. Plot showing the average instantaneous growth rate (mean $\pm$ SD shown by the solid black curve and grey shaded region, respectively) of a cell population ($n = 2629$ cell division cycles) undergoing two rounds of rifampicin treatment in a microfluidic device supplemented with M9gluCAAT. The distribution of the average cell cycle growth rate of unperturbed populations is also shown on the right ($n = 4122$ cell division cycles from a different microfluidics experiment). The solid horizontal line indicates the average growth rate. **B.** Plot showing the average distance between nucleoid peaks for a population of cells (squares, 114 cell division cycles) that were born (-112 to -102 min) and divided before the addition of rifampicin, and for a population of cells (circles, 112 cell division cycles) that were born just before (-22 to -12 min) and divided after the addition of rifampicin. A third-degree polynomial was fitted to the data from the unperturbed population (solid blue curve) and juxtaposed (dashed blue curve) with the data from the interrupted population. **C.** Average 1D profile and 2D projections of the scaled (divided by the whole cell average concentration) RplA-GFP and HupA-mCherry signals for cells before and after rifampicin addition ($n = 112$ cell division cycles). The red dashed horizontal lines in the 1D intensity profiles and the white crosses in the 2D profiles mark the nucleoid peaks. **D.** Plot showing the RplA-GFP accumulation relative to the HupA-mCherry depletion at mid-cell from 0 to 24 min after birth (colormap) for cells that completed their division cycle before the addition of rifampicin ($n = 114$ cell division cycles) and for cells that were subjected to rifampicin 12 min ($n = 112$ cell division cycles) or 3 min ($n = 99$ cell division cycles) after birth. **E.** Average 1D and 2D scaled RplA-GFP and HupA-mCherry intensity profiles for newborn cells (0 to 10 min after birth) before (left, $n = 726$ cell division cycles), just after (middle, $n = 367$ cell division cycles), and much after (right, $n = 235$ cell division cycles) rifampicin addition.

nucleoid compaction in newborn cells (**Figure 5E** [↗](#)). Altogether, these results support the notion that the asymmetric accumulation of polysomes results in an anisotropic force that asymmetrically compacts and segregates nucleoids.

Ectopic polysome production redirects nucleoid dynamics

Our second approach to test causality experimentally was to redirect polysome formation away from the chromosome by over-expressing a useless protein (mTagBFP2) for the cell from a T7 promoter on a multi-copy pET28 plasmid (**Figure 6A** [↗](#)). The resulting CJW7798 strain also carried the ribosome (RplA-msfGFP) and DNA (HupA-mCherry) markers. We reasoned that high expression of BFP2 from the plasmid may slow polysome production within the nucleoid and create polysome accumulations at ectopic cellular locations through recruitment of ribosomes to plasmid transcripts. This, in turn, should affect nucleoid dynamics if our proposed mechanism is correct.

Plasmid expression of mTagBFP2 was induced by addition of 100 μ M IPTG and expression of a chromosomally encoded T7 RNA polymerase (**Figure 6A-B** [↗](#)). The gradual increase of mTagBFP2 fluorescence in the cells was associated with a concomitant decrease in cell growth rate (**Figure 6C** [↗](#)), consistent with reduced gene expression from the chromosome. These cells displayed various patterns of polysome accumulations (Movie S5), presumably due to stochastic clustering of plasmids in DNA-free regions as previously reported for other multi-copy plasmids devoid of DNA partitioning genes (Hsu and Chang, 2019 [↗](#); Reyes-Lamothe et al., 2014 [↗](#); Yao et al., 2007 [↗](#)). Importantly, the ectopic accumulations of polysomes had a drastic effect on nucleoid dynamics. In some cells, polysomes accumulated at one pole instead of the middle of the nucleoid, preventing the nucleoid from splitting (**Figure 6D** [↗](#)). Expansion of the polysome accumulation at a pole effectively pushed the nucleoid toward the opposite pole of the cell. In other cells, polysome accumulation occurred between sister nucleoids, but did not relocate to the segregated nucleoids at the $\frac{1}{4}$ and $\frac{3}{4}$ cell positions. Rather, polysome accumulation persisted and expanded between the sister nucleoids, effectively further pushing them apart (**Figure 6E** [↗](#)). Often, cells filamented and showcased correlated polysome and nucleoid dynamics that changed in time (**Figure 6F** [↗](#)), resulting in transient events of nucleoid fusion, splitting, or changes in migration direction depending on where polysomes accumulated. Movie S5 shows additional examples of such dependency.

All cells displayed diffuse mTagBFP2 fluorescence (**Figure 6B** [↗](#)). We also observed the formation of inclusion bodies (bright phase contrast) that typically remained at a pole or sometimes moved along the edge of a growing polysome accumulation (**Figure 6F** [↗](#)). After a long of period of IPTG induction (> 8 h), polysome accumulation eventually decreased, leading to nucleoid decompaction (**Figure 6 – figure supplement 1** [↗](#)). Altogether, these results support the notion that ectopic polysome accumulation drives nucleoid dynamics.

Radial confinement prevents nucleoid splitting along the cell width and fusion of polysome accumulations from distinct DNA-free regions

In the model (**Figure 4** [↗](#)), phase separation between polysomes and nucleoid occurs when the cell and nucleoid elongate. As *E. coli* cells grow by increasing their length and not their width, we reasoned that radial confinement forces nucleoids to grow along the cell length, thereby promoting nucleoid segregation along this cellular dimension. Keeping nucleoids close to the membrane across the cell width may act as a diffusion barrier for polysomes from distinct DNA-free regions, preventing polysome accumulations to fuse into larger phase condensates.

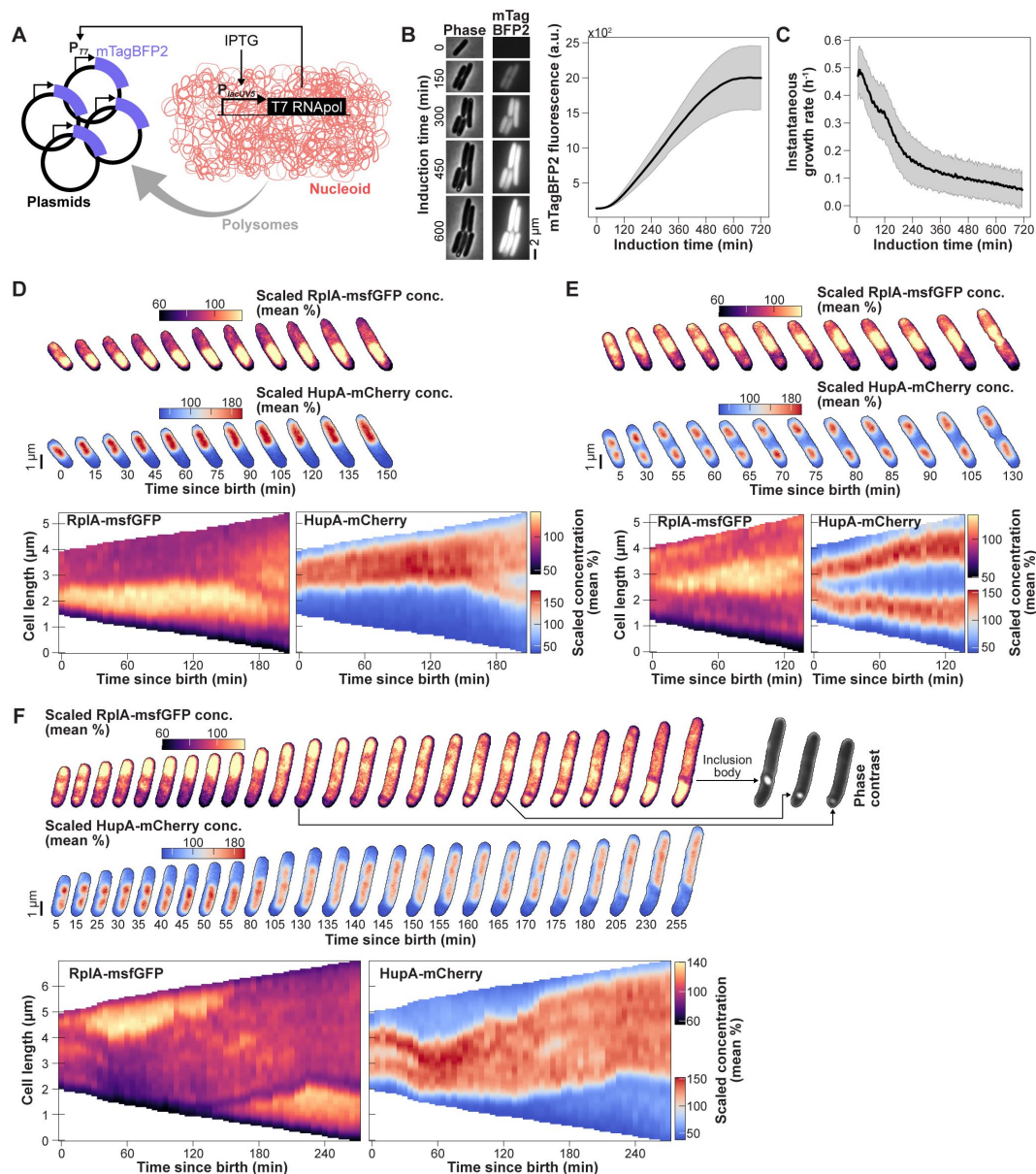


Figure 6

Effects of ectopic polysome accumulation on nucleoid dynamics.

A. Schematic summarizing the experiment. **B.** Representative phase contrast and mTagBFP2 fluorescence images at different times after induction with IPTG (100 μ M) are shown, next to a plot showing the mTagBFP2 fluorescence of the entire population (mean $\pm$ SD, $n = 3624$ cell trajectories) over time. **C.** Plot showing how instantaneous growth rate (mean $\pm$ SD, $n = 3624$ mTagBFP2 induction trajectories) decreases following induction of mTagBFP2 synthesis. **D-F.** Representative kymographs and images of the normalized (divided by the whole cell average) RplA-msfGFP and HupA-mCherry fluorescence signals in cells (CJW7749) born during mTagBFP2 over-expression. **F.** Phase contrast images are shown to illustrate the formation of inclusion bodies (see also **Figure 6 – figure supplement 1**). Additional cell examples are shown in Movie S5.

To test these ideas, we first treated cultures with A22 to inhibit cell width control through the inactivation of MreB (Bean et al., 2009; Iwai et al., 2002). This resulted in cells with a polysome phase at the cell center surrounded by a nucleoid phase around the cell periphery (Figure 7 – figure supplement 1, top). To further increase the cell width (> 2.5-fold), we exposed cells to the cell division inhibitor cephalixin in addition to A22 (Figure 7A and Figure 7 – figure supplement 1, bottom). As a control, we showed that cephalixin treatment alone resulted in filamentous cells (of constant cell width) with multiple nucleoids separated by polysome accumulations (Figure 7B), consistent with previous reports (Chai et al., 2014; Gray et al., 2019; Thappeta et al., 2024). In cells treated with both drugs, we observed two types of subcellular rearrangements. In smaller cells, which started with a single nucleoid, drug treatment resulted in a single large polysome accumulation at the cell center, displacing the nucleoid into a toroidal shape at the cell periphery (Figure 7C and Movie S6). In longer cells with two segregated nucleoids, the nucleoids expanded and often aberrantly segregated along the cell width concomitant with polysome accumulation at the site of nucleoid splitting (white arrowheads, Figure 7D). Consequently, a polysome “bridge” was formed between the polysome accumulations flanking the nucleoid. These polysome bridges resulted in a characteristic cross-like polysome pattern, marking the two axes (longitudinal and radial) of nucleoid segregation (Figure 7E). The loss of radial confinement led to coalescence of polysome accumulations toward the cell center and fusion of nucleoids around the cell periphery (Figure 7C-E and Figure 7 – figure supplement 1). As a result, the nucleoids and polysome accumulations decreased in number while increasing in size in cephalixin/A22-treated cells compared to cephalixin-treated cells with the same cell area distribution but normal cell width (Figure 7F). These results suggest a critical role for cell width regulation in limiting the diffusion of polysomes around the nucleoid and thereby spatially controlling the phase separation between polysomes and nucleoids to promote longitudinal nucleoid segregation.

Discussion

The flow of genetic information intrinsically couples nucleoid segregation to cell growth

Polysomes, a term in which we include monosomes for simplicity, are made of mRNAs in complex with ribosomes. This study provides experimental and theoretical evidence (Figures 1–7) that polysome production within the nucleoid—an inherent product of chromosomal gene expression—contributes to nucleoid segregation and positioning in *E. coli* cells. This may also be true in other bacteria, as reduced or abrogated transcription via gene deletion or antibiotic treatment causes chromosome segregation defects in *Streptococcus pneumoniae* (Kjos and Veening, 2014) and *Bacillus subtilis* (Dworkin and Losick, 2002).

Due to their anabolic footprint (Belliveau et al., 2021; Chure and Cremer, 2023; Dai et al., 2017; Dourado and Lercher, 2020; Hu et al., 2020; Molenaar et al., 2009; Scott et al., 2014; Si et al., 2017), polysomes integrate the rate of nucleoid segregation with that of gene expression and cell growth. The concentration of polysomes and their rate of accumulation in the cell are a direct reflection of transcriptional and translational activities in the cell (Balakrishnan et al., 2022). The higher the concentration of polysomes, the faster is the growth rate. Thus, in our proposed model, an increase in polysome concentration not only leads to more protein synthesis and faster cell growth, but also results in faster nucleoid segregation. The reverse is true for a decrease in polysome concentration, inherently coupling these processes without the help of a dedicated regulatory system. Such coupling was seen across isogenic cells with variable growth rates in the same nutrient condition (Figures 1F, 1H and 2C-E) as well as across nutrient conditions that led to a wide range of growth rates (Figure 2A-B and Figure 2 – figure supplement 1).

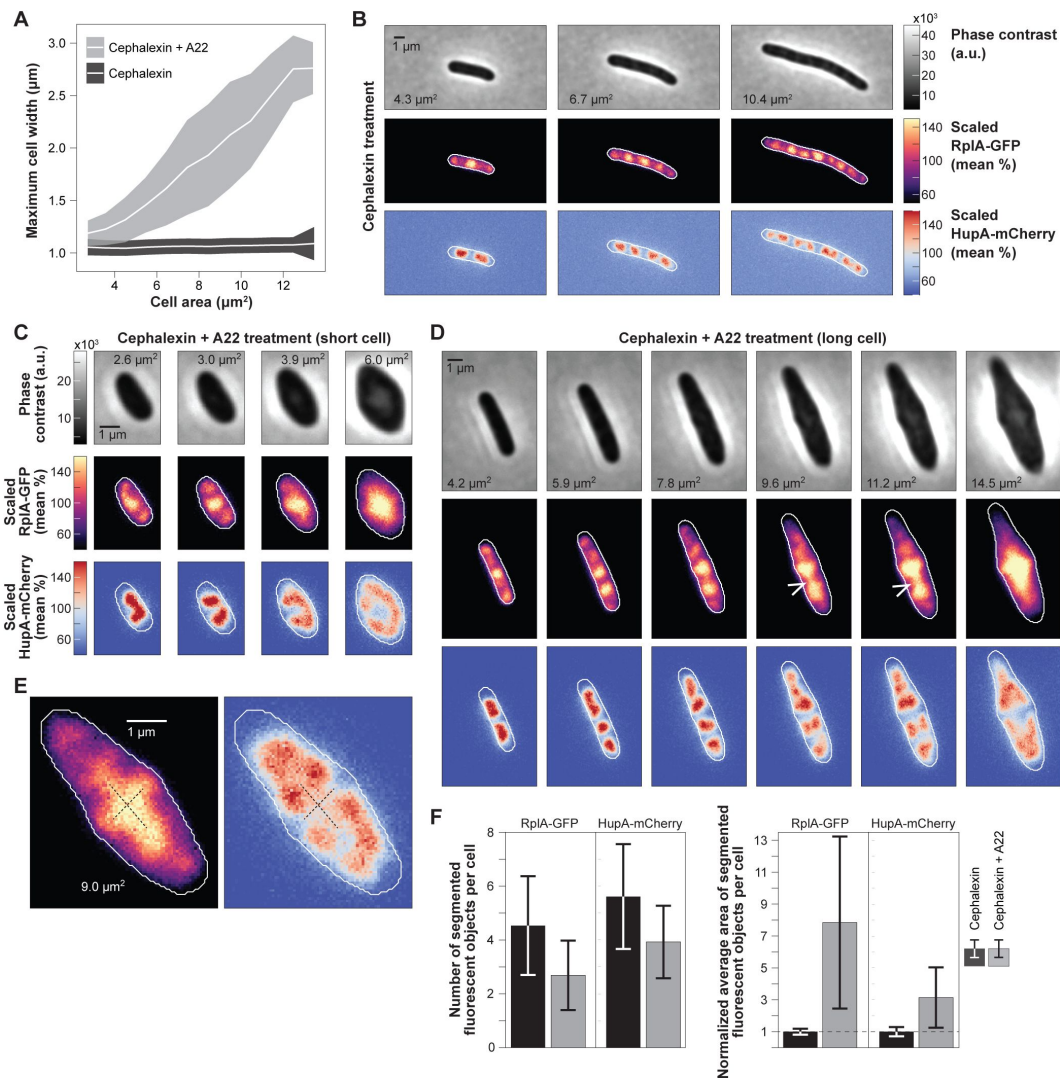


Figure 7

Effects of cell width increase on polysome and nucleoid dynamics.

A. Comparison of the cell width increase during cell growth between CJW7323 cells treated with cephalexin (mean $\pm$ SD, 360 cell growth trajectories, 418 to 1511 segmented cells per bin) and cells treated with both cephalexin (50 μ g/mL) and A22 (4 μ g/mL) (grey, mean $\pm$ SD, 309 cell growth trajectories, 51 to 1684 segmented cells per bin). The same cell area bins are compared between the two populations. **B.** Phase contrast and fluorescence images of a representative cephalexin-treated cell expressing RplA-GFP and HupA-mCherry. **C.** Same as B but for a short cell growing in the presence of A22 and cephalexin. **D.** Same as C but for a longer cell. The white arrowheads indicate the polysome bridges that connect polysome accumulations between two DNA-free regions. Additional examples are shown in Movie S6 **E.** Representative fluorescence images of RplA-GFP and HupA-mCherry in a cell treated with A22 and cephalexin. The dotted lines indicate the representative cross-like polysome accumulation, which forms during the fusion of the polysome accumulation towards the center (see also Movie S6). **F.** Comparison of the segmented polysome accumulations and nucleoid objects between A22+cephalexin (150 sampled segmented cells from 47 growth trajectories) and cephalexin-treated (150 sampled segmented cells from 100 growth trajectories) cells. The polysome and nucleoid areas per cell were normalized by the population-average statistic from cephalexin-treated cells. All differences between the two populations are statistically significant (Mann-Whitney p-value $< 10^{-10}$).

Directional nucleoid splitting requires phase separation and radial confinement

A key aspect of our proposed mechanism for nucleoid segregation is that polysomes form within nucleoids due to chromosomal gene expression, which, together with polysome turnover due to mRNA decay, creates an out-of-equilibrium system (**Figure 4**) (Miangolarra et al., 2021). Redirecting polysome formation to plasmid gene expression led to ectopic polysome accumulations that dramatically affected nucleoid dynamics (**Figure 6** and Movie S5).

In normal cells, the effective force that segregates nucleoids appears to be linked to the propensity of the chromosomal meshwork and polysomes to phase separate. Mutual exclusion is, at least in part, caused by the steric repulsion between the two macromolecules. While ribosomes or ribosomal subunits freely diffuse across the cell unobstructed by the presence of the nucleoid (Bakshi et al., 2012; Sanamrad et al., 2014), the larger polysomes are impeded by the chromosomal mesh based on size considerations alone (Xiang et al., 2021). Modeling studies have suggested that such steric hinderance between large crowdors (polysomes) and a polymeric meshwork (chromosome) can result in phase separation and polymer compaction (Bakshi et al., 2014; Castellana et al., 2016; Miangolarra et al., 2021; Mondal et al., 2011; Wu et al., 2019b). It is also possible that phase separation between nucleoids and polysomes involves non-steric interactions such as electrostatic repulsion between the negatively charged DNA and RNA (mRNA and rRNA), as previously hypothesized (Joyeux, 2015). Such steric and non-steric interactions may contribute to the effective poor solvent quality of the polysome-rich cytoplasm for the chromosome (Xiang et al., 2021). Future research will be necessary to elucidate the precise nature of the interaction between chromosomes and mRNAs, whether individually or in complex with ribosomes.

We found that the width of the cell controls the dynamics of phase separation between polysomes and nucleoids by radially confining the nucleoid and directing its growth and segregation along a single cellular dimension, cell length (**Figure 7**). Radial confinement effectively prevents the diffusion and fusion of polysome accumulations from distinct DNA-free regions, which leads to alternating enrichments of polysomes and DNA along the length of elongating cells (**Figure 7B**). This radial confinement explains why differences in RplA-GFP signal between DNA-free regions (**Figure 3A-B** and **Figure 2 – figure supplement 2**) do not equilibrate rapidly, validating our use of a 1D model. Diffusion restriction around nucleoids is consistent with our previous report that polysomes diffuse much faster over short distances (within DNA-free domains) than long distances (across DNA-free domains) (Gray et al., 2019). Loss of radial confinement results in apparent fusion of polysome accumulations near the cell center and displacement of nucleoids toward the cell periphery (**Figure 7C-E**). This phase separation between the polysomes and the nucleoid likely contributes to the torus topology of nucleoids previously observed in A22-treated *E. coli* (Wu et al., 2019a). Our findings highlight the importance of cell width regulation and suggests that nucleoid segregation may have imposed an evolutionary constraint on cell morphology.

Nucleoid segregation likely involves multiple factors

Our models show that the most trivial case of uniform production of polysomes (i.e., uniform mRNA synthesis) within the nucleoid is sufficient to cause an enrichment of polysomes at mid-nucleoid (**Figure 4**) (Miangolarra et al., 2021). Inside cells, this polysome enrichment at mid-nucleoid may be enhanced by a bias in mRNA synthesis across the nucleoid. For instance, the chromosomal region close to the origin of replication has been shown to be more highly expressed per gene copy than other regions on the chromosome (Scholz et al., 2019) and this region is located in the middle of the nucleoid prior to DNA replication (Bates and Kleckner, 2005; Cass et al., 2016; Fisher et al., 2013; Kuwada et al., 2013; Mäkelä et al., 2021; Sadhir and Murray, 2023; Wang et al., 2006). In addition, this highly expressed chromosomal region is the

first one to replicate, which should lead to further local increase in mRNA expression due to a doubling in gene dosage (Pountain et al., 2022). However, it is important to reiterate that such localized mRNA synthesis is not expected to be a prerequisite for polysome-based nucleoid segregation since it is not included in our reaction-diffusion model (Figure 4).

Other factors are likely involved in nucleoid segregation. In fact, our data revives the abandoned sixty-year-old hypothesis by Jacob et al. (1963) that cell growth and surface expansion separate the sister nucleoids, but with two notable differences. First, the contribution of cell growth to nucleoid splitting would be partial (Figure 1I and Figure 1 – figure supplement 3). Second, cell growth would contribute predominantly near the end of the division cycle (Figure 1I). This late timing makes sense for two reasons. First, it corresponds to the time when polysomes stop accumulating between the separated sister nucleoids and polysome accumulations emerge at the middle of these nucleoids (i.e., at the $\frac{1}{4}$ and $\frac{3}{4}$ cell positions) to start the next round of segregation (Figure 1D and Movie S2). Second, this is also when *E. coli* switches its cell wall growth pattern from dispersed along the cell body to zonal and divisome-dependent at mid-cell (Cooper and Hsieh, 1988; Gray et al., 2015; Navarro et al., 2022; Wientjes and Nanninga, 1989; Woldringh et al., 1987). Indeed, zonal cell growth between the sister nucleoids was a key assumption of the original 1963 model (Jacob et al., 1963). Altogether, we propose a working model in which gene expression, which results in polysome production within the nucleoid, promotes nucleoid splitting and accelerates the migration of the resulting sister nucleoids apart, with cell elongation facilitating this latter function near the end of the cell division cycle. DNA replication and processes that contribute to chromosome condensation and organization (such as DNA loop extrusion, supercoiling, and preferential loading of DNA remodeling complexes) are also likely to be important for robust chromosome segregation across all cells (Danilova et al., 2007; Harju et al., 2023; Hofmann et al., 2019; Holmes and Cozzarelli, 2000; Lemon and Grossman, 2000; Mäkelä et al., 2021; Minnen et al., 2011; Sawitzke and Austin, 2000; Weitao et al., 1999).

We note that in our time-lapse experiments, the accumulation of polysome signal appeared to slightly precede the depletion of DNA signal that marked the initiation of nucleoid splitting (Figure 1E). Polysome enrichment in the middle of unconstricted nucleoids was also occasionally observed in snapshot images of cells growing on glycerol, a slow growth condition that results in a single nucleoid segregation event late during the cell division cycle (Figure 1 – figure supplement 4A-D). This is not seen in our model in which polysome accumulation and nucleoid splitting occur at the same time (Figure 4A and Movie S3). This small discrepancy may reflect a limitation of our experimental or modeling approach. For example, it is possible that the point spread function of our fluorescent DNA marker slightly delays the moment at which we can detect signal depletion at mid-nucleoid and thereby the initiation of nucleoid splitting. Alternatively, the small difference in timing may be associated with a model simplification. In our model, the nucleoid is effectively a solution of DNA fragments. In reality, the nucleoid consists of a circular polymer, crosslinked by nucleoid-associated proteins. These DNA crosslinks may cause a small resistance that marginally delays the initiation of nucleoid splitting relative to the polysome enrichment at mid-nucleoid.

E. coli is an asymmetric organism

The spatial macromolecular asymmetries uncovered in our study contrast with the common perception of *E. coli* as a symmetric organism. Under our experimental conditions, the distribution of polysomes at the new pole in newborn cells was, on average, higher than at the old pole through inheritance of the large mid-cell accumulation of polysomes from their mother cells (Figure 3A and D). We also observed an asymmetric distribution in DNA density within nucleoids, which correlated with the availability of polysome-free space along the cell length and width (Figure 3E-G). This asymmetry emerged before cell division (Figure 3D and Movie S2). Simulations of our reaction-diffusion model suggest that slower diffusing/relaxing nucleoids are more likely to reproduce these nucleoid position and compaction asymmetries during the finite

course of the cell division cycle (**Figure 4D** [↗](#)). A reduction in the apparent nucleoid diffusivity has been linked to nucleoid-associated proteins, which bridge and thus stiffen the DNA polymer (Subramanian and Murray, 2023 [↗](#)). The physiological significance for these dynamic cellular asymmetries is not clear at this time, though it is conceivable that a difference in DNA compaction within the nucleoid may affect gene expression. Regardless, our study illustrates how spatial and temporal asymmetries in the cytoplasm can emerge from the interactions between two of the most important macromolecules.

Methods

Strains and constructs

Strains and plasmids used for this study are listed in Tables S2 and S3, respectively, while the sequences of the oligonucleotides used to make constructs can be found in Table S4.

To measure the concentration and spatial heterogeneity of ribosomes inside *E. coli*, we used strains in which the RplA 50S ribosome subunit protein (strains CJW7323, CJW6768, CJW7020 and CJW7651) or the RpsB 30S ribosome subunit protein (strains CJW6769 and CJW7021) are fused with mEos2 (strains CJW6768 and CJW6769), msfGFP (strains CJW7020, CJW7021, CJW7651, and CJW7798) or GFP (CJW7323) (Gray et al., 2019 [↗](#); Xiang et al., 2021 [↗](#)). Nucleoid characteristics were measured using strains in which HupA, a nucleoid-associated protein, is fused with mCherry (strains CJW7323, CJW6723 and CJW7798) (Xiang et al., 2021 [↗](#)). Alternatively, DAPI was used to stain the DNA (strains CJW6768, CJW6769, CJW7020, CJW7021, and CJW7651).

The mCherry- μ NS (CJW7651) particles were chromosomally expressed from the native Lac promoter after induction with 150 μ M IPTG (for 3 h for agarose pad experiments, or continuously for microfluidics experiments). Strain CJW7651 was constructed as follows. The *gfp* coding sequence in the pER12 (pBAD322A-gfp- μ NS) plasmid (kind gift from Dr. A. Janakiraman, City College of New York) was swapped with the mCherry-coding sequence using megaprimer whole plasmid (MEGAWHOP) cloning (Bryksin and Matsumura, 2010 [↗](#)) and the primer pair ER12-MCR-fwd/ER12-MCR-rev2 to generate plasmid pER12-mCherry. The mCherry- μ NS coding sequence and the *rrnB* transcriptional terminator were amplified from the pER12-mCherry plasmid (primer pair μ NSmCherry fwd/rev) and assembled with the *frt* site flanked with a kanamycin resistance cassette (amplified from pKD13 (Datsenko and Wanner, 2000 [↗](#)) using the primer pair FRT_KanR fwd/rev) and the ColE1 origin of replication (PCR amplified from the pBAD22A plasmid (Guzman et al., 1995 [↗](#)) using the primer pair ColE1 fwd/rev) using Gibson DNA assembly (Gibson et al., 2009 [↗](#)) to form the pAPG1 plasmid. The pAPG1 plasmid also included two 50 bp sites homologous to the *attB* region of the *E. coli* chromosome, introduced as overhangs in the primers used for Gibson DNA assembly. The *attB* homologous regions allowed for the integration of the arabinose inducible mCherry- μ NS expression cassette into the respective site, though this was not used in this study. The pAPG1 plasmid was verified by sequencing using the pAPG1 seq1-5 primers. The mCherry- μ NS coding sequence was then PCR amplified from the pAPG1 plasmid using primer pair lacZYA_red μ NS fwd/rev, which includes 50-bp overhangs homologous to the region upstream and downstream of the *lacZYA* operon, and integrated downstream of the *lac* promoter in the MG1655 strain (Guyer et al., 1981 [↗](#); Jensen, 1993 [↗](#)) using lambda red recombination and the pKD46 plasmid (Datsenko and Wanner, 2000 [↗](#)) for the construction of the CJW7144 strain. Correct insertion of the mCherry- μ NS coding sequence to substitute the *lacZYA* operon coding sequences was confirmed by colony PCR using primer pairs LacI_fwd/CynX_rev and LacI_fwd/mCherry_rev. Transduction with a P1 phage lysate of the CJW7144 strain was used to transfer the mCherry- μ NS expression cassette into the CJW7020 strain using kanamycin as a selection marker. As a result, the CJW7145 strain was constructed, which was verified using colony PCR and the primer pairs LacI_fwd/CynX_rev and LacI_fwd/mCherry_rev, KanR_fwd/CynX_rev. The CJW7145 strain was then

transformed using the pCP20 plasmid (Datsenko and Wanner, 2000 [↗](#)), which encodes the FLP recombinase (Cherepanov and Wackernagel, 1995 [↗](#)) to remove the kanamycin resistance cassette, yielding the CJW7651 strain.

The CJW7798 strain was derived from the MG1655 (DE3) strain (Tseng et al., 2010a [↗](#)), which carries the T7 RNA polymerase-encoding gene under *lacUV5* control. Transduction with a P1 phage lysate of the CJW5158 strain (Gray et al., 2019 [↗](#)) was used to transfer the gene encoding the HupA-mCherry fusion into the MG1655 (DE3) strain using kanamycin as a selection marker. The pCP20 plasmid (Datsenko and Wanner, 2000 [↗](#)), which encodes the FLP recombinase (Cherepanov and Wackernagel, 1995 [↗](#)), was used to remove the kanamycin resistance cassette, yielding the CJW7466 strain. Transduction with a P1 phage lysate of the CJW7019 strain, a kanamycin-resistant intermediate of strain CJW7020 (Gray et al., 2019 [↗](#)), was used to transfer the gene encoding the RplA-msfGFP fusion into the CJW7466 strain using kanamycin as a selection marker. The resulting strain (CJW7766) was grown in the presence of kanamycin since the cells tend to lose msfGFP fluorescence in the absence of the antibiotic. The sequence of the CJW7766 strain was confirmed by whole genome sequencing.

The pET28:mTagBFP2 plasmid variant was derived from the pET28:GFP plasmid (Shis and Bennett, 2013 [↗](#)), which was a gift from Mathew Bennett (Addgene plasmid # 60733; <http://n2t.net/addgene:60733> [↗](#); RRID:Addgene_60733). First, the GFP coding sequence was substituted with the mTagBFP2 coding sequence from the pBAD-mTagBFP2 plasmid (Subach et al., 2011 [↗](#)), a gift from Vladislav Verkhusha (plasmid # 34632; <http://n2t.net/addgene:34632> [↗](#); RRID:Addgene_34632). Specifically, the mTagBFP2 coding sequence was amplified using the primer pair mTagBFP2 fwd/rev and assembled (Gibson DNA assembly) with the pET28 backbone, which was amplified in two pieces using the pET28_one fwd/rev and the pET28_two fwd/rev primer pairs to derive the pET28:mTagBFP2 plasmid. Then, the kanamycin resistance cassette that was originally present in the pET28 backbone was substituted with the chloramphenicol resistance cassette from the pSB3C5-proA-B0032-E0051 plasmid (Davis et al., 2011 [↗](#)) which was a gift from Joseph Davis and Robert Sauer (Addgene plasmid # 107244; <http://n2t.net/addgene:107244> [↗](#); RRID:Addgene_107244). Specifically, the backbone of the pET28:mTagBFP2 plasmid excluding the kanamycin resistance cassette was amplified using the pET28mTagBFP2 fwd/rev primers and assembled (Gibson DNA assembly) with the coding sequence of the chloramphenicol resistance cassette that was amplified using the cmR fwd/rev primer pair. As a result, the pET28:mTagBFP2-CmR plasmid was created. The proper assembly of the pET28:mTagBFP2 and pET28:mTagBFP2-CmR plasmids were confirmed by sequencing.

The pET28:mTagBFP2-CmR plasmid was introduced into CJW7766 by electroporation to create the CJW7798 strain, which was used to redirect the ribosomes away from the nucleoid onto the plasmid-expressed mTagBFP2 mRNAs after inducing the expression of T7 RNA polymerase with IPTG.

Growth conditions

Strains CJW6769, CJW7020 and CJW7021 used in **Figure 2** [↗](#) and **Figure 2 – figure supplement 1** [↗](#) were grown as previously described (Gray et al., 2019 [↗](#)) using a basic M9 medium formulation without trace elements and supplemented with 0.2% (w/v) carbon source and when specified with 0.1% (w/v) casamino acids (CAA) and 1 µg/mL thiamine (T). The strain CJW6768 (used in **Figure 2** [↗](#) and **Figure 2 – figure supplement 1** [↗](#)) was grown using the same medium formulation. The abbreviations of the medium growth conditions presented in **Figure 2A** [↗](#) and **Figure 2 – figure supplement 1** [↗](#) are defined in Table S1. For the rest of our experiments on agarose pads or in a microfluidic device, strains CJW7323, CJW7651 and 7798 were grown in M9 salts (final concentrations: 33.7 mM Na₂HPO₄, 22 mM KH₂PO₄, 8.55 mM NaCl, 9.35 mM NH₄Cl, 1 mM MgSO₄, 0.3 mM CaCl₂) supplemented with trace elements (Fe, Zn, Cu, Co, B, Mn), 0.1% (w/v) thiamine, and, when specified, 0.4% (w/v) casamino acids. The pH of the 10x M9 salts

was adjusted to 7.2 with NaOH. The trace elements were added at a final concentration of 13.4 mM ethylene-diamine-tetra-acetic-acid, 3.1 mM of $\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$, 0.62 mM of ZnCl_2 , 76 μM of $\text{CuCl}_2 \cdot 2\text{H}_2\text{O}$, 42 μM of $\text{CoCl}_2 \cdot 2\text{H}_2\text{O}$, 162 μM of $\text{HBO}_3 \cdot 2\text{H}_2\text{O}$, 8.1 μM of $\text{MnCl}_2 \cdot 4\text{H}_2\text{O}$.

To achieve steady-state exponential growth prior to imaging, a stationary phase liquid culture in the appropriate growth medium was diluted at least 10,000 times and grown to an optical density at 600 nm (OD_{600}) between 0.1 and 0.3. These cells were either loaded in a mother-machine-type microfluidics device (Lin and Jacobs-Wagner, 2022; Wang et al., 2010) where they were grown under the constant flow of medium ($\sim 0.5 \mu\text{L}/\text{sec}$), or spotted on 1% agarose pads prepared with the same growth medium. In the microfluidic device, cells were grown for at least 3 h prior to image acquisition. All precultures and experiments were performed at 30°C.

The mCherry- μNS particles (CJW7651 strain) were imaged on agarose pads after inducing exponentially growing cells with 150 μM IPTG for 3 h (Figure 4E-G). Before spotting on the 1% agarose pad, the induced cells were stained with DAPI (1 $\mu\text{g}/\text{mL}$) for 5 min. In both experiments, the cells were growing in M9gluCAAT.

To redirect ribosomes to plasmid-expressed, T7 promoter-driven mTagBFP2 mRNAs, exponentially growing CJW7798 cells grown in M9glyCAAT to an $\text{OD} \sim 0.2$ were spotted on a 1% agarose pad containing M9glyCAAT and 100 μM IPTG, the latter to induce the expression of the T7 RNA polymerase.

Rifampicin treatment

The antibiotic rifampicin (see Table S5) was used to block transcription, deplete mRNAs, and release the mRNA-bound ribosomes (polysomes). Rifampicin treatment was performed either in batch culture (Figure 1 – figure supplement 1F) or in microfluidics (Figure 5 and Movie S4).

For the microfluidic experiment, two rounds of rifampicin treatment were performed by switching between M9gluCAAT containing antibiotic (100 $\mu\text{g}/\text{mL}$) and antibiotic-free medium at 30°C. For the medium switches, solenoid valves were used in a custom-built pressurized perfusion system, achieving fast (within 1 min) changes in the cellular environment. The first switch to rifampicin occurred 2 h after normal growth in antibiotic-free medium (Figure 5A and Movie S4). Rifampicin treatment lasted 2 h until the system switched back to antibiotic-free medium, where cells were left to recover for 8 h. Twelve hours into the experiment, the system switched again to rifampicin-containing medium.

A22 and cephalixin treatment

Cephalixin (50 $\mu\text{g}/\text{mL}$) and A22 (4 $\mu\text{g}/\text{mL}$) (see Table S5) were added to exponentially growing CJW7323 cell cultures ($\text{OD}_{600} \sim 0.1$) in M9gluCAAT just prior to spotting on a 1% agarose pad, which was made with the same growth medium and contained the same antibiotic concentrations. The radial expansion of the cells and the polysome and nucleoid signals were tracked over time in a time-lapse experiment. The selected concentration of A22 has previously been shown to not affect cell growth (Takacs et al., 2010). All precultures and time-lapse imaging experiments were performed at 30°C.

Ectopic polysome accumulation experiment

Exponentially growing CJW7798 cells in M9glyCAAT supplemented with kanamycin (50 $\mu\text{g}/\text{mL}$) and chloramphenicol (35 $\mu\text{g}/\text{mL}$) were washed one time with M9glyCAAT minimal medium lacking antibiotics but supplemented with 100 μM IPTG (see Table S5). The washed cells were spotted on a 1% agarose pad prepared with M9glyCAAT also supplemented with IPTG (100 μM) for time-lapse imaging. All pre-cultures and time-lapse imaging experiments were performed at 30°C. Since chloramphenicol affects protein synthesis and thus polysome formation, the agarose pads lacked

chloramphenicol, resulting in a fraction that loss the plasmids based on the absence of blue fluorescence. Therefore, only cells that expressed blue fluorescence (and thus carried the pET28:mTagBFP2-CmR plasmid) were analyzed.

Microscopy

Strains CJW6769, CJW7020, and CJW7021 used in **Figure 2A-B** and **Figure 2 – figure supplement 1**, were imaged on agarose pads using the same microscopy set-up and optical configurations as previously described (Gray et al., 2019). Snapshots of the CJW6768 strain (used in **Figure 2A-B** and **Figure 2 – figure supplement 1**) were taken using a Nikon Ti-E microscope equipped with a 100x Plan Apo 1.45NA Ph3 oil objective, a Hamamatsu Orca-Flash4.0 V2 CMOS camera (16-bit, Slow Scan sensor mode) and a Lumencor Spectra X LED (Light Emitting Diode) engine. The 395/25nm LED was used to excite DAPI and the 470/24nm LED was used to excite GFP. DAPI fluorescence was acquired using an ET350/50x (excitation filter), RT400lp (dichroic mirror), ET460/50m (emission filter) filter cube from Chroma. The ET470/40x (excitation filter), T495lpxr (dichroic mirror), ET525/50m (emission filter) configuration from Chroma was used for GFP fluorescence acquisition. The microscope was controlled using the NIS Elements software by Nikon and the ND acquisition module.

For the time-lapse observation of cells (strain CJW7323) treated with A22 and/or cephalixin, images were taken using a Nikon Ti-E microscope, equipped with a 100x Plan Apo 1.45NA Ph3 oil objective, a Hamamatsu Orca-Flash4.0 V2 CMOS camera (16-bit, Slow Scan sensor mode), and a Sola solid state white light source (Lumencor), in a temperature-controlled enclosure (Okolabs). The AT470/40x excitation filter, combined with a T495LPXR beam-splitter and an ET525/50m emission filter, was used for GFP. For mCherry visualization, the ET560/40x excitation filter, combined with a T585lp beam-splitter and a ET630/75m emission filter, was used. A neutral density filter (32x) was applied in the excitation path to reduce phototoxicity and photobleaching. Images were taken every 2.5 min in the brightfield channel (phase contrast) and every 5 min in the fluorescence channels (RplA-GFP and HupA-mCherry).

For the rest of the brightfield and epi-fluorescence wide-field microscopy experiments (strains CJW7323, CJW7651, and CJW7798), snapshots or time-lapse images were taken using a Nikon Ti2-E inverted microscope, equipped with a 100x Plan Apo 1.45NA Ph3 oil objective, a Photometrics Prime BSI back-illuminated sCMOS camera (2048×2048 pixels sensor with a pixel size of 6.5 μm), and a Lumencor Spectra III LED (Light Emitting Diode) engine, in a temperature-controlled enclosure (Okolabs). The HDR 16-bit sensor mode was used to acquire images. The microscope was controlled using the NIS Elements software by Nikon, and the JOBS or the ND acquisition module were used to acquire snapshots or time-lapse images. The Perfect Focus System (by Nikon) was used to maintain focus in microfluidics experiments. The auto-focus function in NIS Elements was used to locate the optimal z-position in agarose pad experiments (snapshots or time-lapse). For the DAPI, BFP, GFP, or mCherry channels, a polychroic mirror (FF-409/493/596-Di02 by Shemrock) combined with a triple-pass emitter (FF-1-432/523/702-25 by Shemrock) was used. For DAPI, BFP, and GFP imaging, additional emission filters were applied in the optical path (FF01-432-36 by Shemrock, FF01-432-36 by Shemrock and ET525/50M by Chroma, respectively). DAPI, BFP, GFP, and mCherry were excited using a 390/22nm, 390/22nm, 475/28nm and 575/25nm LED, respectively.

For the microfluidic experiments (strain CJW7323), the excitation light intensity was reduced to 20% for the 1-min interval and 30% for the 3-min interval imaging of the RplA-GFP and HupA-mCherry. An exposure time of 120-ms was used for both markers. In all time-lapse experiments, a neutral density filter (absorptive ND filter, OD:1.3 / 5% transmission, NE13B by Thorlabs) was also applied to reduce the LED power and minimize phototoxicity and photobleaching. The LED powers (factored by the neutral density filters) were calibrated using a microscope slide power meter with an 18×18 mm sensor size (S170C by Thorlabs). The light intensity was measured for each excitation wavelength (475/28nm and 575/25nm) at the end of the objective (without immersion oil) for different LED intensities (% of maximum intensity). From the generated

calibration curves, the RplA-GFP excitation light power was estimated to be 524 μW and 629 μW for the 1-min and 3-min fluorescence interval imaging, respectively. HupA-mCherry was excited with 109 μW light power for the 1-min and 191 μW for the 3-min fluorescence interval imaging.

The type N immersion oil (by Nikon) was used in all experiments except for the time-lapse observation of A22+cephalexin-treated cells where the type F immersion oil (by Nikon) was used. All imaging (time-lapse and snapshots) was done at 30°C.

Analysis software and code availability

Cropping and alignment of the microfluidic channels were performed with MATLAB (www.mathworks.com) using a previously published pipeline from our lab (Lin and Jacobs-Wagner, 2022). Supervised classification and curation of the Oufti cell meshes was also implemented in MATLAB (Campos et al., 2018; Govers et al., 2024; Gray et al., 2019). For the T7 experiments, segmentation and tracking of the CJW7798 cells was performed using the Omnipose deep neural network architecture (Cutler et al., 2022), using a previously trained model (Thappeta et al., 2024) and the SuperSegger MATLAB-based package (Stylianidou et al., 2016). The remaining analyses were performed using Python 3.9 (www.python.org), that included the *numpy* (Harris et al., 2020), *scipy* (Virtanen et al., 2020), *pandas* (McKinney, 2010), *scikit-mage* (Van Der Walt et al., 2014), *scikit-learn* (Pedregosa et al., 2012), *shapely* (Gillies, Sean et al., 2023), *statsmodels* (Seabold and Perktold, 2010) and *pytorch* (Paszke et al., 2019) libraries. The *matplotlib* (Hunter, 2007) and *seaborn* (Waskom, 2021) libraries were used for plotting. The analysis pipeline and functions (summarized in Table S6) are available in the Jacobs-Wagner lab GitHub repository: www.github.com/JacobsWagnerLab/published/tree/master/Papagiannakis_2024.

Cell segmentation and tracking

For the CJW6768, CJW6769, CJW7020, and CJW7021 strains used in **Figure 2** and **Figure 2 – figure supplement 1**, Oufti (Paintdakhi et al., 2016) was used to draw cell meshes on the phase contrast snapshots and a MATLAB-based support vector machine model was used to remove the badly segmented cells as previously described (Campos et al., 2018; Govers et al., 2024; Gray et al., 2019).

The Omnipose deep neural network architecture (Cutler et al., 2022) with a previously trained model (Thappeta et al., 2024) was used to segment the CJW7798 cells during T7 RNA polymerase induction on an agarose pad. The segmented cells were then tracked using SuperSegger (Stylianidou et al., 2016). A custom class (*omnipose_to_python_ghv.py*) was developed to transfer the SuperSegger segmentation and tracking data into Python for post-processing.

Due to the unusual and variable morphology of the A22/cephalexin-treated cells, a custom Python class was developed for their segmentation and tracking. The *Otsu_phase_segmentation_ghv.py* class segments the cells by applying an Otsu-threshold (Otsu, 1979) on the inverted phase images, followed by binary dilation (*scikit-image.morphology.binary_dilation* Python function) hole filling (*scipy.ndimage.binary_fill_holes* Python function), and binary closing (*scikit-image.morphology.binary_closing* Python function). The segmentation masks were tracked between subsequent timepoints, using cell distance as well as cell area constraints, and linked into cell growth trajectories. Cell morphology criteria such as the maximum pixel distance from the medial axis and the medial axis sinuosity were used to remove bad segmentations. The remaining segmentation masks were manually curated.

For the other experiments on agarose pads, a neural network (Wiktor et al., 2021; Zhou et al., 2020) with a previously designed and trained U-net architecture (Mäkelä et al., 2024) was used to segment single cells based on phase contrast snapshots. The generated segmentation

masks were further processed by watershed separation, filling the holes within masks and removing unusually small masks. Finally, a graphical interface was used to manually remove the badly curated cells (less than 5% of the segmented cell population).

In contrast to the Oufi software (Paintdakhi et al., 2016) that generated sub-pixel meshes around the cell boundaries, the other applied segmentation algorithms returned pixel-based cell masks. To deal with this discrepancy, the sub-pixel cell meshes from Oufi were converted into pixel-based cell masks by collecting the pixels within the circumscribed single cell area in Python (*oufti_snapshots_GrayGovers* class, included in the *snapshots_analysis_OUFTI_GrayGovers* Python script). After this conversion, the same Python-based functions were applied for the analysis of the cell fluorescence and morphology statistics regardless of the segmentation method used.

To segment cells growing in the microfluidic device, a custom library of image analysis functions (*mother_machine_segmentation* class, included in the *microfluidics_segmentation_ghv* Python script) was developed in Python. This algorithm was applied on the cropped, aligned, background-subtracted, and inverted phase contrast images that were produced using a previously published pipeline (Lin and Jacobs-Wagner, 2022) in MATLAB. Cell segmentation was performed in three steps. First, all the cells within the microfluidics channel, which were brighter than the microfluidic channel background in the inverted and background-corrected phase-contrast channel, were segmented using an Otsu intensity threshold (Otsu, 1979). This crude thresholding step, which separated the cell (brighter: 1) from the background (darker: 0) pixels, yielded at least one masked label for the entire row of stacked cells in each microfluidic channel. A watershed segmentation was then applied to define the boundaries between individual cells and split the Otsu-based binary mask(s). The watershed algorithm was guided by the number of cells and their relative positions in the microfluidic channel, which were determined using an adaptive filter combined with the local decrease of the inverted phase intensity between the poles of adjacent cells. Finally, a graphical interface was used to curate the cell masks by manually splitting or merging cell labels.

After segmentation, the curated single-cell masks were tracked over time and linked into trajectories from birth to division using the centroid distance and the relative area difference between cells at consecutive time-points (*mother_machine_tracking* class, included in the *microfluidics_segmentation_ghv* Python script). Examples of cell segmentation and tracking in microfluidics are shown in Movie S1 and **Figure 1 – figure supplement 1A**. Regardless of the image acquisition time intervals for the fluorescence channels, phase-contrast images were acquired every minute, which allowed for accurate cell tracking.

Fluorescence background correction was different between the microfluidic and agarose-pad experiments. In microfluidics, the average background was measured within two 8-pixel (0.528 μm) wide areas, one on the left side and one on the right side of the channel, at least 10 pixels (0.66 μm) away from the channel boundaries from top to bottom. The estimated background was then subtracted along the channel length. The background correction was integrated in the *get_fluorescence_image* function, located in the *microfluidics_analysis_functions_ghv* Python script.

In agarose-pad experiments, an Otsu threshold (Otsu, 1979) was used on the inverted phase-contrast image to segment all cells. A binary dilation was then performed on these crude cell masks before estimating the local average fluorescence of the unmasked pixels (pixels outside the dilated cell masks). The locally-averaged background was then used to fill in the cell areas and reconstruct the background of the entire field of view in the absence of cells. The smoothed (Gaussian smoothing) reconstructed background fluorescence was subtracted from each fluorescence image. This background correction pipeline (*back_sub* function) is integrated in both the *unet_snapshots* class and the *oufti_snapshots_GoversGray* class. For the time-lapse imaging of A22-treated CJW7323 cells (**Figure 7**) and CJW7798 cells during induction of T7 RNA polymerase expression (**Figure 6**), the fluorescence background was not subtracted since we did not need to

quantify the raw pixel values. The normalized polysome and nucleoid fluorescence normalized by the average whole cell fluorescence is shown instead, marking the relative macromolecular rearrangements.

mCherry-μNS particle localization and tracking

For the localization of the diffraction-limited mCherry-μNS particles (**Figure 4E-G** [↗](#)), a set of custom functions were developed in Python and implemented in the *particle_positions_snapshots* class, included in the *snapshots_analysis_UNET_ghv* Python script.

The fluorescent particles were segmented in the background corrected mCherry-μNS fluorescence images using a Laplace of Gaussian (LoG) and an adaptive filter combined. The particle masks that had an area larger than a specified threshold (80 pixels for the microfluidics and 90 pixels for the agarose-pad experiments), which usually included two particles from the same or adjacent cells, were further processed by applying a relative fluorescence threshold (90th percentile of masked pixel intensity) to find the local fluorescence maxima and separate the two objects. Additional minimum area and aspect ratio constraints were applied. To accurately estimate the particle position with sub-pixel resolution, a 2D Gaussian function (**eq. 1** [↗](#)) with rotation (**eq. 2** [↗](#)) was fitted (least square method: *scipy.optimize.leastsq*) to an area of 7×7 pixels centered at the centroid of the particle mask:

$$g(x, y) = A \exp \left(-\frac{1}{2} \left(\left(\frac{x_0 - x_{rot}}{\sigma_x} \right)^2 + \left(\frac{y_0 - y_{rot}}{\sigma_y} \right)^2 \right) \right) \quad [\text{eq. 1}]$$

and

$$\begin{aligned} x_{rot} &= x \cos(\theta) - y \sin(\theta) \\ y_{rot} &= x \sin(\theta) + y \cos(\theta) \end{aligned} \quad [\text{eq. 2}]$$

where x and y are the coordinates in the 2D imaging plane, A is the amplitude of the fitted Gaussian, σ is the standard deviation in each dimension, and θ is the rotation angle in radians, which was used to rotate the Gaussian distribution (x_{rot}, y_{rot}) around the particle center or the Gaussian mean (x_0, y_0) .

Two-dimensional cell mapping and alignment

Ribosome and nucleoid fluorescence statistics, as well as particle positions within cells, were mapped in relative 2D cellular coordinates from pole to pole and across the cell width. Such mapping allowed us to project the fluorescence or position statistics in 1 or 2D for specific cell length ranges and cell division cycle intervals. For this analysis, it was necessary to draw the medial axis for each of the segmented cells, which was obtained differently in microfluidic and agarose-pad images.

In the microfluidic experiments, where all the cells were stacked in a vertically oriented channel, the medial axis was drawn by fitting a second-degree polynomial to the most distant coordinates from the cell boundaries, excluding the cell caps at the poles where the medial axis was linearly extrapolated. The length of the cells, which was parallel to the length of the microfluidic channel, was used as the independent variable for the fitting. This medial axis estimation is implemented via the *all_medial_axis* function in the *microfluidics_analysis_functions_ghv* Python script.

However, in the agarose-pad experiments, cells were randomly oriented in the imaging plane. Thus, it was impossible to use one of the coordinates (x or y) as the independent variable as this would bias our medial axis estimation toward the same coordinate for cells that were not diagonally oriented. Instead, we developed an algorithm that scanned through the middle of the cells with a fixed sub-pixel step and directionality constraints to identify nodes at the most distant

locations from the cell boundaries. These ordered (from pole to pole) and numbered nodes were used to fit the x and y coordinates of the medial axis separately, using the number of the node as an independent variable and its position coordinates as dependent variables. The degree of the fitted polynomials (d) scaled linearly with the total number of nodes (N) based on the relation $d = 0.1N - 5$. Similar to the microfluidic experiments, the medial axis was linearly extrapolated at the cell caps. This medial axis estimation is implemented using the *get_medial_axis* function in both the *UNET_snapshots* and *oufti_snapshots_GoversGray* classes located in the *snapshots_analysis_UNET_ghv* and *snapshots_analysis_OUFTI_GrayGovers_ghv* Python scripts respectively. The medial axis estimation function is also provided as a separate function in the *Bivariate_medial_axis_estimation.py* Python script.

The medial axis of each cell, with a resolution of 0.1 pixel, was used to map the cell pixels and particle positions along the cell length and width. The projection of each pixel or particle position on the medial axis, defined as the most proximal node on the central line, was used to determine its cell length coordinate. The distance between the pixel or particle position and its medial axis projection was used to determine its absolute cell width coordinate. The sign of the cross product $\overrightarrow{PL} \times \overrightarrow{PC}$ (plus or minus), where L was the location of the particle, P was its projection on the medial axis, and C was the center of mass of the cell mask, was used to determine the position of particles or cell pixels on the sagittal plane.

Cell polarity and ages

In time-lapse experiments, the old and new poles of the cells were determined as follows. The pole of the daughter cell that was closer to the mid-cell position of its predivisional mother cell, where cell constriction occurs, was defined as the new pole. The medial axis coordinates were adjusted based on this polarity. As a result, the relative cell length extended from -1 to 1, with -1 corresponding to the old pole, 1 to the new pole, and zero to the cell center (see **Figure 3 – figure supplement 1B**, **D** and **E**). The implementation of this method is included in the *get_polarity* function in the *microfluidics_analysis_functions_ghv* Python script.

Cell lineages and ages relative to the oldest mother cells (first lineage) at the closed end of the microfluidic channel were determined using the orientation of the cell poles. The daughter cells that had the same polarity as the oldest mother cells were assigned the same age and lineage as their mothers. The daughter cells with the opposite polarity, which inherited the new pole of their mother cells, were considered younger and belonged to the second lineage. The daughter cells of mother cells belonging to the second lineage were classified as lineages three and four depending on their polarity. Cells from the third lineage had opposite polarity and those from the fourth lineage had the same polarity as their second lineage mother cells. Finally, mother cells from the third lineage, which inherited the new pole of their second-lineage mother cells, divided to yield the fifth and sixth cell lineages. The fifth lineage had the same polarity as its third-lineage mother and its sister lineage (sixth) had the opposite polarity and inherited the new pole from its third-lineage mother cell. The implementation of this method is included in the *get_ages* function in the *microfluidics_analysis_functions_ghv* Python script. For the analysis of the data presented in **Figure 3**, only the third, fourth, fifth and sixth lineages were considered to avoid age-related effects at the cell poles (Chao et al., 2023; Coquel et al., 2013; Koleva and Hellweger, 2015; Łapińska et al., 2019; Lindner et al., 2008; Proenca et al., 2019).

Construction of intensity profiles, 2D cell projections, demographs, and kymographs

With the ribosome and nucleoid fluorescence pixels as well as the μ NS particle positions inside the cells having been mapped, the fluorescence and particle statistics from pole to pole (medial axis projection) and across the cell width (distance from the medial axis) were plotted for different cell division cycle intervals or cell length ranges.

An intensity profile represents the change in a fluorescence statistic along the cell from one pole to the other. Such a fluorescence statistic can be the average fluorescence intensity (as in **Figures 1D**, 3G, 4G, and **Figure 3-figure supplement 1B**), which corresponds to the concentration of the reported protein per cell area (sum of pixels divided by the number of pixels). In other analysis (as in **Figure 5C** and **Figure 1 – figure supplement 4C**), the protein concentration along the cell length was divided by the whole cell average. This scaled statistic is expressed as a percentage change relative to average concentration (mean %). A value above 100 indicates increase above the average, whereas a value below 100 corresponds to a decrease below the average concentration. This scaled statistic is insensitive to the RplA and HupA concentration variability between cells, or to the maturation of the GFP or mCherry fluorophores. As a result, the scaled concentrations showcase rearrangements of the tagged proteins and do not imply changes in protein synthesis relative to cell growth.

Intensity profiles were plotted for single cells (**Figure 3 – figure supplement 1B** and **Figure 1 – figure supplement 4A**) or populations (as in **Figures 1D**, 3G, 4G, 5C, 5E). In the second instance, the intensity profiles represent the average 1D projection of the fluorescence statistic for a specific cell division cycle interval, cell length range, growth rate range, polysome and nucleoid asymmetry group, or time range (e.g., during antibiotic treatment). To generate a single-cell intensity profile, the medial axis length and its projected pixels were binned, and the average value of the selected fluorescence statistic was calculated per bin. If the number of bins was higher than the length of the medial axis in pixels, the missing values were filled in by linear interpolation. To avoid artefacts from the cell boundaries, the intensity profile was estimated for a box with a width of six pixels (~0.4 μm) centered on the medial axis (i.e., three pixels on each side of the medial axis). The intensity profile was occasionally smoothed by a moving average of a specified cell-length window, without excluding the bins at the edges which were not smoothed. Alternatively, a univariate spline (*scipy.interpolate.UnivariateSpline*) was fitted to the cell-length-binned data (as in **Figure 4G**). The univariate splines were fitted to the binned fluorescence intensity along the cell length. The number of cell length bins (N_{bin}) linearly scaled with the cell length in μm (l_{cell}) using the first-order relationship: $N_{bin} = \frac{100}{4.3} l_{cell}$ (only the whole part of the decimal was considered). Similar to the intensity profile where a fluorescence statistic was plotted along the cell length, the density profiles (histograms of the particle positions along the cell) for the mCherry- μNS (**Figure 4G**) particles along the cell were also plotted. Note that the intensity profiles shown in **Figures 1D**, 3G, 4G, 5C, 5E, **Figure 1 – figure supplement 1D**, and **Figure 2 – figure supplement 2D** are averages of many segmented cells or cell division cycles. In these averaged profiles, the variability in the timing of nucleoid segregation across cells attenuates the appearance of polysome accumulation. Furthermore, the point spread function of fluorescent ribosome markers causes an underestimation of the magnitude of the polysome accumulations in epi-fluorescence micrographs (Bakshi et al., 2012).

The intensity profiles were also used to construct demographs from agarose-pad experiments (**Figure 2A**, **Figure 2 – figure supplement 1**, and **Figure 1 – figure supplement 4B**) where cells were sorted by cell length from the shorter newborn to the longer predivisional cells. Intensity profiles were also used to construct ensemble kymographs from microfluidic experiments where the cells were sorted by their relative position in the cell division cycle (**Figures 1C** and **2D**) or single-cell kymographs from agarose pad time-lapse measurements (**Figure 6D-F**). Specifically, the average intensity profile was estimated for each cell length or cell division cycle bin, and the average intensity profiles were stacked from birth to division (left to right) oriented according to the cell polarity (top to bottom). In the ensemble kymographs, the fluorescence intensities were projected along the relative cell length (cell length %), whereas in the single-cell kymographs the length of the projection linearly scales with the cell length. A Gaussian smoothing (*skimage.filters.gaussian*) was applied to smooth the demographs and ensemble kymographs.

In addition to the intensity or density profiles for a fluorescence statistic or the density of the μ NS particles along the medial axis, 2D projections of the ribosome or nucleoid signal distribution (**Figures 1D**, **3D**, **3G**, **4F**, **5C**, **5E**, and **Figure 2 – figure supplement 2A**) as well as the μ NS particle positions (**Figure 4F**) were also constructed. These 2D intensity or density maps were constructed by binning the cell pixels or particle positions not only by cell length, but also by cell width. The average fluorescence statistic or the particle density was then shown per bin, including data from many single cells within a specified cell division cycle or cell length range. A Gaussian smoothing (*skimage.filters.gaussian*) was applied to smooth the 2D projections.

Extraction of population-level statistics of polysomes and nucleoids

To quantify the correlation between polysome accumulation and nucleoid segregation at the population level (**Figure 2A-B** and **Figure 2 – figure supplement 1**), the average polysome accumulation at mid-cell was extracted from cell snapshot images. This statistic was obtained by averaging the scaled (divided by the whole cell average concentration) RplA-GFP and HupA-mCherry intensity profiles of all the cells in the population. Therefore, this statistic describes the average “behavior” of the population under a specific growth condition. We reasoned that if, in a specific nutrient condition, nucleoid segregation happens earlier during the division cycle, then a higher fraction of the population will have segregated nucleoids. If so, the scaled HupA-mCherry concentration should exhibit a stronger depletion at mid-cell on average. This depletion was measured as the relative concentration difference between the HupA-mCherry peaks, which corresponded to the two lobes of the constricting nucleoid, and the trough at mid-cell, which corresponds to the point of nucleoid splitting. Similarly, the average polysome accumulation was measured as the RplA-GFP relative concentration difference between the mid-cell accumulation (peak) and the polysome depleted regions near the quarter cell positions, or at the centers of the segregating sister nucleoids (troughs). The same statistics were used to quantify the relative polysome accumulation at mid-cell and the relative nucleoid depletion in the same region between cell division cycles with different growth rates in the same nutrient condition (**Figure 2C-E**).

Nucleoid segregation cycle tracking

In this work, the nucleoid segregation cycle was defined as the period from the end of a nucleoid splitting event (i.e., when a segregating nucleoid is segmented by local thresholding (custom Python function: *LoG_adaptive_image_filter.py*) as a separate object from its sister nucleoid) until the end of the splitting of the same nucleoid object (**Figure 1 – figure supplement 2D**), usually after cell division but sometimes in the same cell division cycle (**Figure 1 – figure supplement 2A**).

To track the nucleoid objects through the nucleoid segregation cycle from the mother cells to their daughters, we used the cell polarity, i.e., based on pole identity (**Figure 1 – figure supplement 2B**). The nucleoids of daughter cells that had opposite polarity to their mother cells were inherited from the new pole of the mother cells (groups 1 and -1 in **Figure 1 – figure supplement 2B-C**). The nucleoids in the daughter cells that had the same polarity as their mother cells were inherited from the old pole of the mother cells (groups 2 and -2 in **Figure 1 – figure supplement 2B-C**). During a complete nucleoid segregation cycle, the nucleoid was tracked from the quarter position of the mother cell to the center of its daughter cells (**Figure 1 – figure supplement 2C**). This method is implemented in the *track_nucleoids* function, included in the *microfluidics_analysis_functions_glv* Python script.

To quantify the polysome accumulation and nucleoid splitting during the nucleoid segregation cycle (**Figure 1E-F**), we measured the RplA-GFP and HupA-mCherry concentration within the cell length region of 2.5 pixels adjacent to the center of the tracked nucleoid object (5 pixels or 0.33 μ m in total).

Quantification and segmentation of polysomes and nucleoids segmentation in cells treated with cephalalexin or cephalalexin + A22

The polysome accumulations and nucleoids objects were also segmented in A22 and cephalalexin-treated cells using an adaptive filter (custom Python function: *LoG_adaptive_image_filter.py*) on the masked cell fluorescence. The number of the detected polysome accumulations or nucleoid objects corresponds to the number of the segmented labels in the RplA-GFP or HupA-mCherry channel respectively. Similarly, the average polysome accumulation or nucleoid area corresponds to the average area of all segmented labels per channel. For a fair comparison between cells treated with A22 + cephalalexin and cells treated with cephalalexin alone, cells were randomly sampled without substitution from 12 cell area bins between 10 and 13 μm^2 (bin width = 0.25 μm^2) such that the compared populations in **Figure 7F** had the same cell area distributions (same number of cells per cell area bin).

Linear mixed-effects models to determine the relative contribution of polysome accumulation and cell elongation to the migration of the sister nucleoids

To estimate the relative contribution of the polysome accumulation at mid-cell and that of cell elongation to the migration of the separated sister nucleoids (first seen in **Figure 1H**), linear mixed-effects regressions were fitted to the scaled single cell data shown in **Figure 1** – figure supplement 3B (*statsmodel* package in Python, *statsmodels.formula.api.mixedlm* function). One model was fitted for each relative time interval between the end of nucleoid splitting and cell division, describing the rate of distance increase between the separated sister nucleoids (σ , response variable) as a function of the polysome concentration increase rate (ρ), the rate of cell elongation (λ), their interaction ($\rho\lambda$), and noise (ϵ) (*statsmodels.formula.api.ols*) as follows:

$$\sigma = \beta_0 + \beta_1\rho + \beta_2\lambda + \beta_3(\rho\lambda) + \epsilon \quad [\text{eq. 3}]$$

The single cell data were scaled by subtracting the population mean and then dividing by the standard deviation prior to model fitting for each of the four relative time intervals (in **Figure 1** – figure supplement 3B). The coefficients of the linear regression model (shown in **Figure 1I**) provide a measure of the relative effect of the independent variables on the dependent variable. The cell elongation (λ) had a statistically significant effect to the response variable (σ) $\text{Prob}(> |Z|) < 10^{-9}$) across all relative time intervals. The polysome accumulation coefficients (ρ) had a statistically significant effect to the response variable (σ) ($\text{Prob}(> |Z|) < 10^{-14}$) for the first three relative time intervals (0-25%, 25-50% and 50-75%) and a marginally significant one $\text{Prob}(> |Z|) = 0.02$ for the last quartile (75-100%). The interaction term ($\rho\lambda$) had a marginally significant effect ($\text{Prob}(> |Z|) = 0.002$) for the first quartile (0-25%) yet with a very small coefficient of 0.06 and did not present any significance ($\text{Prob}(> |Z|) > 0.05$) for the remaining time intervals (25-50%, 50-75% and 75-100%). Hence, it is not shown in **Figure 1I**. The most significant effect on the migration of the sister nucleoids (σ) was presented by the rate of polysome accumulation at mid-cell (ρ), 0-25% from the end of nucleoid splitting to cell division, with a coefficient of 0.46 and $\text{Prob}(> |Z|) < 10^{-84}$.

Calculation of rates

The rates of the polysome accumulation ($\frac{\Delta RplA_{mid-nuc} \text{ conc.}}{\Delta T}$) or nucleoid depletion

($\frac{\Delta HupA_{mid-nuc} \text{ conc.}}{\Delta T}$) in the middle of the nucleoid (2.5 pixels adjacent the nucleoid center) (**Figure 1F**), correspond to the slope of a linear regression (Python fitting function: *numpy.polyfit*) fitted to the change of the mid-nucleoid RplA-GFP or HupA-mCherry concentration over time, 40-90%

into the nucleoid segregation cycle (as in **Figure 1E**). This nucleoid cycle interval corresponds to the average time from the initiation of nucleoid splitting to just before its completion. Choosing the appropriate range to correlate the two statistics was important since polysomes appeared to accumulate in the middle of the nucleoid before the onset of nucleoid splitting (**Figures 1E** and **Figure 1** – figure supplement 4D).

The rate of RplA-GFP concentration increase at mid-cell ($\frac{\Delta RplA_{mid-cell}^{conc.}}{\Delta T}$) was calculated (**Figure 1H**) within a region covering 10% of the total cell length at the cell center. The same results were obtained when the rates were calculated within a cell region covering 2.5 pixels adjacent to the cell center. The rate of nucleoid migration ($\frac{\Delta Distance_{nuc}}{\Delta T}$) corresponds to the slope of a linear regression that describes the natural log-transformed distance increase between the separated sister nucleoids over time (**Figure 1H**). Similarly, the rate of cell elongation ($\frac{\Delta Length_{cell}}{\Delta T}$) was calculated considering the natural log-transformed cell length increase over time (**Figure 1H**). The rate of RplA-GFP concentration increase at mid-cell ($\frac{\Delta RplA_{mid-cell}^{conc.}}{\Delta T}$), the rate of nucleoid migration ($\frac{\Delta Distance_{nuc}}{\Delta T}$), and the rate of cell elongation ($\frac{\Delta Length_{cell}}{\Delta T}$) were calculated within four relative time intervals from the end of nucleoid splitting to cell division (**Figure 1G** and **Figure 1** – figure supplement 3B). Only single nucleoid migration intervals with more than four timepoints were considered to ensure a reliable fitting. The small fraction (~8%) of cells that were born with two separately detected nucleoid objects, which indicates that the end of nucleoid splitting occurred in the previous cell division cycle, were removed from the analysis. The cell division cycles with a negative cell elongation rate (<1%) were also excluded from the analysis.

Reaction-diffusion model

Our theoretical model is an extension of previous work (Miangolarra et al., 2021), which showed that steric effects between the DNA and the transcription-translation machinery (including polysomes) are sufficient to drive nucleoid segregation. To simply describe these interactions, we employed the Flory-Huggins theory for regular solutions (Flory, 1942; Huggins, 1941) and modeled the non-dimensionalized free-energy density f as a function of the local volume fractions of the nucleoid (n) and polysomes (p):

$$f(n, p) = \frac{p}{v_p} \ln p + \frac{n}{v_n} \ln n + (1 - p - n) \ln(1 - p - n) + \chi_p p(1 - p - n) + \chi_n n(1 - p - n) + \chi_{np} np - \kappa_p \nabla p \nabla(1 - p - n) - \kappa_n \nabla n \nabla(1 - p - n) - \kappa_{np} \nabla n \nabla p. \quad [\text{eq. 4}]$$

Here, $(1 - n - p)$ is the volume fraction of the cytoplasm not occupied by nucleoid or polysomes, whereas v_n and v_p are proportional to the molecular volumes of the elementary translational degrees of freedom (i.e., of the nucleoid element and single polysome, respectively). χ_p , χ_n and χ_{np} are the Flory-Huggins interaction parameters, and k_p , k_n and k_{np} are the interfacial tensions. We take $X_{ij} = \lambda^2 X_{ij}$, with λ being a characteristic interface width.

The time evolution of the volume fractions is described by a combination of Cahn-Hilliard theory (Cahn and Hilliard, 1958) and the reactions kinetics of the polysomes. We assume that polysomes are produced inside the nucleoid at a constant rate k_1 and degraded uniformly along the cell at a rate k_{-1} :

$$\frac{\partial p}{\partial t} = \nabla(M_p \nabla \mu_p) + k_1 n - k_{-1} p, \quad [\text{eq. 5}]$$

$$\frac{\partial n}{\partial t} = \nabla(M_n \nabla \mu_n), \quad [\text{eq. 6}]$$

where $\mu_p = \frac{\delta F}{\delta p}$ and $\mu_n = \frac{\delta F}{\delta n}$ are the local chemical potentials of polysomes and nucleoid elements respectively, with $F = \int f[n(x), p(x)]dx$ representing the total free energy. $M_p = v_p D_p p$ and $M_n = v_n D_n n$ are the mobility coefficients of the polysomes and the nucleoid, respectively, which depend on the diffusion coefficients of the polysomes (D_p) and the nucleoid (D_n). This recovers the Fick law of diffusion in the noninteracting limit.

Since *E. coli* grows by elongation while maintaining its cylindrical shape and cell width, we reduce the problem to one dimension, with x representing the position along the long axis of the cell. We assume that the cell length grows exponentially such that the length at time t is $L(t) = L(0)e^{\gamma t}$, with γ being the growth rate and $L(0)$ the initial cell length at birth. Exponential growth dilutes the existing polysomes with rate γ . On the other hand, we assume that the nucleoid is not diluted since DNA replication occurs continuously, and the nucleoid length is known to grow in proportion to cell length (Gray et al., 2019). Letting $\tilde{x} = x/L(t)$ be the relative position along the cell's long axis, the time evolution is given by:

$$\frac{\partial p}{\partial t} = \partial_{\tilde{x}}(\tilde{M}_p \partial_{\tilde{x}} \mu_p) + k_1 n - (k_{-1} + \gamma)p, \quad [\text{eq. 7}]$$

$$\frac{\partial n}{\partial t} = \partial_{\tilde{x}}(\tilde{M}_n \partial_{\tilde{x}} \mu_n), \quad [\text{eq. 8}]$$

where $\partial_{\tilde{x}} = \frac{\partial}{\partial \tilde{x}} = L(t) \frac{\partial}{\partial x}$, and $\tilde{M}_{n,p} = M_{n,p} L(t)^{-2}$. The chemical potentials are:

$$\mu_p = v_p^{-1} \ln p - \ln(1 - p - n) + \chi_p(1 - 2p - n) + (\chi_{np} - \chi_n)n - \chi_p \tilde{\lambda}^2 \partial_{\tilde{x}}^2(2p + n) + (\chi_{np} - \chi_n) \tilde{\lambda}^2 \partial_{\tilde{x}}^2 n, \quad [\text{eq. 9}]$$

$$\mu_n = v_n^{-1} \ln n - \ln(1 - p - n) + \chi_n(1 - 2n - p) + (\chi_{np} - \chi_p)p - \chi_n \tilde{\lambda}^2 \partial_{\tilde{x}}^2(2n + p) + (\chi_{np} - \chi_p) \tilde{\lambda}^2 \partial_{\tilde{x}}^2 p, \quad [\text{eq. 10}]$$

where $\tilde{\lambda} = \lambda/L(t)$.

We take the non-dimensionalized Flory-Huggins interaction parameters to be $X_p = 0.2$, $X_n = 0.4$ and $X_{np} = 1.2$. The X_{np} parameter captures the steric repulsion between the nucleoid and the polysomes (Miangolarra et al., 2021). The diffusion coefficient of the polysome was set to $D_p = 0.015 \mu\text{m}^2/\text{s}$, which lies within the same order of magnitude as was previously measured by single-molecule tracking (Bakshi et al., 2012). Different diffusion coefficients were tested for the nucleoid ($D_n = 0.0003, 0.0005, 0.001$ or $0.005 \mu\text{m}^2/\text{s}$) to capture different levels of nucleoid stiffness and hence response time to the local changes in polysome concentration. The polysome degradation rate was set to $k_{-1} = 0.003 \text{s}^{-1}$, which corresponds to a half-life of around 5 min (Bernstein et al., 2004). We take the polysome production rate to be $k_1 = k_{1,0} \left(1 + \frac{\gamma}{k_{-1}}\right)$ with $k_{1,0} = 0.002 \text{s}^{-1}$, to match the observation that the nucleoid length is proportional to cell length (Gray et al., 2019). The initial cell length $L(0)$ matched the cell length at birth, which has been previously shown to increase exponentially with the growth rate (Govers et al., 2024). Hence, each of the simulated growth rates ($\gamma = 0.25, 0.4, 0.55, 0.7, 0.85$ or 1h^{-1}) was matched to a cell length at birth (Figure 4 – figure supplement 1A) by fitting a linear regression ($L(0) = l_0 e^{\gamma/\gamma_0}$, with fitted parameters $l_0 = 1.61 \mu\text{m}$ and $\gamma_0 = 1.84 \text{h}^{-1}$) to the experimental data (Govers et al., 2024). Other parameters are chosen as follows: $\lambda = 0.03 \mu\text{m}$, $v_n = 10$, $v_p = 5$.

The dynamic equations (eqs. 7–8) were solved numerically in a fixed 1D domain $\tilde{x} \in (0,1)$, with no-flux boundary conditions. Space is discretized into $N = 128$ grid points, and time is discretized to steps of Δt . Time stepping was implemented with an implicit-explicit scheme as previously demonstrated (Mao et al., 2020, 2019). The simulations were initialized from the (symmetric) steady-state solution of the system at the initial cell length $L(0)$ (Figure 4A, dashed curves in Figure 4D, and Movie S3), or using the (asymmetric) macromolecular distribution of the

predivisional cell of the former simulation as the initial condition (**Figure 4C** [↗](#), solid curves in **Figure 4D** [↗](#)) which allowed us to capture cell polarity (new versus old pole). The cell length $L(t)$ was updated and recorded at each time step during the simulation. The nucleoid splitting time (shown in **Figure 4C** [↗](#)) was determined by thresholding the relative decrease of the nucleoid concentration at the cell center (as shown in **Figure 4 – figure supplement 1B** [↗](#)). The code for the model is available at <https://github.com/qiweiyuu/polysome> [↗](#).

Polysome and nucleoid analysis based on fitted Gaussian functions

Quantification of polysome and nucleoid asymmetries within cells along the cell division cycle was achieved by fitting Gaussian functions (Python least squares optimization: `scipy.optimize.curve_fit`) to the RplA-GFP or the HupA-mCherry intensity profiles of cells growing in M9gluCAAT in the microfluidic device. Three Gaussian functions were fitted to the polysome signal concentration early in the cell division cycle. Two of them captured the nucleoid-excluded polysomes accumulating at the poles and the third captured the polysome accumulation in the middle of nucleoid (**Figure 3** [↗](#) – figure supplement 1A-B):

$$Poly(l) = A_{old} \exp\left(-\frac{1}{2}\left(\frac{l_{old}-l}{\sigma_{old}}\right)^2\right) + A_{mid} \exp\left(-\frac{1}{2}\left(\frac{l_{mid}-l}{\sigma_{mid}}\right)^2\right) + A_{new} \exp\left(-\frac{1}{2}\left(\frac{l_{new}-l}{\sigma_{new}}\right)^2\right) \quad [\text{eq. 11}],$$

where A_{old} , A_{new} and A_{mid} are the amplitudes, l_{old} , l_{new} and l_{mid} are the means, and σ_{old} , σ_{new} and σ_{mid} are the standard deviations of the respective fitted functions along the relative cell length (l).

Two Gaussian functions were fitted to the concentration of constricting nucleoids (**Figure 3** [↗](#) – figure supplement 1A-B):

$$Nuc(l) = A_{old} \exp\left(-\frac{1}{2}\left(\frac{l_{old}-l}{\sigma_{old}}\right)^2\right) + A_{new} \exp\left(-\frac{1}{2}\left(\frac{l_{new}-l}{\sigma_{new}}\right)^2\right) \quad [\text{eq. 12}]$$

All Gaussian functions were fitted over the cell background fluorescence, which corresponds to the baseline fluorescence within the masked cell boundaries and along the relative cell length coordinates from the old pole (-1) via the cell center (0) to the new pole (1). The Gaussian functions were fitted either to single newborn cells (0-2.5% into the cell division cycle) as shown in **Figure 3** [↗](#) – figure supplement 1A-B and as applied in **Figures 3A** [↗](#) and **3E** [↗](#), or to the less noisy average intensity profile of the segmented cells within the 0-10% range of their cell division cycle, as shown in **Figure 3** [↗](#) – figure supplement 1C-D and used in **Figures 3B** [↗](#) and **3F** [↗](#). The areas of the Gaussian functions fitted to the polysomes (used in **Figure 3A-B** [↗](#)) were estimated using the function:

$$Area = A \sigma \sqrt{2\pi} \quad [\text{eq. 13}],$$

where A is the amplitude and σ is the standard deviation.

The Gaussian areas were used to quantify the polysome asymmetries between the poles:

$$Poly_{asym} = \log_{10}\left(\frac{Area_{new}}{Area_{old}}\right) \quad [\text{eq. 14}],$$

where $Area_{new}$ and $Area_{old}$ are the areas of the fitted Gaussians to the nucleoid-excluded polysomes accumulating in the new and the old pole, respectively. The statistic in **eq. 14** [↗](#) corresponds to the x-axis in **Figure 3B** [↗](#). It is a positive value when the new pole contains more polysomes than the old one, and a negative value when the old pole contains more polysomes than the new one.

The nucleoid Gaussian statistics were used to quantify the nucleoid position asymmetry defined as:

$$Nuc_{pos} = \frac{l_{new} + l_{old}}{2} \frac{l_{cell}}{2} \quad [\text{eq. 15}],$$

and the nucleoid compaction asymmetry was calculated as:

$$Nuc_{comp} = \frac{A_{new}}{A_{old}} \quad [\text{eq. 16}],$$

where l_{old} and l_{new} are the means of the nucleoid Gaussians in relative cell coordinates (as in [eq. 1](#) [1](#) [2](#)) and l_{cell} is the cell length in μm . The nucleoid position was first estimated in relative cell coordinates from the old pole (-1) via the cell center (0) to the new pole (1), and then multiplied by half the cell length to convert to absolute cell length units. A_{old} and A_{new} are the amplitudes of the fitted nucleoid Gaussians (as in [eq. 1](#) [1](#) [2](#)) in arbitrary fluorescence units. The nucleoid position corresponds to the mid-point between the two Gaussian means ([eq. 1](#) [1](#) [5](#)) and is a positive number when the nucleoid is shifted toward the new pole and a negative number when it is positioned closer to the old pole (y-axis in [Figure 3B](#) [1](#)).

The nucleoid compaction asymmetry was calculated as the relative nucleoid Gaussian amplitude ([eq. 1](#) [1](#) [6](#)). It had a positive value greater than 1 when the nucleoid concentration was higher toward the new pole and had a positive value smaller than 1 if the nucleoid was more concentrated toward the old pole (y-axis in [Figure 3F](#) [1](#)).

To determine which polysome Gaussian statistics and related asymmetries may contribute to nucleoid compaction asymmetry ([eq. 1](#) [1](#) [6](#)), a linear mixed-effects model was used (*statsmodel* package in Python, *statsmodels.formula.api.mixedlm* function). The polysome Gaussian statistics were used as independent variables and the nucleoid compaction asymmetry as a dependent variable:

$$Nuc_{comp} = \beta_0 + \beta_1 l_{old} + \beta_2 l_{mid} + \beta_3 l_{new} + \beta_4 \sigma_{old} + \beta_5 \sigma_{mid} + \beta_6 \sigma_{new} + \beta_7 A_{old} + \beta_8 A_{mid} + \beta_9 A_{new} + \varepsilon \quad [\text{eq. 17}],$$

where β_0 is the constant (intercept), β_n ($n=1,2,\dots,9$) are the coefficients for each of the polysome Gaussian statistics (same as in [eq. 4](#) [1](#) and plotted in [Figure 3](#) [1](#) – figure supplement 1E-F), and ε is the error. The entire dataset was treated as a single group. Before fitting the linear mixed-effects model, all data were normalized by subtracting the average and dividing the resulting value by their standard deviation ([Figure 3](#) [1](#) – figure supplement 1F). The relative cell length positions of the polysomes at the old pole (l_{old}) and in the middle of the nucleoid (l_{mid}) were found to significantly influence the asymmetric nucleoid compaction ([Figure 3](#) [1](#) – figure supplement 1G). We reasoned that the available spaces between the polysome positions ([Figure 3](#) [1](#) – figure supplement 1E) may influence the compaction of the nucleoid and devised a new statistic that compares the available polysome-free space toward the new pole to that of toward the old pole:

$$Poly_{space} = \frac{l_{new} - l_{mid}}{l_{mid} - l_{old}} \quad [\text{eq. 18}],$$

where l_{new} , l_{mid} and l_{old} are the polysome Gaussian means (as in [eq. 1](#) [1](#) [1](#)), which represent the positions of the polysomes at the new pole, mid-cell region, and old pole, respectively, expressed as relative cell coordinates from the old pole (-1) via the cell center (0) to the new pole (1). The fraction above ([eq. 1](#) [1](#) [8](#)) has a value greater than 1 when the polysome-free space is greater toward the new pole and a value smaller than 1 when the polysome-free space is greater toward the old pole (x-axis in [Figure 3F](#) [1](#)).

Linear regressions

Two types of linear regressions were used in this work. In **Figure 2B**, a first-degree polynomial was fitted using the ordinary least squares method and the numpy library in Python (*numpy.polyfit*), always using the x-axis as the independent variable and the y-axis as the dependent variable. In **Figures 1F** and **3B**, a principal component regression was fitted using the scikit-learn library in Python (*sklearn.decomposition.PCA*). A principal component analysis was first applied on the two-dimensional z-transformed data to find the linear regressor that explained most of the variance (the first principal component). This linear regressor was then rescaled to the original 2D plane. The principal component regression eliminates the dependent variable bias associated with a traditional univariate fit. This bias is introduced when calculating the prediction error (sum of squares) only along the dependent variable axis during the least squares optimization. The elimination of this univariate bias is particularly important for the fitted regressions in **Figures 1F** and **3B** since their parameters were used to estimate the rate of nucleoid splitting in the absence of polysome accumulation in the middle (y-intercept in **Figure 1F**), the position of the nucleoid with equal polysomes at the poles (y-intercept in **Figure 3B**), or the polar asymmetry required for a centered nucleoid (x-intercept in **Figure 3B**).

Kernel density estimations

One-dimensional (**Figures 3A**, E, and **Figure 1-figure supplement 2C**) or two-dimensional (**Figures 1F**, 1H, **Figure 1** – figure supplement 3A-B, and **Figure 3** – figure supplement 1F) Gaussian kernel density estimations were fitted using the *scipy.stats.gaussian_kde* method. The bandwidth was determined using Scott's rule (Scott, 1992).

Correlation coefficient calculations

All Spearman correlation coefficients (referred to as Spearman ρ) were estimated using the *scipy.stats* Python library and the *spearmanr* function. Only those Spearman correlation coefficients with a p-value below 0.05 are shown. Otherwise, the correlation is annotated as not-significant (NS) (as in **Figure 1** – figure supplement 3B).

Supplementary tables

Table S1.

Growth medium abbreviation and composition. Abbreviation Composition

Abbreviation	Composition
M9acet	1x M9 salts, 0.2% acetate (sodium salt)
M9acetCAAT	1x M9 salts, 0.2% acetate, 0.1% casamino acids, 1µg/mL thiamine
M9fuc	1x M9 salts, 0.2% fucose
M9fucCAAT	1x M9 salts, 0.2% fucose, 0.1% casamino acids, 1µg/mL thiamine
M9fum	1x M9 salts, 0.2% fumarate (disodium salt)
M9fumCAAT	1x M9 salts, 0.2% fumarate, 0.1% casamino acids, 1µg/mL thiamine
M9GlcNac	1x M9 salts, 0.2% N-acetylglucosamine
M9GlcNacCAAT	1x M9 salts, 0.2% N-acetylglucosamine, 0.1% casamino acids, 1µg/mL thiamine
M9glu	1x M9 salts, 0.2% glucose
M9gluCAAT	1x M9 salts, 0.2% glucose, 0.1% casamino acids, 1µg/mL thiamine
M9gly	1x M9 salts, 0.2% glycerol
M9glyCAAT	1x M9 salts, 0.2% glycerol, 0.1% casamino acids, 1µg/mL thiamine
M9Lala	1x M9 salts, 0.2% L-alanine
M9LalaCAAT	1x M9 salts, 0.2% L-alanine, 0.1% casamino acids, 1µg/mL thiamine
M9Lara	1x M9 salts, 0.2% L-arabinose
M9LaraCAAT	1x M9 salts, 0.2% L-arabinose, 0.1% casamino acids, 1µg/mL thiamine
M9mala	1x M9 salts, 0.2% malate (sodium salt)
M9malaCAAT	1x M9 salts, 0.2% malate, 0.1% casamino acids, 1µg/mL thiamine
M9malt	1x M9 salts, 0.2% maltose
M9maltCAAT	1x M9 salts, 0.2% maltose, 0.1% casamino acids, 1µg/mL thiamine
M9mann	1x M9 salts, 0.2% mannose
M9mannCAAT	1x M9 salts, 0.2% mannose, 0.1% casamino acids, 1µg/mL thiamine
M9pyr	1x M9 salts, 0.2% pyruvate (sodium salt)
M9pyrCAAT	1x M9 salts, 0.2% pyruvate, 0.1% casamino acids, 1µg/mL thiamine
M9sorb	1x M9 salts, 0.2% sorbitol
M9sorbCAAT	1x M9 salts, 0.2% sorbitol, 0.1% casamino acids, 1µg/mL thiamine
M9succ	1x M9 salts, 0.2% succinate (disodium salt)
M9succCAAT	1x M9 salts, 0.2% succinate, 0.1% casamino acids, 1µg/mL thiamine
M9treh	1x M9 salts, 0.2% trehalose
M9trehCAAT	1x M9 salts, 0.2% trehalose, 0.1% casamino acids, 1µg/mL thiamine
M9xyl	1x M9 salts, 0.2% xylose
M9xylCAAT	1x M9 salts, 0.2% xylose, 0.1% casamino acids, 1µg/mL thiamine

Table S2

Escherichia coli strains used in this study.

Strain	Genotype	Source
MG1655	<i>E. coli</i> MG1655 (F-lambda- <i>ilvG</i> - <i>rfb</i> -50 <i>rph</i> -1)	(Guyer et al., 1981; Jensen, 1993)
MG1655 (DE3)	<i>E. coli</i> MG1655 (DE3)	Kind gift from Dr K. Prather (Massachusetts Institute of Technology), (Tseng et al., 2010b)
CJW5158	<i>E. coli</i> BW25113 <i>hupA::hupA-mcherry-frt-kanR-frt</i>	(Gray et al., 2019)
CJW6723	<i>E. coli</i> MG1655 Δ <i>lacZYA::P_{lac}-egfp-µNS hupA::hupA-mcherry</i>	(Gray et al., 2019)
CJW6768	<i>E. coli</i> MG1655 <i>rplA::rplA-meos2</i>	(Sanamrad et al., 2014)
CJW6769	<i>E. coli</i> MG1655 <i>rspB::rpsB-meos2</i>	(Sanamrad et al., 2014)
CJW7019	<i>E. coli</i> MG1655 <i>rplA::rplA-msfgfp-frt-kanR-frt</i>	(Gray et al., 2019)
CJW7020	<i>E. coli</i> MG1655 <i>rplA::rplA-msfgfp</i>	(Gray et al., 2019)
CJW7021	<i>E. coli</i> MG1655 <i>rpsB::rpsB-msfgfp</i>	(Gray et al., 2019)
CJW7144	<i>E. coli</i> MG1655 Δ <i>lacZYA::P_{lac}-mcherry-µNS-frt-kanR-frt</i>	This study
CJW7145	<i>E. coli</i> MG1655 Δ <i>lacZYA::P_{lac}-mcherry-µNS-frt-kanR-frt</i> <i>rplA::rplA-msfgfp</i>	This study
CJW7323	<i>E. coli</i> MG1655 <i>rplA::rplA-msfgfp hupA::hupA-mcherry</i>	(Xiang et al., 2021)
CJW7466	<i>E. coli</i> MG1655 (DE3) <i>hupA::hupA-mcherry</i>	This study
CJW7651	<i>E. coli</i> MG1655 Δ <i>lacZYA::P_{lac}-mcherry-µNS rplA::rplA-msfgfp</i>	This study
CJW7766	<i>E. coli</i> MG1655 (DE3) <i>hupA::hupA-mcherry rplA::rplA-msfgfp-frt-kanR-frt</i>	This study
CJW7798	<i>E. coli</i> MG1655 (DE3) <i>hupA::hupA-mcherry rplA::rplA-msfgfp-frt-kanR-frt pET28:mTagBFP2-CmR</i>	This study

Plasmid name	Relevant genetic elements	Source
pKD13	<i>frt-kanR-frt-R6Kori-ampR</i>	(Datsenko and Wanner, 2000)
pKD46	<i>araC-bet-exo-A101(Ts)ori-ampR</i>	(Datsenko and Wanner, 2000)
pCP20	<i>cmR-A101(Ts)ori-ampR-flp-λ-repressor(Ts)</i>	(Datsenko and Wanner, 2000)
pER12	<i>pBAD322A-gfp-μNS</i>	Kind gift from Dr A. Janakiraman (City College of New York), (Broering et al., 2005)
pER12-mcherry	<i>pBAD322A-mcherry-μNS</i>	This study
pAPG1	<i>attB-pBAD322A-mcherry-μNS-attB-kanR-ColE1ori</i>	This study
pSB3C5-proA-B0032-E0051	<i>PproA-lacZα-GFP-p15Aori-cmR</i>	(Davis et al., 2011)
pET28:GFP	<i>PT7-lacO-GFP-kanR-pMB1ori-lacI</i>	(Shis and Bennett, 2013)
pBAD:TagBFP2	<i>araC-ParaBAD-mTagBFP2-ampR-pBAD322ori</i>	(Subach et al., 2011)
pET28:mTagBFP2	<i>PT7-lacO-mTagBFP2-kanR-pMB1ori-lacI</i>	This study
pET28:mTagBFP2-CmR	<i>PT7-lacO-mTagBFP2-cmR-pMB1ori-lacI</i>	This study

Table S3

Plasmids used in this study.

Table S4

DNA oligonucleotides used in this study.

Oligo name	Sequence	Source
ER12-MCR-fwd	5' CTCCATACCCGTTTTTTTGGGCTAGCAGGAGGAATTCATGGTGAGCAAGGGCG AGGAG 3'	This study
ER12-MCR-rev2	5' CACTGGAGGAGCCTGCTTTTTGTACAACTTGTGACTTGACAGCTCGTCCA TG 3'	This study
μNSmCherry fwd	5' CACAGGTTGCTCCGGGCTATGAAATAGAAAAATGAATCCGTTGAAGCCTGATC GATGCATAATGTGCC 3'	This study
μNSmCherry rev	5' AGCTCCAGCCTACACAGAGTTTGTAGAAACGCAAAAAG 3'	This study
FRT_KanR fwd	5' GTTCTACAACTCTGTGTAGGCTGGAGCTGC 3'	This study
FRT_KanR Rev	5' TTAAAGGTATAAAAACAACCTTTTGTCTTTTACCTTCCGTTTCGCTC CTGTCAAACATGAGAAATTAATCC 3'	This study
ColE1 fwd	5' GAGCGAAACGGGAAGGTAAAAAGACAAAAAGTTGTTTTAATACCTTTAA CGTTCCACTGAGCGTC 3'	This study
ColE1 rev	5' CAGGCTTCAACGGATTCTTTTCTATTTTCATAGCCCGGAGCAACCTGTG GGTATCCACAGAATCAGG 3'	This study
pAPG1 seq1	5' GATCAAGCAGAGGCTGAAG 3'	This study
pAPG1 seq2	5' CCAGGCATCAAATTAAGC 3'	This study
pAPG1 seq3	5' TCTACGTGTTCCGCTTCC 3'	This study
pAPG1 seq4	5' TTGAAGCCTGATCGATGC 3'	This study
pAPG1 seq5	5' AAGATTAGCGGATCCTACC 3'	This study
lacZYA_redμNS fwd	5' TATGTTGTGTGGAATTGTGAGCGGATAACAATTCACACAGGAAACAGCT ATGGTGAGCAAGGGCGA 3'	This study
lacZYA_redμNS rev	5' CAATTTTATAATTTAAACTGACGATTCAACTTTTATAATCTTTGAAATAA GGATCCGTCGACCTGCAG 3'	This study
Lacl fwd	5' GGCCGATTCAATTAATGCAGCTGGC 3'	This study
CynX rev	5' GGCCTGATAAGCGCAGCGTATC 3'	This study
mCherry rev	5' GGTGCTTCACGTAGGCCCTGG 3'	This study
KanR fwd	5' CGGAGAACCTGCGTGCAATCC 3'	This study
mTagBFP2 fwd	5' CGGAGCTCGAATTCGGATCCTTAATTAAGCTTGTGCCCCAGTTTG 3'	This study
mTagBFP2 fwd	5' CTTTAAGAAGGAGATATACCATGAGCGAGCTGATTAAGGAGAAC 3'	This study
pET28_one fwd	5' TCCTTAATCAGCTCGCTCATGGTATATCCTTCTTAAAGTTAAAC 3'	This study
pET28_one rev	5' GGATAACCGTATTACCGCCTTTGAGTGAGCTGATACCG 3'	This study
pET28_two fwd	5' AGCGGTATCAGCTCACTCAAAGGCGGTAATACGGTTATCC 3'	This study
pET28_two rev	5' TGGGGCACAAAGCTTAATTAAGGATCCGAATTCGAGCTCC 3'	This study
pET28mTagBFP2 fwd	5' CTTAGTGACTCGAATTCGCGCGCCAATCCGGATATAGTTCC 3'	This study
pET28mTagBFP2 rev	5' ACTTCTGGCTGGATGATGGACGTGAGTTTTCGTTCCACTG 3'	This study
cmR fwd	5' AGTGAACGAAACTCACGTCCATCCAGCCAGAAAGTG 3'	This study
cmR rev	5' GAACTATATCCGGATTGGCGCGCAATTCGAGTCACTAAGG 3'	This study

Table S5

Chemicals used in this study.

Chemical	Source	Catalog number
Agarose	AmericanBio	Cat#AB00972-00500
4',6-diamidine-2'-phenylindole (DAPI), dihydrochloride fluorescent dye	Thermo Fisher Scientific	Cat#D1306
Isopropyl β-D-1-thiogalactopyranoside (IPTG)	Sigma Aldrich	Cat#I5502
Rifampicin	Sigma Aldrich	Cat#R3501
Cephalexin	Sigma Aldrich	Cat#C4895
A22	Sigma Aldrich	Cat#SML0471
4',6-Diamidine-2'-phenylindole dihydrochloride (DAPI)	Thermo Fisher Scientific	Cat#D1306

Program / library / script	Package	Source
Oufiti	www.oufiti.org	(Paintdakhi et al., 2016)
MATLAB	www.mathworks.com	Mathworks
Python	www.python.org	Python Software Foundation
Numpy	www.numpy.org	(Harris et al., 2020)
Scipy	www.scipy.org	(Virtanen et al., 2020)
Pytorch	www.pytorch.org	(Paszke et al., 2019)
Scikit-image	www.scikit-image.org	(Van Der Walt et al., 2014)
Scikit-learn	www.scikit-learn.org/stable/	(Pedregosa et al., 2012)
Statsmodels	www.statsmodels.org/stable/index.html	(Seabold and Perktold, 2010)
Shapely	www.pyapi.org/project/shapely/	(Gillies, Sean et al., 2023)
Matplotlib	www.matplotlib.org	(Hunter, 2007)
Seaborn	www.seaborn.pydata.org	(Waskom, 2021)
Pandas	www.pandas.pydata.org	(McKinney, 2010)
Python	Omnipose neural network	(Cutler et al., 2022)
MATLAB	SuperSegger	(Stylianidou et al., 2016)
Python	Unet neural network	(Wiktor et al., 2021; Zhou et al., 2020)

Table S6

Software used in this study.

snapshots_analysis_UNET_ghv.py	Snapshot image analysis (from UNET masks) – custom Python class	This study (www.github.com/JacobsWagnerLab/published/tree/master/Papagiannakis_2024)
snapshots_analysis_OUFTI_GrayGovers_ghv.py	Snapshot image analysis (from Oufti masks) – custom Python class	This study (www.github.com/JacobsWagnerLab/published/tree/master/Papagiannakis_2024)
snapshots_analysis_functions.py	Extraction of fluorescence and morphology statistics from cell snapshots – custom Python functions	This study (www.github.com/JacobsWagnerLab/published/tree/master/Papagiannakis_2024)
microfluidics_segmentation_ghv.py	Cell segmentation and tracking from time-lapse images in microfluidics – custom Python class	This study (www.github.com/JacobsWagnerLab/published/tree/master/Papagiannakis_2024)
microfluidics_analysis_functions_ghv.py	Extraction of fluorescence and morphology statistics from time-lapse images in microfluidics – custom Python functions	This study (www.github.com/JacobsWagnerLab/published/tree/master/Papagiannakis_2024)
Omnipose_to_python_ghv.py	Extraction of segmentation masks and tracked lineages from Omnipose/SuperSegger to Python for further analysis	This study (www.github.com/JacobsWagnerLab/published/tree/master/Papagiannakis_2024)
Otsu_phase_segmentation_ghv.py	Cell segmentation and tracking using Otsu thresholding (Otsu, 1979)	This study (www.github.com/JacobsWagnerLab/published/tree/master/Papagiannakis_2024)
Bivariate_medial_axis_estimation.py	Function that draws the medial axis in randomly oriented cells on agarose pads	This study (www.github.com/JacobsWagnerLab/published/tree/master/Papagiannakis_2024)
LoG_adaptive_image_filter.py	Image filter that applies relative and local thresholding to segment fluorescence objects and particles	This study (www.github.com/JacobsWagnerLab/published/tree/master/Papagiannakis_2024)
Microfluidics_segmentation_and_tracking_example.pdf	Example for cell segmentation and tracking in microfluidics	This study (www.github.com/JacobsWagnerLab/published/tree/master/Papagiannakis_2024)
Analysis_of_cell_morphology_and_fluorescence_in_microfluidics.pdf	Example for the extraction of fluorescence and morphology statistics from time-lapse images in microfluidics	This study (www.github.com/JacobsWagnerLab/published/tree/master/Papagiannakis_2024)
Analysis_of_cell_morphology_and_fluorescence_in_agarose_pads.pdf	Example for the extraction of fluorescence and morphology statistics from cell snapshots	This study (www.github.com/JacobsWagnerLab/published/tree/master/Papagiannakis_2024)
Example_of_using_Otsu_based_segmentation_ghv.pdf	Example for the segmentation of antibiotic-treated cells using the Otsu threshold	This study (www.github.com/JacobsWagnerLab/published/tree/master/Papagiannakis_2024)
tune_diffusion.ipynb	Simulations of the minimal reaction-diffusion model that describes nucleoid segregation in <i>E. coli</i> .	This study (https://github.com/qiweiyuu/polysome)

Table S6 (continued)

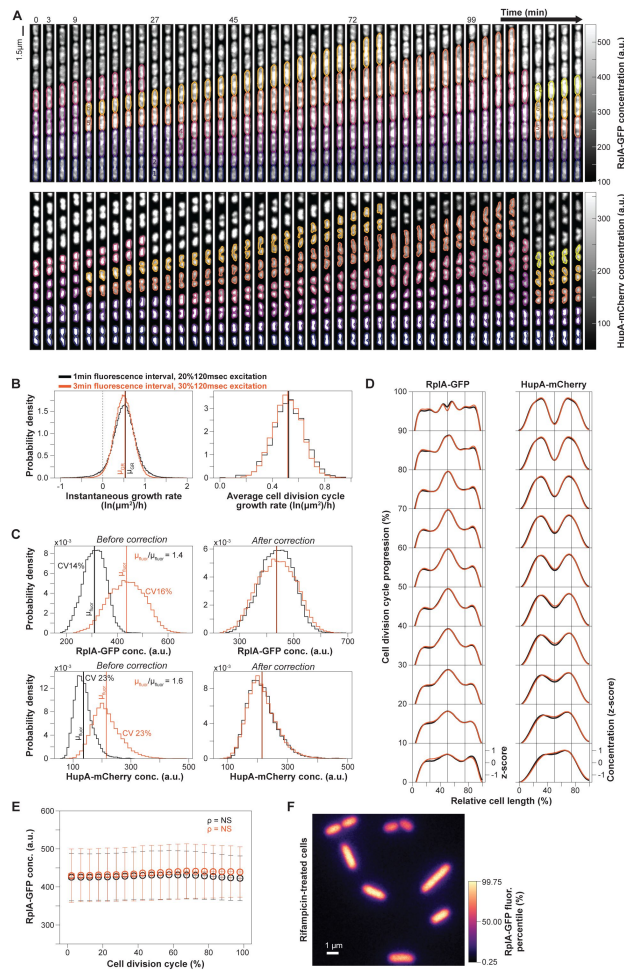


Figure 1 – figure supplement 1

Reproducibility analysis of the dynamic ribosome and nucleoid distributions between microfluidic experiments.

While phase-contrast images were acquired every minute in all microfluidics experiments, two different intervals (1 min and 3 min) were used for fluorescence image acquisition. **A.** Kymographs showing the RplA-GFP and HupA-mCherry concentration in CJW7323 cells growing in microfluidic channels in M9gluCAAT. Cell and nucleoid contours are shown in each channel using a different color for each cell lineage (from dark purple to bright orange). **B-E.** Two biological replicate experiments were performed for this strain and nutrient condition using different intervals of fluorescence image acquisition (1 or 3 min). **B.** Plots showing that the instantaneous and average division cycle growth rates were nearly identical between the two experiments. **C.** Plots showing the distributions of the indicated fluorescence signals for the 1-min acquisition interval (black) compared to the 3-min frame rate interval (orange) before and after correction. The difference in the excitation power between the two experiments was 50% (240 %ms vs. 360 %ms), which was also reflected in the ratio of their average fluorescence values (1.4 for RplA-GFP and 1.6 for HupA-mCherry). The fluorescence values from the 1-min interval experiment were corrected by multiplying by these ratios, resulting in a near-perfect overlap between fluorescence distributions after correction. **D.** Plots showing that the scaled (z-score) RplA-GFP and HupA-mCherry intensity profiles from birth to division were almost identical between the two experiments. The intensity profiles are shown for 10 cell division cycle intervals (1489 and 2633 cell division cycles for the 1-min and 3-min interval experiments, respectively). The z-score was calculated for each segmented cell by subtracting the whole cell average fluorescence and then dividing the difference by the standard deviation. Each intensity profile corresponds to the average z-score of all segmented cell instances within the corresponding cell cycle interval. **E.** Plot showing that the average RplA-GFP concentration from birth to division (averages $\pm$ SD across 20 cell division cycle bins) remained constant and was virtually identical between the two experiments. **F.** Representative image of RplA-GFP fluorescence in cells (CJW7323) treated with rifampicin (100 μ g/mL). The cells were treated for 45 min, washed, and spotted on an M9gluCAAT agarose pad without antibiotic. The presented snapshot corresponds to the first time point immediately after cell spotting showing diffuse distribution of RplA-GFP signal.

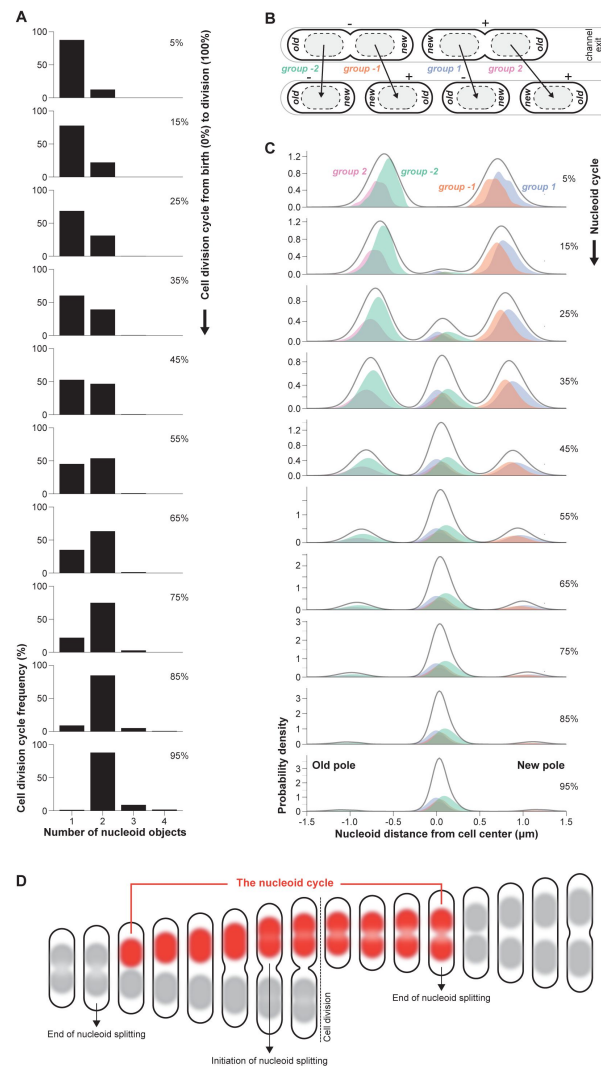


Figure 1 – figure supplement 2

Tracking nucleoid segregation cycles.

Since the timing of nucleoid segregation varied between cell division cycles, the nucleoid segregation cycles were tracked independently of the cell division cycles for each cell lineage to measure the relative timing of RplA-GFP accumulation and HupA-mCherry depletion in the middle of the nucleoid (as shown in **Figure 1E**). **A**. Plots showing the frequency of cells with one, two, three, or four detected nucleoid objects across 10 cell division cycle bins from birth (top) to division (bottom). Data from 4122 cell division cycles are shown. **B**. The polarity of the cells (+/-) and the relative position of the nucleoid mask (toward the new or old pole) was used to track the nucleoid segregation cycle. This strategy was used to identify four groups of nucleoid segregation cycles (group -2, -1, 1 and 2). Group -2 includes nucleoids that were "born" toward the old cell pole in a mother cell with negative (-) polarity and were inherited at the center of a daughter cell with a negative polarity. Group -1 includes nucleoids that were "born" toward the new cell pole in a mother cell with negative (-) polarity and were inherited at the center of a daughter cell with a positive (+) polarity. Group 1 includes nucleoids that were "born" toward the new cell pole in a mother cell with positive (+) polarity and were inherited at the center of a daughter cell with negative (-) polarity. Group 2 includes nucleoids that were "born" toward the old cell pole in a mother cell with positive (+) polarity and were inherited at the center of a daughter cell with positive (+) polarity. **C**. Plots showing the distributions of nucleoid positions around the cell center for 10 nucleoid segregation cycle bins, from the time a nucleoid was "born" (top) until it split (bottom). These plots show the inheritance of the nucleoids from the quarter cell positions of the mother cells to the middle of their daughters for the four groups of nucleoid segregation cycles. The solid gray line represents the average density of the nucleoid positions for all four groups. Data from 2286 complete nucleoid cycles are shown. **D**. Schematic that explains the definition of a nucleoid cycle. The nucleoid cycle ranges from the end of a nucleoid splitting event, until the next splitting of the sister nucleoids. It usually extends beyond cell division, into the next cell division cycle.

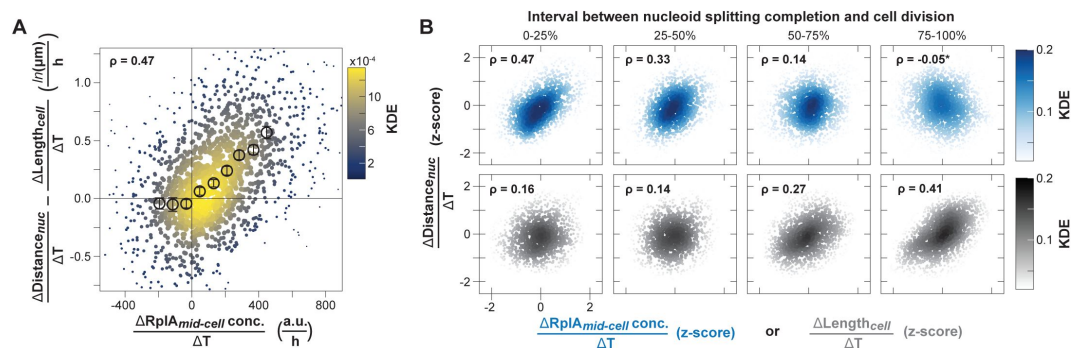


Figure 1 – figure supplement 3

Correlations used to calculate the relative contribution of polysome accumulation and cell elongation to nucleoid migration.

A. Correlation (Spearman $\rho = 0.47$, p -value $< 10^{-10}$) between the rate of RplA-GFP accumulation at mid-cell ($\frac{\Delta \text{RplA}_{\text{mid-cell conc.}}}{\Delta T}$) and the rate of distance increase between the sister nucleoids minus the rate of cell elongation ($\frac{\Delta \text{Distance}_{\text{nuc}}}{\Delta T} - \frac{\Delta \text{Length}_{\text{cell}}}{\Delta T}$) across cells. The colormap corresponds to a Gaussian kernel density estimation. Binned data are also shown (mean $\pm$ SEM, 75 to 177 cell division cycles per bin, 9 bins in total) within the 5th-95th percentiles of the x-axis range. **B.** Scaled correlations (z-scores) between $\frac{\Delta \text{RplA}_{\text{mid-cell conc.}}}{\Delta T}$ (blue) or $\frac{\Delta \text{Length}_{\text{cell}}}{\Delta T}$ (grey) and $\frac{\Delta \text{Distance}_{\text{nuc}}}{\Delta T}$ (y-axis) during four relative time bins (1335 to 1957 cell division cycles per bin) covering the period from the end of nucleoid splitting until cell division. Shown are the Spearman correlations (ρ), all with a p -value below 10^{-6} , except for the one marked with an asterisk (p -value = 0.02).

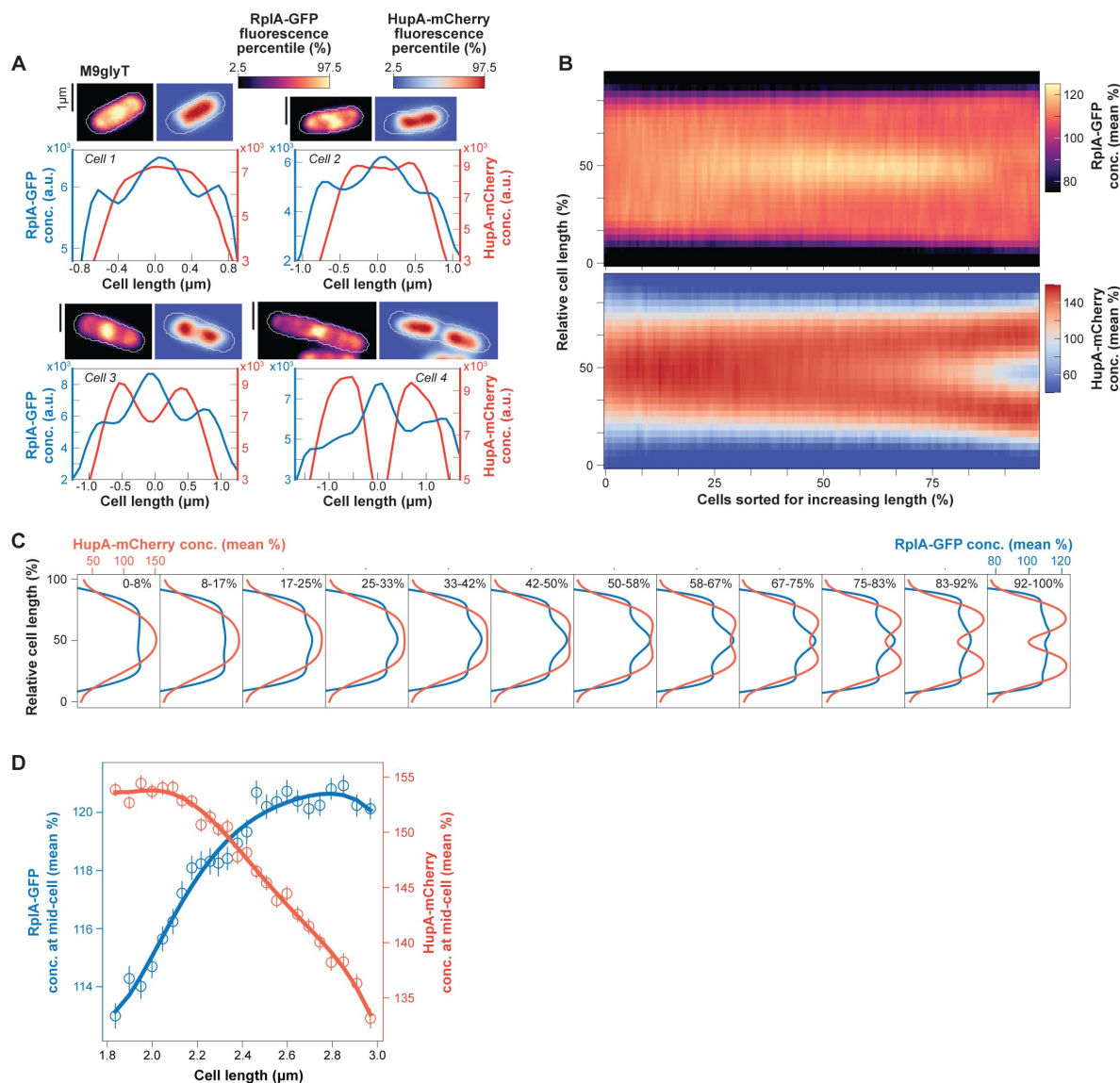


Figure 1 – figure supplement 4

Examination of the relative timing of the initiation of nucleoid constriction and the accumulation of polysomes at mid-nucleoid.

A. RplA-GFP and HupA-mCherry concentration (fluorescence arbitrary units) images of four representative single cells (CJW7323) growing in M9glyT and their fluorescence intensity profiles along the cell length. **B.** Demographs of the scaled RplA-GFP and HupA-mCherry concentration from 13554 *E. coli* cells grown in M9glyT. **C.** Average 1D intensity profiles of the scaled (divided by the whole cell average) RplA-GFP and HupA-mCherry concentration for 12 bins of cells lengths (>1100 cells per intensity profile). **D.** Plot showing the scaled RplA-GFP and HupA-mCherry concentration in the middle of the cells for increasing cell length (~370 cells per bin, 25 bins). Cells longer than 3 μm were excluded to avoid the effects of cell constriction. The arrow and vertical dashed line indicate the minimum cell length bin with an apparent HupA-mCherry depletion at mid-cell, marking the initiation of nucleoid splitting. The error bars indicate mean $\pm$ standard error of the mean (SEM).

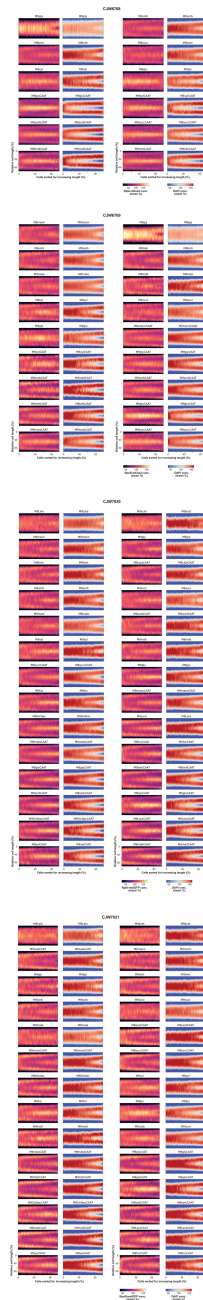


Figure 2 – figure supplement 1

Demographs of scaled ribosome and nucleoid fluorescence for strains with different ribosome markers and under various nutrient conditions.

Demographs are ordered according to increasing average cell area, which scales with growth rate (Schaechter et al., 1958 [\[link\]](#)), from left to right and top to bottom. These demographs were constructed from snapshots of the following strains: **A.** CJW6768 (RplA-mEos2 ribosomal marker, 747 to 2446 cells per condition), **B.** CJW6769 (RpsB-mEos2 ribosomal marker, 690 to 3169 cells per condition), **C.** CJW7020 (RplA-msfGFP ribosomal marker, 657 to 2432 cells per condition) and **D.** CJW7021 (RpsB-msfGFP ribosomal marker, 788 to 1950 cells per condition), using the cell length as a proxy for the cell division cycle, with the signal intensity profile sorted from the shortest newborn cells to the longest predivisional cells. Note that the demographs for the CJW6768 strain (panel A) in the nutrient conditions M9mann, M9mala, M9malt, M9mannCAAT, M9glyCAAT and M9malaCAAT are not shown here, as they are presented in **Figure 2A** [\[link\]](#). The nutrient abbreviations and compositions are explained in Table S1.

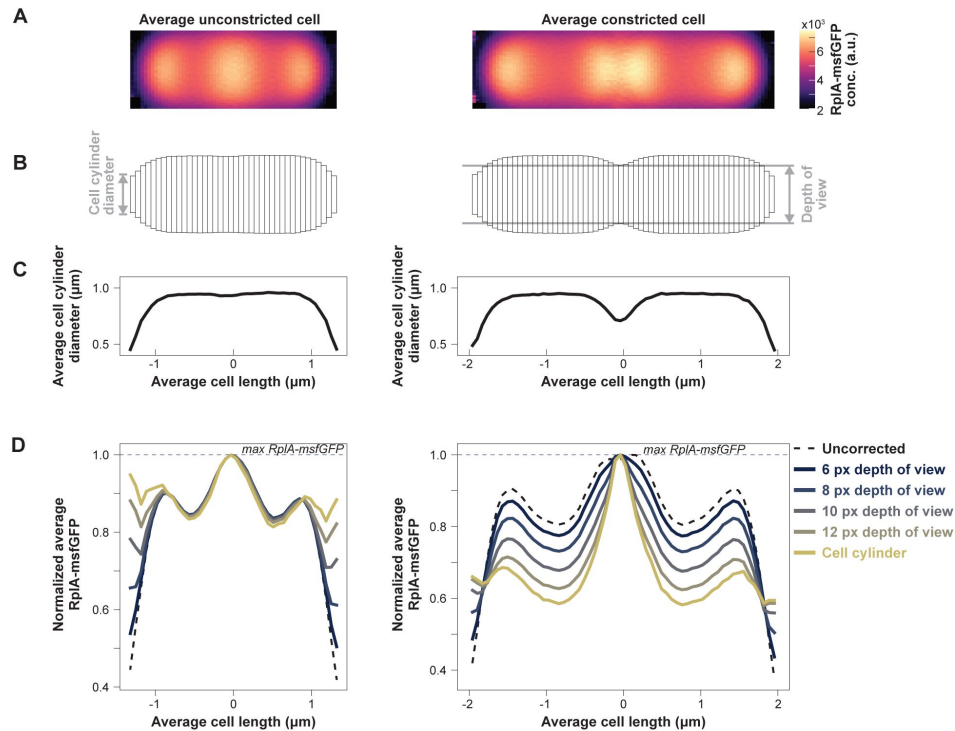


Figure 2 – figure supplement 2

Calculation of RplA-msfGFP concentration after cell curvature correction.

A. Average 2D RplA-msfGFP projections for short (2.5 to 3 μm – left), unstricted cells and long (3.5 and 5.5 μm - right) constricted cells. The average RplA-msfGFP signal from 1000 sampled cells (CJW7651) is shown for each population. **B.** The 2D cell areas were divided in cylindrical sectors with a height (h) of a single pixel. The number of cylindrical sectors corresponds to the average cell length in pixels (40 and 60 sectors for the unstricted and constricted cells, respectively). The radius (r) of each cylindrical segment corresponds to half the cell diameter for that specific cell length position. The cylindrical segments at the poles or constriction site have a smaller radius due to the curvature of the cell boundaries. The division of the cell area into cylindrical segments allowed us to calculate the volume for each segment ($V_{cylinder} = \pi r^2 h$). The sum of the fluorescence per segment was corrected for the 3D curvature of the cell boundaries by dividing by the volume of the cylinder, either considering the entire cell width or different depths of view narrower than the maximal cell width. **C.** 1D profiles of the cylinder diameter are also indicative of the cell curvature at the poles and constriction site. **D.** Comparisons of the 1D average RplA-msfGFP projection between the poles before (dashed lines) or after correcting for the cell curvature (solid lines) for different depths of view (colormap). The uncorrected signal (dashed lines) corresponds to the average fluorescence along each cylindrical segment (mean of pixel intensities). The 1D profiles were normalized by dividing by the maximal RplA-msfGFP concentration at mid-cell. This correction shows that the pronounced decrease of the RplA-msfGFP fluorescence at the poles is the result of the curvature of the cell boundaries. The observed over-correction at the cell poles for larger depths of view (> 8 pixels or px) is likely due to the over-estimation of the cell segmentation mask that was determined based on the phase contrast snapshot images of cells on agarose pads.

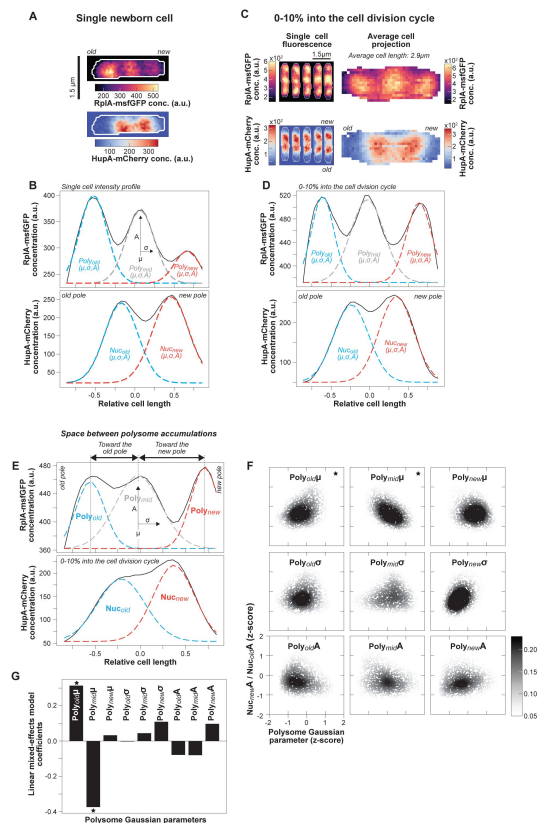


Figure 3 – figure supplement 1

Analysis of the polysome and nucleoid asymmetries using fitted Gaussian functions.

A. Example images of RplA-GFP and HupA-mCherry fluorescence signals in a single newborn cell instance. **B.** Plots showing the Gaussian-function fitting on RplA-GFP and HupA-mCherry intensity profiles from the single cell snapshot shown in panel A. Three Gaussian functions were fitted to the RplA-GFP fluorescence in newborn cells (eq. 1 in Methods), capturing the accumulation of polysomes at mid-cell and the poles. Two Gaussian functions were fitted to the HupA-mCherry fluorescence (eq. 2 in Methods) capturing the two lobes of the segregating sister nucleoids. The parameters of the fitted Gaussian functions provided information about the position (μ : mean) and the concentration (A: amplitude) of each fluorescence statistic, as well as the cell length range occupied by it (σ : standard deviation). The Gaussian area (eq. 3 in Methods) corresponds to the abundance of each macromolecule. The RplA-GFP Gaussians were fitted above the cellular background, which presumably corresponds to the uniform fluorescence of free ribosomes or ribosomal subunits. For the HupA-mCherry Gaussian fittings, the cellular background corresponds to the DNA-free cell regions. The Gaussian parameters were also used to describe the polysome asymmetries between the poles (eq. 4 in Methods) as well as the position of the nucleoid around the cell center (eq. 5 in Methods). These statistics were used in Figure 3B. **C.** Gaussian functions were also fitted to the ensemble average RplA-GFP and HupA-mCherry fluorescence (right) calculated from all the cell segmentation instances of a single cell division cycle, 0-10% from birth to division (left). **D.** The Gaussian fitting to the ensemble RplA-GFP and HupA-mCherry intensity profiles (concentration in arbitrary fluorescence units) early in the cell division cycle for the 2D projection shown in panel C (right). **E.** Plots showing the nucleoid and polysome density axial asymmetry determined using the parameters of the fitted Gaussians on the HupA-mCherry and RplA-GFP fluorescence (eq. 6 and eq. 8 in Methods). **F.** Plots showing the scaled correlations between the polysome Gaussian parameters (μ , A, and σ for each polysome accumulation) and the nucleoid density asymmetry (eq. 6 in Method; 2103 cell division cycles). Each marker corresponds to the data for a single cell division cycle (0-10% into the cell division cycle) as in panel C (right). A Gaussian kernel density estimation (KDE) was used to illustrate the density of the scatter plots (see Methods). **G.** Bar graph showing the coefficients of a linear mixed-effects model (see eq. 7 in Methods) used to identify the polysome statistics that contribute to the nucleoid density asymmetry. The highly contributing polysome statistics were combined to a compound polysome statistic (eq. 8 in Methods) that describes the available DNA space between polysome accumulations. This compound statistic correlates with the nucleoid density asymmetry (eq. 6 in Methods) as shown in Figure 3F. The stars indicate the polysome statistics that most significantly correlate with the nucleoid density asymmetry (absolute coefficient above 0.3).

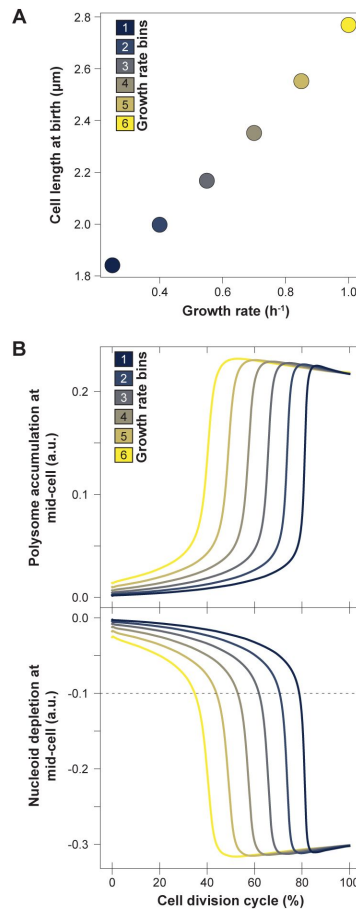


Figure 4 – figure supplement 1

Statistics of polysome accumulation and nucleoid depletion during simulated cell growth.

Polysome and nucleoid dynamics were simulated for six different growth rates. The predivisional polysome and nucleoid profiles of simulated trajectories initialized from the steady state (infinite relaxation time) were used as initial conditions ($D_n = 10^{-3} \mu m^2/s$). **A.** Plot showing cell length at birth increasing with growth rate according to published measurements (Govers et al., 2024). **B.** Plot showing the dynamics of polysome accumulation (top) and nucleoid depletion (bottom) at mid-cell from birth to division for six simulated growth rates (colormap). A nucleoid depletion threshold of 0.1 arbitrary units (a.u., dashed horizontal line) was used to determine the relative time of nucleoid splitting that is presented in Figure 4C.

Figure 5 – figure supplement 1

Phenotypic effects of transcription inhibition.

Plots showing cellular parameters (instantaneous growth rate, cell area, nucleoid area, and the ratio between nucleoid area and cell area, or NC ratio) affected by the rifampicin treatment. The instantaneous growth rate was estimated using the log-transformed cell area, using a rolling window of 10 min. The black line corresponds to the averages from 1859 cell division cycles and the gray-shaded area indicates the range of one standard deviation around the mean. The two cycles of rifampicin treatment (as shown in [Figure 5A](#)) were overlaid for this plot, using the time of rifampicin addition as $t = 0$ min.

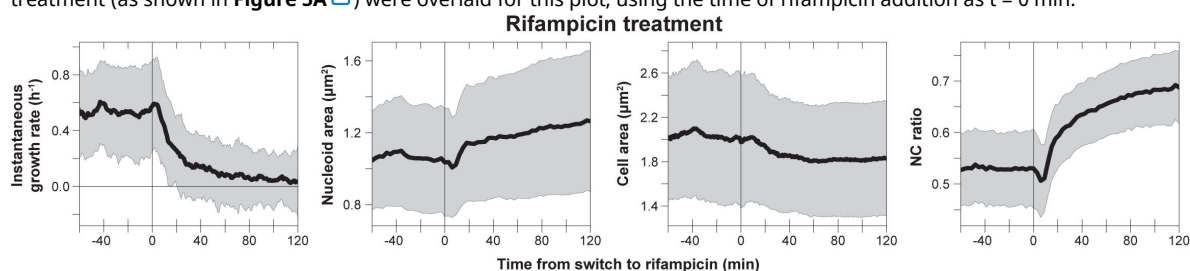
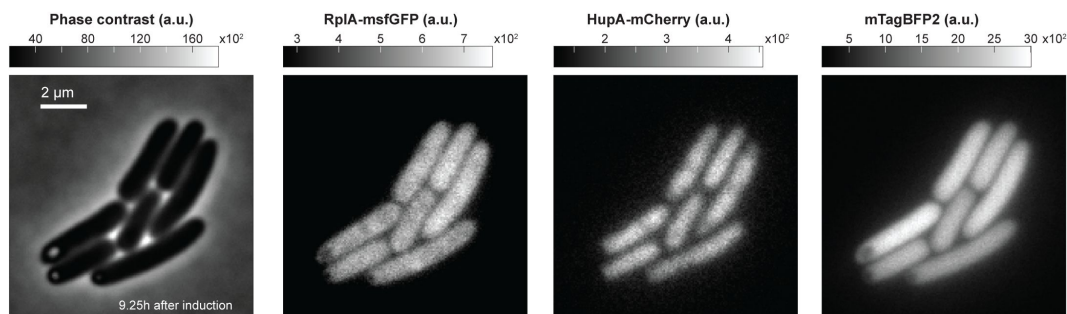


Figure 6 – figure supplement 1

Phenotypic effects of prolonged protein over-expression from plasmids.

Representative phase contrast and fluorescence images of cells (CJW7798) after prolonged induction (100 μ M IPTG, for 9 h and 15 min) of mTagBFP2 expression from aT7 promoter on a multi-copy plasmid.

mTagBFP2 over-expression from a plasmid



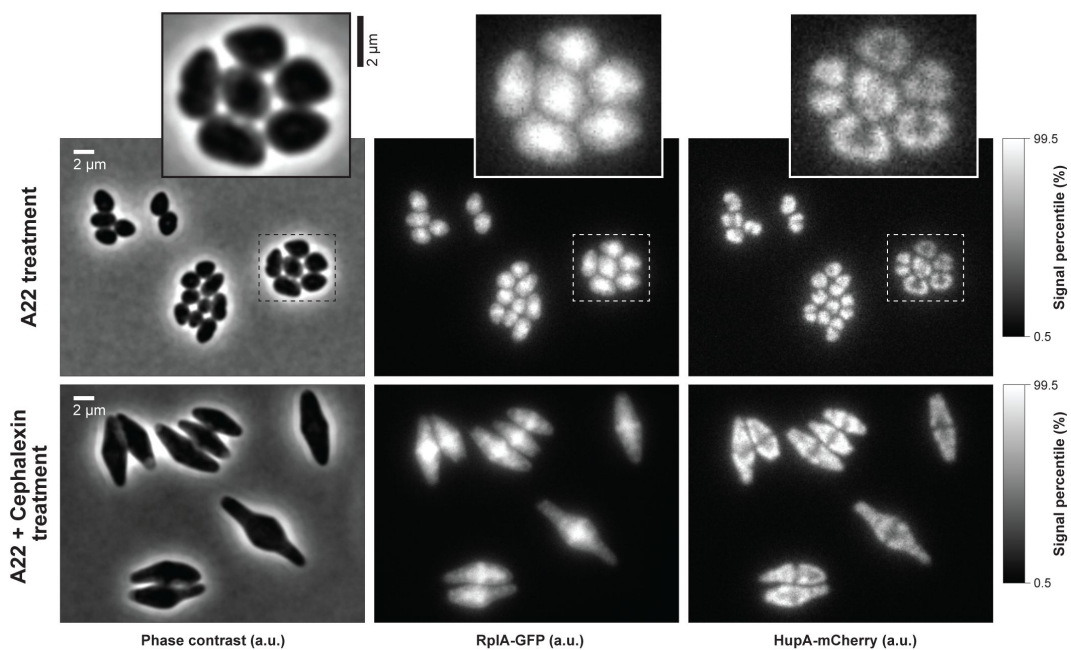


Figure 7 – figure supplement 1

Phenotypic of effects of cell growth under A22 and cephalalexin treatment.

Representative phase contrast and fluorescence images of cells (CJW7323) treated with A22 (4 $\mu\text{g/mL}$) alone or with both A22 (4 $\mu\text{g/mL}$) and cephalalexin (50 $\mu\text{g/mL}$) for 130 min.

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Additional information

Author contributions

Conceptualization, A.P. and C.J.-W.; methodology, A.P.; software, A.P. and Q.Y.; formal analysis, A.P.; investigation, A.P., Q.Y. and S.K.G.; resources, A.P., Q.Y., and W.-H.L.; data curation, A.P.; writing – original draft, A.P. and C.J.-W.; writing – review and editing, A.P., C.J.-W, N. W., Q.Y., S.K.G., and W.-H.L.; visualization, A.P. and C.J.-W.; supervision, N.W. and C.J.-W.; project administration, C.J.-W.; funding acquisition, A.P., N.W. and C.J.-W.

Additional files

Movie S1. Example time-lapse sequence showing ribosome and nucleoid dynamics during cell growth under steady state condition in a microfluidic channel. Shown are corresponding inverted phase contrast signal, RplA-GFP signal, HupA-mCherry signal, cell contours (based on phase contrast signals) and nucleoid contours (based on nucleoid signal segmentation) of *E. coli* cells (CJW7323) grown in a microfluidic channel supplemented with M9gluCAAT. The white circles indicate the centroid of the cell segmentation masks and the extending lines indicate the tracked cell traces. [🔗](#)

Movie S2. Video showing the average subcellular distribution of ribosome and nucleoid signals from birth to division. Shown are 2D average RplA-GFP (top) and HupA-mCherry (bottom) projections from birth to division, with corresponding average intensity profiles (right). The cell projections are oriented from the old pole on the left to the new pole on the right. The average cell contour is also drawn. Ensemble data from 4122 division cycles of CJW7323 cells growing in M9gluCAAT are shown. [🔗](#)

Movie S3. Video showing simulated 1D profiles of polysome and nucleoid concentration during slow and fast cell growth. Simulations during slow (left) and fast (right) growth are shown. The simulations were initialized from the equilibrium configuration, with a compact symmetric nucleoid ($Dn = 10^{-3} \mu\text{m}^2/\text{sec}$) at the cell center. [🔗](#)

Movie S4. Video showing the effects of rifampicin addition on cell growth, ribosome signal heterogeneity, nucleoid segregation, and nucleoid compaction. Examples of five microfluidic channels showing the corresponding inverted phase contrast, RplA-GFP and HupA-mCherry signals (from left to right) of cells (CJW7323) growing within microfluidic channels. Rifampicin was added at 120 and 720 min. Each antibiotic treatment lasted 120 min. [🔗](#)

Movie S5. Video showing the effects of ectopic polysome formation on nucleoid dynamics over time. The RplA-GFP and HupA-mCherry fluorescence normalized (norm.) by the average cell fluorescence are shown, together with their corresponding phase contrast image, for multiple cells (CJW7798) in succession

following induction of mTagBFP2 expression from a T7 promoter on a multi-copy plasmid. The scale bar indicates 1 μm , and the time since cell birth is shown in minutes. The cell contours indicate the boundaries of the cell masks obtained by cell segmentation of the corresponding phase contrast images. [🔗](#)

Movie S6. Video showing how the loss of cell width confinement due to cephalixin and A22 treatment affects ribosome and nucleoid distributions over time. Fluorescence images of RplA-GFP and HupA-mCherry fluorescence (fluor.) normalized by the average cellular fluorescence images, together with their corresponding phase contrast images, are shown for cells (CJW7323) following treatment with A22 (4 $\mu\text{g}/\text{mL}$) and cephalixin (50 $\mu\text{m}/\text{mL}$). The scale bar indicates 1 μm . The time since drug addition is shown in minutes. [🔗](#)

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Reviewer #1 (Public review):

Summary:

This paper is an elegant, mostly observational work, detailing observations that polysome accumulation appears to drive nucleoid splitting and segregation. Overall I think this is an insightful work with solid observations.

Strengths:

The strengths of this paper are the careful and rigorous observational work that leads to their hypothesis. They find the accumulation of polysomes correlates with nucleoid splitting, and that the nucleoid segregation occurring right after splitting correlates with polysome segregation. These correlations are also backed up by other observations:

- (1) Faster polysome accumulation and DNA segregation at faster growth rates.
- (2) Polysome distribution negatively correlating with DNA positioning near asymmetric nucleoids.
- (3) Polysomes form in regions inaccessible to similarly sized particles.

These above points are observational, I have no comments on these observations leading to their hypothesis.

Weaknesses:

It is hard to state weaknesses in any of the observational findings, and furthermore, their two tests of causality, while not being completely definitive, are likely the best one could do to examine this interesting phenomenon.

Points to consider / address:

Notably, demonstrating causality here is very difficult (given the coupling between transcription, growth, and many other processes) but an important part of the paper. They do two experiments toward demonstrating causality that help bolster - but not prove - their hypothesis. These experiments have minor caveats, my first two points.

(1) First, "Blocking transcription (with rifampicin) should instantly reduce the rate of polysome production to zero, causing an immediate arrest of nucleoid segregation". Here they show that adding rifampicin does indeed lead to polysome loss and an immediate halting of segregation - data that does fit their model. This is not definitive proof of causation, as rifampicin also (a) stops cell growth, and (b) stops the translation of secreted proteins. Neither of these two possibilities is ruled out fully.

1a) As rifampicin also stops all translation, it also stops translational insertion of membrane proteins, which in many old models has been put forward as a possible driver of nucleoid segregation, and perhaps independent of growth. This should at last be mentioned in the discussion, or if there are past experiments that rule this out it would be great to note them.

1b) They address at great length in the discussion the possibility that growth may play a role in nucleoid segregation. However, this is testable - by stopping surface growth with antibiotics. Cells should still accumulate polysomes for some time, it would be easy to see if nucleoids are still segregated, and to what extent, thereby possibly decoupling growth and polysome production. If successful, this or similar experiments would further validate their model.

(2) In the second experiment, they express excess TagBFP2 to delocalize polysomes from midcell. Here they again see the anticorrelation of the nucleoid and the polysomes, and in some cells, it appears similar to normal (polysomes separating the nucleoid) whereas in others the nucleoid has not separated. The one concern about this data - and the differences between the "separated" and "non-separated" nuclei - is that the over-expression of TagBFP2 has a huge impact on growth, which may also have an indirect effect on DNA replication and termination in some of these cells. Could the authors demonstrate these cells contain 2 fully replicated DNA molecules that are able to segregate?

(3) What is not clearly stated and is needed in this paper is to explain how polysomes do (or could) "exert force" in this system to segregate the nucleoid: what a "compaction force" is by definition, and what mechanisms causes this to arise (what causes the "force") as the "compaction force" arises from new polysomes being added into the gaps between them caused by thermal motions.

They state, "polysomes exert an effective force", and they note their model requires "steric effects (repulsion) between DNA and polysomes" for the polysomes to segregate, which makes sense. But this makes it unclear to the reader what is giving the force. As written, it is unclear if (a) these repulsions alone are making the force, or (b) is it the accumulation of new polysomes in the center by adding more "repulsive" material, the force causes the nucleoids to move. If polysomes are concentrated more between nucleoids, and the polysome

concentration does not increase, the DNA will not be driven apart (as in the first case) However, in the second case (which seems to be their model), the addition of new material (new polysomes) into a sterically crowded space is not exerting force - it is filling in the gaps between the molecules in that region, space that needs to arise somehow (like via Brownian motion). In other words, if the polysome region is crowded with polysomes, space must be made between these polysomes for new polysomes to be inserted, and this space must be made by thermal (or ATP-driven) fluctuations of the molecules. Thus, if polysome accumulation drives the DNA segregation, it is not "exerting force", but rather the addition of new polysomes is iteratively rectifying gaps being made by Brownian motion.

The authors use polysome accumulation and phase separation to describe what is driving nucleoid segregation. Both terms are accurate, but it might help the less physically inclined reader to have one term, or have what each of these means explicitly defined at the start. I say this most especially in terms of "phase separation", as the currently huge momentum toward liquid-liquid interactions in biology causes the phrase "phase separation" to often evoke a number of wider (and less defined) phenomena and ideas that may not apply here. Thus, a simple clear definition at the start might help some readers.

(4) Line 478. "Altogether, these results support the notion that ectopic polysome accumulation drives nucleoid dynamics". Is this right? Should it not read "results support the notion that ectopic polysome accumulation inhibits/redirects nucleoid dynamics"?

(5) It would be helpful to clarify what happens as the RplA-GFP signal decreases at midcell in Figure 1- is the signal then increasing in the less "dense" parts of the cell? That is, (a) are the polysomes at midcell redistributing throughout the cell? (b) is the total concentration of polysomes in the entire cell increasing over time?

(6) Line 154. "Cell constriction contributed to the apparent depletion of ribosomal signal from the mid-cell region at the end of the cell division cycle (Figure 1B-C and Movie S1)" - It would be helpful if when cell constriction began and ended was indicated in Figures 1B and C.

(7) In Figure 7 they demonstrate that radial confinement is needed for longitudinal nucleoid segregation. It should be noted (and cited) that past experiments of *Bacillus* l-forms in microfluidic channels showed a clear requirement role for rod shape (and a given width) in the positing and the spacing of the nucleoids.

Wu et al, Nature Communications, 2020 . "Geometric principles underlying the proliferation of a model cell system" <https://dx.doi.org/10.1038/s41467-020-17988-7>

(8) "The correlated variability in polysome and nucleoid patterning across cells suggests that the size of the polysome-depleted spaces helps determine where the chromosomal DNA is most concentrated along the cell length. This patterning is likely reinforced through the displacement of the polysomes away from the DNA dense region"

It should be noted this likely functions not just in one direction (polysomes dictating DNA location), but also in the reverse - as the footprint of compacted DNA should also exclude (and thus affect) the location of polysomes

(9) Line 159. Rifampicin is a transcription inhibitor that causes polysome depletion over time. This indicates that all ribosomal enrichments consist of polysomes and therefore will be referred to as polysome accumulations hereafter". Here and throughout this paper they use the term polysome, but cells also have monosomes (and 2 somes, etc). Rifampicin stops the assembly of all of these, and thus the loss of localization could occur from both. Thus, is it accurate to state that all transcription events occur in polysomes? Or are they grouping all of the n-somes into one group?

<https://doi.org/10.7554/eLife.104276.1.sa3>

Reviewer #2 (Public review):**Summary:**

The authors perform a remarkably comprehensive, rigorous, and extensive investigation into the spatiotemporal dynamics between ribosomal accumulation, nucleoid segregation, and cell division. Using detailed experimental characterization and rigorous physical models, they offer a compelling argument that nucleoid segregation rates are determined at least in part by the accumulation of ribosomes in the center of the cell, exerting a steric force to drive nucleoid segregation prior to cell division. This evolutionarily ingenious mechanism means cells can rely on ribosomal biogenesis as the sole determinant for the growth rate and cell division rate, avoiding the need for two separate 'sensors,' which would require careful coupling.

Strengths:

In terms of strengths; the paper is very well written, the data are of extremely high quality, and the work is of fundamental importance to the field of cell growth and division. This is an important and innovative discovery enabled through a combination of rigorous experimental work and innovative conceptual, statistical, and physical modeling.

Weaknesses:

In terms of weaknesses, I have three specific thoughts.

Firstly, my biggest question (and this may or may not be a bona fide weakness) is how unambiguously the authors can be sure their ribosomal labeling is reporting on polysomes, specifically. My reading of the work is that the loss of spatial density upon rifampicin treatment is used to infer that spatial density corresponds to polysomes, yet this feels like a relatively indirect way to get at this question, given rifampicin targets RNA polymerase and not translation. It would be good if a more direct way to confirm polysome dependence were possible.

Second, the authors invoke a phase separation model to explain the data, yet it is unclear whether there is any particular evidence supporting such a model, whether they can exclude simpler models of entanglement/local diffusion (and/or perhaps this is what is meant by phase separation?) and it's not clear if claiming phase separation offers any additional insight/predictive power/utility. I am OK with this being proposed as a hypothesis/idea/working model, and I agree the model is consistent with the data, BUT I also feel other models are consistent with the data. I also very much do not think that this specific aspect of the paper has any bearing on the paper's impact and importance.

Finally, the writing and the figures are of extremely high quality, but the sheer volume of data here is potentially overwhelming. I wonder if there is any way for the authors to consider stripping down the text/figures to streamline things a bit? I also think it would be useful to include visually consistent schematics of the question/hypothesis/idea each of the figures is addressing to help keep readers on the same page as to what is going on in each figure. Again, there was no figure or section I felt was particularly unclear, but the sheer volume of text/data made reading this quite the mental endurance sport! I am completely guilty of this myself, so I don't think I have any super strong suggestions for how to fix this, but just something to consider.

<https://doi.org/10.7554/eLife.104276.1.sa2>

Reviewer #3 (Public review):**Summary:**

Papagiannakis et al. present a detailed study exploring the relationship between DNA/polysome phase separation and nucleoid segregation in *Escherichia coli*. Using a combination of experiments and modelling, the authors aim to link physical principles with biological processes to better understand nucleoid organisation and segregation during cell growth.

Strengths:

The authors have conducted a large number of experiments under different growth conditions and physiological perturbations (using antibiotics) to analyse the biophysical factors underlying the spatial organisation of nucleoids within growing *E. coli* cells. A simple model of ribosome-nucleoid segregation has been developed to explain the observations.

Weaknesses:

While the study addresses an important topic, several aspects of the modelling, assumptions, and claims warrant further consideration.

Major Concerns:**Oversimplification of Modelling Assumptions:**

The model simplifies nucleoid organisation by focusing on the axial (long-axis) dimension of the cell while neglecting the radial dimension (cell width). While this approach simplifies the model, it fails to explain key experimental observations, such as:

(1) Inconsistencies with Experimental Evidence:

The simplified model presented in this study predicts that translation-inhibiting drugs like chloramphenicol would maintain separated nucleoids due to increased polysome fractions. However, experimental evidence shows the opposite-separated nucleoids condense into a single lobe post-treatment (Bakshi et al 2014), indicating limitations in the model's assumptions/predictions. For the nucleoids to coalesce into a single lobe, polysomes must cross the nucleoid zones via the radial shells around the nucleoid lobes.

(2) The peripheral localisation of nucleoids observed after A22 treatment in this study and others (e.g., Japaridze et al., 2020; Wu et al., 2019), which conflicts with the model's assumptions and predictions. The assumption of radial confinement would predict nucleoids to fill up the volume or ribosomes to go near the cell wall, not the nucleoid, as seen in the data.

(3) The radial compaction of the nucleoid upon rifampicin or chloramphenicol treatment, as reported by Bakshi et al. (2014) and Spahn et al. (2023), also contradicts the model's predictions. This is not expected if the nucleoid is already radially confined.

(4) Radial Distribution of Nucleoid and Ribosomal Shell:

The study does not account for well-documented features such as the membrane attachment of chromosomes and the ribosomal shell surrounding the nucleoid, observed in super-resolution studies (Bakshi et al., 2012; Sanamrad et al., 2014). These features are critical for understanding nucleoid dynamics, particularly under conditions of transcription-translation coupling or drug-induced detachment. Work by Yongren et al. (2014) has also shown that the radial organisation of the nucleoid is highly sensitive to growth and the multifork nature of DNA replication in bacteria.

The omission of organisation in the radial dimension and the entropic effects it entails, such as ribosome localisation near the membrane and nucleoid centralisation in expanded cells, undermines the model's explanatory power and predictive ability. Some observations have been previously explained by the membrane attachment of nucleoids (a hypothesis proposed by Rabinovitch et al., 2003, and supported by experiments from Bakshi et al., 2014, and recent super-resolution measurements by Spahn et al.).

Ignoring the radial dimension and membrane attachment of nucleoid (which might coordinate cell growth with nucleoid expansion and segregation) presents a simplistic but potentially misleading picture of the underlying factors.

This reviewer suggests that the authors consider an alternative mechanism, supported by strong experimental evidence, as a potential explanation for the observed phenomena: Nucleoids may transiently attach to the cell membrane, possibly through transertion, allowing for coordinated increases in nucleoid volume and length alongside cell growth and DNA replication. Polysomes likely occupy cellular spaces devoid of the nucleoid, contributing to nucleoid compaction due to mutual exclusion effects. After the nucleoids separate following ter separation, axial expansion of the cell membrane could lead to their spatial separation.

Incorporating this perspective into the discussion or future iterations of the model may provide a more comprehensive framework that aligns with the experimental observations in this study and previous work.

Simplification of Ribosome States:

Combining monomeric and translating ribosomes into a single 'polysome' category may overlook spatial variations in these states, particularly during ribosome accumulation at the mid-cell. Without validating uniform mRNA distribution or conducting experimental controls such as FRAP or single-molecule measurements to estimate the proportions of ribosome states based on diffusion, this assumption remains speculative.

<https://doi.org/10.7554/eLife.104276.1.sa1>

Author response:

Public Reviews:

Reviewer #1 (Public review):

Summary:

This paper is an elegant, mostly observational work, detailing observations that polysome accumulation appears to drive nucleoid splitting and segregation. Overall I think this is an insightful work with solid observations.

Thank you for your appreciation and positive comments. In our view, an appealing aspect of this proposed biophysical mechanism for nucleoid segregation is its self-organizing nature and its ability to intrinsically couple nucleoid segregation to biomass growth, regardless of nutrient conditions.

Strengths:

The strengths of this paper are the careful and rigorous observational work that leads to their hypothesis. They find the accumulation of polysomes correlates with nucleoid

splitting, and that the nucleoid segregation occurring right after splitting correlates with polysome segregation. These correlations are also backed up by other observations:

(1) Faster polysome accumulation and DNA segregation at faster growth rates.

(2) Polysome distribution negatively correlating with DNA positioning near asymmetric nucleoids.

(3) Polysomes form in regions inaccessible to similarly sized particles.

These above points are observational, I have no comments on these observations leading to their hypothesis.

Thank you!

Weaknesses:

It is hard to state weaknesses in any of the observational findings, and furthermore, their two tests of causality, while not being completely definitive, are likely the best one could do to examine this interesting phenomenon.

It is indeed difficult to prove causality in a definitive manner when the proposed coupling mechanism between nucleoid segregation and gene expression is self-organizing, i.e., does not involve a dedicated regulatory molecule (e.g., a protein, RNA, metabolite) that we could have depleted through genetic engineering to establish causality. We are grateful to the reviewer for recognizing that our two causality tests are the best that can be done in this context.

Points to consider / address:

Notably, demonstrating causality here is very difficult (given the coupling between transcription, growth, and many other processes) but an important part of the paper. They do two experiments toward demonstrating causality that help bolster - but not prove - their hypothesis. These experiments have minor caveats, my first two points.

(1) First, "Blocking transcription (with rifampicin) should instantly reduce the rate of polysome production to zero, causing an immediate arrest of nucleoid segregation". Here they show that adding rifampicin does indeed lead to polysome loss and an immediate halting of segregation - data that does fit their model. This is not definitive proof of causation, as rifampicin also (a) stops cell growth, and (b) stops the translation of secreted proteins. Neither of these two possibilities is ruled out fully.

That's correct; cell growth also stops when gene expression is inhibited, which is consistent with our model in which gene expression within the nucleoid promotes nucleoid segregation and biomass growth (i.e., cell growth), inherently coupling these two processes. This said, we understand the reviewer's point: the rifampicin experiment doesn't exclude the possibility that protein secretion and cell growth drive nucleoid segregation. We are assuming that the reviewer is envisioning an alternative model in which sister nucleoids would move apart because they would be attached to the membrane through coupled transcription-translation-protein secretion (transertion) and the membrane would expand between the separating nucleoids, similar to the model proposed by Jacob et al in 1963 (doi:10.1101/SQB.1963.028.01.048). There are several observations arguing against this cell elongation/transertion model.

(1) For this alternative mechanism to work, membrane growth must be localized at the middle of the splitting nucleoids (i.e., midcell position for slow growth and $\frac{1}{4}$ and $\frac{3}{4}$ cell positions for fast growth) to create a directional motion. To our knowledge, there is no

evidence of such localized membrane incorporation. Furthermore, even if membrane growth was localized at the right places, the fluidity of the cytoplasmic membrane (PMID: 6996724, 20159151, 24735432, 27705775) would be problematic. To circumvent the membrane fluidity issue, one could potentially evoke an additional connection to the rigid peptidoglycan, but then again, peptidoglycan growth would have to be localized at the middle of the splitting nucleoid. However, peptidoglycan growth is dispersed early in the cell division cycle when the nucleoid splitting happens in fast growing cells and only appears to be zonal after the onset of cell constriction (PMID: 35705811, 36097171, 2656655).

(2) Even if we ignore the aforementioned caveats, Paul Wiggins's group ruled out the cell elongation/transertion model by showing that the rate of cell elongation is slower than the rate of chromosome segregation (PMID: 23775792). In the revised manuscript, we will clarify this point and provide confirmatory data showing that the cell elongation rate is indeed slower than the nucleoid segregation rate, indicating that it cannot be the main driver.

(3) Furthermore, our correlation analysis comparing the rate of nucleoid segregation to the rate of either cell elongation or polysome accumulation argues that polysome accumulation plays a larger role than cell elongation in nucleoid segregation. These data were already shown in Figure 1H and Figure 1 – figure supplement 3 of the original manuscript but were not highlighted in this context. We will revise the text to clarify this point.

(4) The asymmetries in nucleoid compaction that we described in our paper are predicted by our model. We do not see how they could be explained by cell growth or protein secretion.

(5) We also show that polysome accumulation at ectopic sites (outside the nucleoid) results in correlated nucleoid dynamics, consistent with our proposed mechanism. These nucleoid dynamics cannot be explained by cell growth or protein secretion (transertion).

(1a) As rifampicin also stops all translation, it also stops translational insertion of membrane proteins, which in many old models has been put forward as a possible driver of nucleoid segregation, and perhaps independent of growth. This should at last be mentioned in the discussion, or if there are past experiments that rule this out it would be great to note them.

It is not clear to us how the attachment of the DNA to the cytoplasmic membrane could alone create a directional force to move the sister nucleoids. We agree that old models have proposed a role for cell elongation (providing the force) and transertion (providing the membrane tether). Please see our response above for the evidence (from the literature and our work) against it. This was mentioned in the introduction and Results section, but we agree that this was not well explained. We will add experimental data and revise the text to clarify these points.

(1b) They address at great length in the discussion the possibility that growth may play a role in nucleoid segregation. However, this is testable - by stopping surface growth with antibiotics. Cells should still accumulate polysomes for some time, it would be easy to see if nucleoids are still segregated, and to what extent, thereby possibly decoupling growth and polysome production. If successful, this or similar experiments would further validate their model.

We reviewed the literature and could not find a drug that stops cell growth without stopping gene expression. Any drug that affects the membrane integrity or potential stops gene expression, which requires ATP. However, our experiment in which we drive polysome accumulation at ectopic sites decouples polysome accumulation from cell growth. In this experiment, by redirecting most of chromosome gene expression to a single plasmid-encoded gene, we reduce the rate of cell growth but still create a large accumulation of polysomes at

an ectopic location. This ectopic polysome accumulation is sufficient to affect nucleoid dynamics in a correlated fashion. In the revised manuscript, we will clarify this point and add model simulations to show that our experimental observations are predicted by our model.

(2) In the second experiment, they express excess TagBFP2 to delocalize polysomes from midcell. Here they again see the anticorrelation of the nucleoid and the polysomes, and in some cells, it appears similar to normal (polysomes separating the nucleoid) whereas in others the nucleoid has not separated. The one concern about this data - and the differences between the "separated" and "non-separated" nuclei - is that the over-expression of TagBFP2 has a huge impact on growth, which may also have an indirect effect on DNA replication and termination in some of these cells. Could the authors demonstrate these cells contain 2 fully replicated DNA molecules that are able to segregate?

We will perform the requested experiment.

(3) What is not clearly stated and is needed in this paper is to explain how polysomes do (or could) "exert force" in this system to segregate the nucleoid: what a "compaction force" is by definition, and what mechanisms causes this to arise (what causes the "force") as the "compaction force" arises from new polysomes being added into the gaps between them caused by thermal motions.

They state, "polysomes exert an effective force", and they note their model requires "steric effects (repulsion) between DNA and polysomes" for the polysomes to segregate, which makes sense. But this makes it unclear to the reader what is giving the force. As written, it is unclear if (a) these repulsions alone are making the force, or (b) is it the accumulation of new polysomes in the center by adding more "repulsive" material, the force causes the nucleoids to move. If polysomes are concentrated more between nucleoids, and the polysome concentration does not increase, the DNA will not be driven apart (as in the first case) However, in the second case (which seems to be their model), the addition of new material (new polysomes) into a sterically crowded space is not exerting force - it is filling in the gaps between the molecules in that region, space that needs to arise somehow (like via Brownian motion). In other words, if the polysome region is crowded with polysomes, space must be made between these polysomes for new polysomes to be inserted, and this space must be made by thermal (or ATP-driven) fluctuations of the molecules. Thus, if polysome accumulation drives the DNA segregation, it is not "exerting force", but rather the addition of new polysomes is iteratively rectifying gaps being made by Brownian motion.

We apologize for the understandable confusion. In our picture, the polysomes and DNA (conceptually considered as small plectonemic segments) basically behave as dissolved particles. If these particles were noninteracting, they would simply mix. However, both polysomes and DNA segments are large enough to interact sterically. So as density increases, steric avoidance implies a reduced conformational entropy and thus a higher free energy per particle. We argue (based on Miangolarra et al. PNAS 2021 PMID: 34675077 and Xiang et al. Cell 2021 PMID: 34186018) that the demixing of polysomes and DNA segments occurs because DNA segments pack better with each other than they do with polysomes. This raises the free energy cost associated with DNA-polysome interactions compared to DNA-DNA interactions. We model this effect by introducing a term in the free energy χ_{np} , which refer to as a repulsion between DNA and polysomes, though as explained above it arises from entropic effects. At realistic cellular densities of DNA and polysomes this repulsive interaction is strong enough to cause the DNA and polysomes to phase separate.

This same density-dependent free energy that causes phase separation can also give rise to forces, just in the way that a higher pressure on one side of a wall can give rise to a net force on the wall. Indeed, the “compaction force” we refer to is fundamentally an osmotic pressure difference. At some stages during nucleoid segregation, the region of the cell between nucleoids has a higher polysome concentration, and therefore a higher osmotic pressure, than the regions near the poles. This results in a net poleward force on the sister nucleoids that drives their migration toward the poles. This migration continues until the osmotic pressure equilibrates. Therefore, both phase separation (due to the steric repulsion described above) and nonequilibrium polysome production and degradation (which creates the initial accumulation of polysomes around midcell) are essential ingredients for nucleoid segregation.

This will be clarified in the revised text, with the support of additional simulation results.

The authors use polysome accumulation and phase separation to describe what is driving nucleoid segregation. Both terms are accurate, but it might help the less physically inclined reader to have one term, or have what each of these means explicitly defined at the start. I say this most especially in terms of "phase separation", as the currently huge momentum toward liquid-liquid interactions in biology causes the phrase "phase separation" to often evoke a number of wider (and less defined) phenomena and ideas that may not apply here. Thus, a simple clear definition at the start might help some readers.

Phase separation means that the DNA-polysome steric repulsion is strong enough to drive their demixing, which creates a compact nucleoid. As mentioned in a previous point, this effect is captured in the free energy by the χ_{np} term, which is an effective repulsion between DNA and polysomes, though as explained above it arises from entropic effects.

In the revised manuscript, we will illustrate this with our theoretical model by initializing a cell with a diffuse nucleoid and low polysome concentration. For the sake of simplicity, we assume that the cell does not elongate. We observe that the DNA-polysome steric repulsion is sufficient to compact the nucleoid and place it at mid-cell.

(4) Line 478. "Altogether, these results support the notion that ectopic polysome accumulation drives nucleoid dynamics". Is this right? Should it not read "results support the notion that ectopic polysome accumulation inhibits/redirects nucleoid dynamics"?

We think that this is correct; the ectopic polysome accumulation drives nucleoid dynamics. In our theoretical model, we can introduce polysome production at fixed sources to mimic the experiments where ectopic polysome production is achieved by high plasmid expression (Fig. 6). The model is able to recapitulate the two main phenotypes observed in experiments. These new simulation results will be added to the revised manuscript.

(5) It would be helpful to clarify what happens as the RplA-GFP signal decreases at midcell in Figure 1- is the signal then increasing in the less "dense" parts of the cell? That is, (a) are the polysomes at midcell redistributing throughout the cell? (b) is the total concentration of polysomes in the entire cell increasing over time?

It is a redistribution—the RplA-GFP signal remains constant in concentration from cell birth to division (Figure 1 – Figure Supplement 1E). This will be clarified in the revised text.

(6) Line 154. "Cell constriction contributed to the apparent depletion of ribosomal signal from the mid-cell region at the end of the cell division cycle (Figure 1B-C and Movie S1)" -

It would be helpful if when cell constriction began and ended was indicated in Figures 1B and C.

Good idea. We will add markers to indicate the start of cell constriction. We will also indicate that cell birth and division correspond to the first and last images/timepoint in Fig. 1B and C, respectively.

(7) In Figure 7 they demonstrate that radial confinement is needed for longitudinal nucleoid segregation. It should be noted (and cited) that past experiments of Bacillus l-forms in microfluidic channels showed a clear requirement role for rod shape (and a given width) in the positing and the spacing of the nucleoids.

Wu et al, Nature Communications, 2020 . "Geometric principles underlying the proliferation of a model cell system" <https://dx.doi.org/10.1038/s41467-020-17988-7>

Good point. We will add this reference. Thank you.

(8) "The correlated variability in polysome and nucleoid patterning across cells suggests that the size of the polysome-depleted spaces helps determine where the chromosomal DNA is most concentrated along the cell length. This patterning is likely reinforced through the displacement of the polysomes away from the DNA dense region"

It should be noted this likely functions not just in one direction (polysomes dictating DNA location), but also in the reverse - as the footprint of compacted DNA should also exclude (and thus affect) the location of polysomes

We agree that the effects could go both ways at this early stage of the story. We will revise the text accordingly.

(9) Line 159. Rifampicin is a transcription inhibitor that causes polysome depletion over time. This indicates that all ribosomal enrichments consist of polysomes and therefore will be referred to as polysome accumulations hereafter". Here and throughout this paper they use the term polysome, but cells also have monosomes (and 2 somes, etc). Rifampicin stops the assembly of all of these, and thus the loss of localization could occur from both. Thus, is it accurate to state that all transcription events occur in polysomes? Or are they grouping all of the n-somes into one group?

In the discussion, we noted that our term “polysomes” also includes monosomes for simplicity, but we agree that the term should have been defined much earlier. This will be done in the revised manuscript.

Thank you for the valuable comments and suggestions!

Reviewer #2 (Public review):

Summary:

The authors perform a remarkably comprehensive, rigorous, and extensive investigation into the spatiotemporal dynamics between ribosomal accumulation, nucleoid segregation, and cell division. Using detailed experimental characterization and rigorous physical models, they offer a compelling argument that nucleoid segregation rates are determined at least in part by the accumulation of ribosomes in the center of the cell, exerting a steric force to drive nucleoid segregation prior to cell division. This evolutionarily ingenious mechanism means cells can rely on ribosomal biogenesis as the sole determinant for the growth rate and cell division rate, avoiding the need for two separate 'sensors,' which would require careful coupling.

Terrific summary! Thank you for your positive assessment.

Strengths:

In terms of strengths; the paper is very well written, the data are of extremely high quality, and the work is of fundamental importance to the field of cell growth and division. This is an important and innovative discovery enabled through a combination of rigorous experimental work and innovative conceptual, statistical, and physical modeling.

Thank you!

Weaknesses:

In terms of weaknesses, I have three specific thoughts.

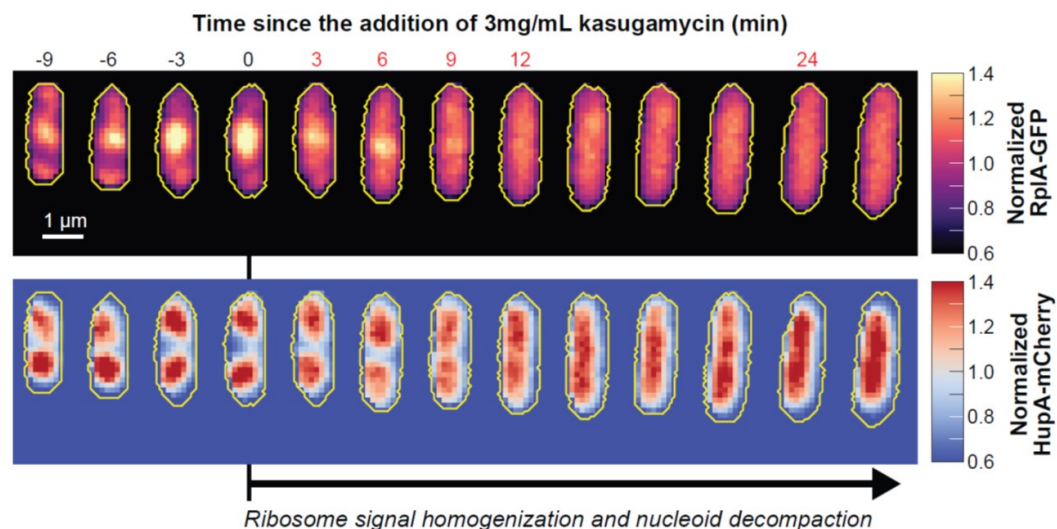
Firstly, my biggest question (and this may or may not be a bona fide weakness) is how unambiguously the authors can be sure their ribosomal labeling is reporting on polysomes, specifically. My reading of the work is that the loss of spatial density upon rifampicin treatment is used to infer that spatial density corresponds to polysomes, yet this feels like a relatively indirect way to get at this question, given rifampicin targets RNA polymerase and not translation. It would be good if a more direct way to confirm polysome dependence were possible.

The heterogeneity of ribosome distribution inside *E. coli* cells has been attributed to polysomes by many labs (PMID: 25056965, 38678067, 22624875, 31150626, 34186018, 10675340). The attribution is also consistent with single-molecule tracking experiments showing that slow-moving ribosomes (polysomes) are excluded by the nucleoid whereas fast-diffusing ribosomes (free ribosomal subunits) are distributed throughout the cytoplasm (PMID: 25056965, 22624875).

Furthermore, inhibition of translation initiation with kasugamycin treatment, which decreases the pool of polysomes, results in a homogenization of ribosomes and expansion of the nucleoid (see Author response image 1). This further supports the rifampicin experiments. Given that the attribution of ribosome heterogeneity to polysomes is well accepted in the field, we would prefer to not include these kasugamycin data in the revised manuscript because long-term exposure to this drug leads to nucleoid re-compaction (PMID: 25250841 and PMID: 34186018). This secondary effect may possibly be due to a dysregulated increase in synthesis of naked rRNAs (PMID: 14460744, PMID: 2114400, and PMID: 2448483) or ribosome aggregation, which we are currently investigating.

Author response image 1.

Effects of kasugamycin treatment on the intracellular distribution of ribosomes and nucleoids. Representative single cell (CJW7323) growing in M9gluCAAT. Kasugamycin (3 mg/mL) was added at time = 0 min. Show is the early response (0-30 min) to the drug characterized by the homogenization of the ribosomal RplA-GFP fluorescence and the expansion of the HupA-mCherry-labeled nucleoids. For each segmented cell, the RplA-GFP and HupA-mCherry signals were normalized by the average fluorescence.



Second, the authors invoke a phase separation model to explain the data, yet it is unclear whether there is any particular evidence supporting such a model, whether they can exclude simpler models of entanglement/local diffusion (and/or perhaps this is what is meant by phase separation?) and it's not clear if claiming phase separation offers any additional insight/predictive power/utility. I am OK with this being proposed as a hypothesis/idea/working model, and I agree the model is consistent with the data, BUT I also feel other models are consistent with the data. I also very much do not think that this specific aspect of the paper has any bearing on the paper's impact and importance.

We appreciate the reviewer's comment, but the output of our reaction-diffusion model is a bona fide phase separation (spinodal decomposition). So, we feel that we need to use the term when reporting the modeling results. Inside the cell, the situation is more complicated. As the reviewer points out, there likely are entanglements (not considered in our model) and other important factors (please see our discussion on the model limitations). This said, we will revise our text to clarify our terms and proposed mechanism.

Finally, the writing and the figures are of extremely high quality, but the sheer volume of data here is potentially overwhelming. I wonder if there is any way for the authors to consider stripping down the text/figures to streamline things a bit? I also think it would be useful to include visually consistent schematics of the question/hypothesis/idea each of the figures is addressing to help keep readers on the same page as to what is going on in each figure. Again, there was no figure or section I felt was particularly unclear, but the sheer volume of text/data made reading this quite the mental endurance sport! I am completely guilty of this myself, so I don't think I have any super strong suggestions for how to fix this, but just something to consider.

We agree that there is a lot to digest. We will add schematics and a didactic simulation. We will also try to streamline the text.

Reviewer #3 (Public review):

Summary:

Papagiannakis et al. present a detailed study exploring the relationship between DNA/polysome phase separation and nucleoid segregation in *Escherichia coli*. Using a combination of experiments and modelling, the authors aim to link physical principles

with biological processes to better understand nucleoid organisation and segregation during cell growth.

Strengths:

The authors have conducted a large number of experiments under different growth conditions and physiological perturbations (using antibiotics) to analyse the biophysical factors underlying the spatial organisation of nucleoids within growing E. coli cells. A simple model of ribosome-nucleoid segregation has been developed to explain the observations.

Weaknesses:

While the study addresses an important topic, several aspects of the modelling, assumptions, and claims warrant further consideration.

Thank you for your feedback. Please see below for a response to each concern.

Major Concerns:

Oversimplification of Modelling Assumptions:

The model simplifies nucleoid organisation by focusing on the axial (long-axis) dimension of the cell while neglecting the radial dimension (cell width). While this approach simplifies the model, it fails to explain key experimental observations, such as:

(1) Inconsistencies with Experimental Evidence:

The simplified model presented in this study predicts that translation-inhibiting drugs like chloramphenicol would maintain separated nucleoids due to increased polysome fractions. However, experimental evidence shows the opposite-separated nucleoids condense into a single lobe post-treatment (Bakshi et al 2014), indicating limitations in the model's assumptions/predictions. For the nucleoids to coalesce into a single lobe, polysomes must cross the nucleoid zones via the radial shells around the nucleoid lobes.

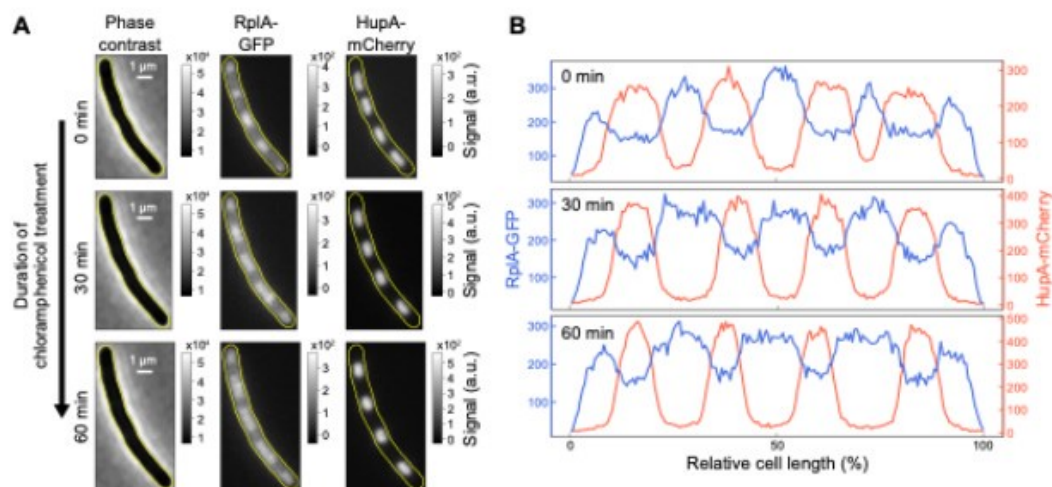
We do not think that the results from chloramphenicol-treated cells are inconsistent with our model. Our proposed mechanism predicts that nucleoids will condense in the presence of chloramphenicol, consistent with experiments. It also predicts that nucleoids that were still relatively close at the time of chloramphenicol treatment could fuse if they eventually touched through diffusion (thermal fluctuation) to reduce their interaction with the polysomes and minimize their conformational energy. Fusion is, however, not expected for well-separated nucleoids since their diffusion is slow in the crowded cytoplasm. This is consistent with our experimental observations: In the presence of a growth-inhibitory concentration of chloramphenicol (70 µg/mL), nucleoids in relatively close proximity can fuse, but well-separated nucleoids condense and do not fuse. Since the growth rate inhibition is not immediate upon chloramphenicol treatment, many cells with well-separated condensed nucleoids divide during the first hour. As a result, the non-fusion phenotype is more obvious in non-dividing cells, achieved by pre-treating cells with the cell division inhibitor cephalixin (50 µg/mL). In these polyploid elongated cells, well-separated nucleoids condensed but did not fuse, not even after an hour in the presence of chloramphenicol (as illustrated in Author response image 2).

In Bakshi et al, 2014, nucleoid fusion was shown for a single cell in which the sister nucleoids were relatively close to each other at the time of chloramphenicol treatment. Population statistics were provided for the relative length and width of the nucleoids, but not for the fusion events. So, it is unclear whether the illustrated fusion was universal or not. Also, we note that Bakshi et al (2014) used a chloramphenicol concentration of 300 µg/mL, which is 20-

fold higher than the minimal inhibitory concentration for growth, compared to 70 $\mu\text{g}/\text{mL}$ in our experiments.

Author response image 2.

Effects of chloramphenicol treatment on the intracellular distribution of ribosomes and nucleoids in non-dividing cells. Exponentially growing cells (M9glyCAAT at 30°C) were pre-treated with cephalixin for one hour before being spotted on an 1% agarose pad for time-lapse imaging. The agarose pad contained M9glyCAAT, cephalixin, and chloramphenicol. (A) Phase contrast, RplA-GFP fluorescence and HupA-mCherry fluorescence images of a representative single cell. Three timepoints are shown, including the first image after spotting on the agarose pad (at 0 min), 30 minutes and one hour of chloramphenicol treatment. (B) One-dimensional profiles of the ribosomal (RplA-GFP) and nucleoid (HupA-mCherry) fluorescence from the cells shown in panel A. These intensity profiles correspond to the average fluorescence along the medial axis of the cell considering a 6-pixel region (0.4 μm) centered on the central line of the cell. The fluorescence intensity is plotted along the relative cell length, scaled from 0 to 100% between the two poles, illustrating the relative nucleoid length ($L_{\text{DNA}}/L_{\text{cell}}$) that was plotted by Bakshi et al in 2014 (PMID: 25250841).



(2) The peripheral localisation of nucleoids observed after A22 treatment in this study and others (e.g., Japaridze et al., 2020; Wu et al., 2019), which conflicts with the model's assumptions and predictions. The assumption of radial confinement would predict nucleoids to fill up the volume or ribosomes to go near the cell wall, not the nucleoid, as seen in the data.

The reviewer makes a good point that DNA attachment to the membrane through transertion likely contributes to the nucleoid being peripherally localized in A22 cells. We will revise the text to add this point. However, we do not think that this contradicts the proposed nucleoid segregation mechanism based on phase separation and out-of-equilibrium dynamics described in our model. On the contrary, by attaching the nucleoid to the cytoplasmic membrane along the cell width, transertion might help reduce the diffusion and thus exchange of polysomes across nucleoids. We will revise the text to discuss transertion over radial confinement.

(3) The radial compaction of the nucleoid upon rifampicin or chloramphenicol treatment, as reported by Bakshi et al. (2014) and Spahn et al. (2023), also contradicts the model's predictions. This is not expected if the nucleoid is already radially confined.

We originally evoked radial confinement to explain the observation that polysome accumulations do not equilibrate between DNA-free regions. We agree that transertion is an alternative explanation. Thank you for bringing it to our attention. However, please note that this does not contradict the model. In our view, it actually supports the 1D model by providing a reasonable explanation for the slow exchange of polysomes across DNA-free regions. The attachment of the nucleoid to the membrane along the cell width may act as diffusion barrier. We will revise the text and the title of the manuscript accordingly.

(4) Radial Distribution of Nucleoid and Ribosomal Shell:

The study does not account for well-documented features such as the membrane attachment of chromosomes and the ribosomal shell surrounding the nucleoid, observed in super-resolution studies (Bakshi et al., 2012; Sanamrad et al., 2014). These features are critical for understanding nucleoid dynamics, particularly under conditions of transcription-translation coupling or drug-induced detachment. Work by Yongren et al. (2014) has also shown that the radial organisation of the nucleoid is highly sensitive to growth and the multifork nature of DNA replication in bacteria.

We will discuss the membrane attachment. Please see the previous response.

The omission of organisation in the radial dimension and the entropic effects it entails, such as ribosome localisation near the membrane and nucleoid centralisation in expanded cells, undermines the model's explanatory power and predictive ability. Some observations have been previously explained by the membrane attachment of nucleoids (a hypothesis proposed by Rabinovitch et al., 2003, and supported by experiments from Bakshi et al., 2014, and recent super-resolution measurements by Spahn et al.).

We agree—we will add a discussion about membrane attachment in the radial dimension. See previous responses.

Ignoring the radial dimension and membrane attachment of nucleoid (which might coordinate cell growth with nucleoid expansion and segregation) presents a simplistic but potentially misleading picture of the underlying factors.

As mentioned above, we will discuss membrane attachment in the revised manuscript.

This reviewer suggests that the authors consider an alternative mechanism, supported by strong experimental evidence, as a potential explanation for the observed phenomena:

*Nucleoids may transiently attach to the cell membrane, possibly through transertion, allowing for coordinated increases in nucleoid volume and length alongside cell growth and DNA replication. Polysomes likely occupy cellular spaces devoid of the nucleoid, contributing to nucleoid compaction due to mutual exclusion effects. After the nucleoids separate following *ter* separation, axial expansion of the cell membrane could lead to their spatial separation.*

This “membrane attachment/cell elongation” model is reminiscent to the hypothesis proposed by Jacob et al in 1963 (doi:10.1101/SQB.1963.028.01.048). There are several lines of evidence arguing against it as the major driver of nucleoid segregation:

(Below is a slightly modified version of our response to a comment from Reviewer 1—see page 3)

(1) For this alternative model to work, axial membrane expansion (i.e., cell elongation) would have to be localized at the middle of the splitting nucleoids (i.e., midcell position for slow

growth and $\frac{1}{4}$ and $\frac{3}{4}$ cell positions for fast growth) to create a directional motion. To our knowledge, there is no evidence of such localized membrane incorporation. Furthermore, even if membrane growth was localized at the right places, the fluidity of the cytoplasmic membrane (PMID: 6996724, 20159151, 24735432, 27705775) would be problematic. To go around this fluidity issue, one could potentially evoke a potential connection to the rigid peptidoglycan, but then again, peptidoglycan growth would have to be localized at the middle of the splitting nucleoid to “push” the sister nucleoid apart from each other. However, peptidoglycan growth is dispersed prior to cell constriction (PMID: 35705811, 36097171, 2656655).

(2) Even if we ignore the aforementioned caveats, Paul Wiggins’s group ruled out the cell elongation/transertion model by showing that the rate of cell elongation is slower than the rate of chromosome segregation (PMID: 23775792). In the revised manuscript, we will provide additional data showing that the cell elongation rate is indeed slower than the nucleoid segregation rate.

(3) Furthermore, our correlation analysis comparing the rate of nucleoid segregation to the rate of either cell elongation or polysome accumulation argues that polysome accumulation plays a larger role than cell elongation in nucleoid segregation. These data were already shown in the original manuscript (Figure 1I and Figure 1 – figure supplement 3) but were not highlighted in this context. We will revise the text to clarify this point.

(4) The membrane attachment/cell elongation model does not explain the nucleoid asymmetries described in our paper (Figure 3), whereas they can be recapitulated by our model.

(5) The cell elongation/transertion model cannot predict the aberrant nucleoid dynamics observed when chromosomal expression is largely redirected to plasmid expression. In the revised manuscript, we will add simulation results showing that these nucleoid dynamics are predicted by our model.

In line of these arguments, we do not believe that a mechanism based on membrane attachment and cell elongation is the major driver of nucleoid segregations. However, we do believe that it may play a complementary role (see “Nucleoid segregation likely involves multiple factors” in the Discussion). We will revise this section to clarify our thoughts and mention the potential role of transertion.

Incorporating this perspective into the discussion or future iterations of the model may provide a more comprehensive framework that aligns with the experimental observations in this study and previous work.

As noted above, we will revise the text to mention about transertion.

Simplification of Ribosome States:

Combining monomeric and translating ribosomes into a single 'polysome' category may overlook spatial variations in these states, particularly during ribosome accumulation at the mid-cell. Without validating uniform mRNA distribution or conducting experimental controls such as FRAP or single-molecule measurements to estimate the proportions of ribosome states based on diffusion, this assumption remains speculative.

Indeed, for simplicity, we adopt an average description of all polysomes with an average diffusion coefficient and interaction parameters, which is sufficient for capturing the fundamental mechanism underlying nucleoid segregation. To illustrate that considering multiple polysome species does not change the physical picture, we consider an extension of our model, which contains three polysome species, each with a different diffusion coefficient ($D_p = 0.018, 0.023, \text{ or } 0.028 \mu\text{m}^2/\text{s}$), reflecting that polysomes with more ribosomes will have a

lower diffusion coefficient. Simulation of this model reveals that the different polysome species have essentially the same concentration distribution, suggesting that the average description in our minimal model is sufficient for our purposes. We will present these new simulation results in the revised manuscript.

<https://doi.org/10.7554/eLife.104276.1.sa0>